

STATE OF CHARGE ESTIMATION FOR ELECTRIC VEHICLE BATTERY PACK USING CUBATURE KALMAN FILTER

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ABSTRACT

Accurate state of charge (SoC) estimation is central to safe, reliable operation of the battery management system (BMS) in a retrofitted electric vehicle, since terminal voltage and current alone do not reveal how much usable energy a lithium-ion pack has left. This dissertation develops and evaluates a real-time SoC estimator for a *38S2P* retrofitted-EV battery pack, built around a second-order resistor–capacitor (*2RC*) equivalent-circuit model whose parameters are identified from Hybrid Pulse Power Characterization (HPPC) data generated in MATLAB/Simulink across an eleven-point temperature sweep from 0°C to 50°C .

The estimator itself is developed as a six-variant cubature Kalman filter (CKF) ablation chain, in which each variant adds exactly one mechanism on top of the last: a square-root covariance formulation for numerical stability, an adaptive process- and measurement-noise update, Huber robust innovation weighting, pack-current and pack-voltage sensor-bias states, and finally a dual information gate that shapes the Kalman gain using both the local OCV slope and the current-excitation level. All six variants are evaluated under an identical protocol on the same real HPPC record, with results scored on a held-out test partition rather than the training data.

The results show that accuracy does not improve monotonically as mechanisms are added. The adaptive-noise variant achieves the lowest test-partition SoC RMSE (*1.163%*) among all six variants, while adding sensor-bias states without the dual information gate causes a sharp regression (RMSE rising to *1.511%*), because the real HPPC excitation used in this study carries no sensor bias for the added states to absorb. The fully-featured proposed variant, with dual gating, partially recovers from this regression and achieves the best full-run RMSE (*0.961%*) and the most favourable numerical conditioning of the six, but it does not lead on test-partition RMSE, voltage-domain fidelity, or innovation calibration, trade-offs that are documented explicitly rather than averaged away.

These findings support carrying the adaptive-noise variant forward as the strongest evidenced candidate on accuracy grounds, with the fully-featured dual-gated variant retained specifically for the low-excitation, bias-fragile conditions under which it performs best. The thesis concludes with an explicit, itemised account of the further

evidence, measured pack data, a multi-temperature sweep, and a dedicated sensor-fault stress test, that any single unconditional estimator recommendation would still depend on.

Keywords: *Battery Management System (BMS); Cubature Kalman Filter (CKF); Equivalent Circuit Model (ECM); Retrofitted Electric Vehicle (EV); State of Charge (SoC) Estimation.*

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
<i>BMS</i>	Battery management system	<i>NIS</i>	Normalised innovation squared
<i>CKF</i>	Cubature Kalman filter	<i>OCV</i>	Open-circuit voltage
<i>SRCKF</i>	Square root cubature Kalman filter	<i>PCHIP</i>	Piecewise cubic Hermite interpolating polynomial
<i>EKF</i>	Extended Kalman filter	<i>PF</i>	Particle filter
<i>UKF</i>	Unscented Kalman filter	<i>AEKPF</i>	Adaptive extended Kalman particle filter
<i>ECM</i>	Equivalent circuit model	<i>RC</i>	Resistor capacitor (branch)
<i>2RC</i>	Second-order resistor-capacitor equivalent circuit	<i>RMSE</i>	Root mean square error
<i>SoC</i>	State of charge	<i>MAE</i>	Mean absolute error
<i>SoH</i>	State of health	<i>RAE</i>	Relative absolute error
<i>HPPC</i>	Hybrid pulse power characterisation	<i>EIS</i>	Electrochemical impedance spectroscopy
<i>38S2P</i>	38 cells in series, 2 strings in parallel	<i>QR</i>	QR matrix decomposition

Symbols

Symbol	Meaning	Symbol	Meaning
SoC_k	State of charge at sample k	x_k	Filter state vector at sample k
U_{1k}, U_{1k}	Fast/slow RC -branch voltages	P_k, S_k	State covariance and its Cholesky/square root factor
bI, bV	Pack current/voltage sensor bias states	Q, R	Process and measurement noise covariance
I_{pack_k}, I_{cell_k}	Pack and cell level current	K_k	Kalman gain at sample k
V_{pack_k}	Simulated/predicted pack terminal voltage	v_k	Innovation (measurement residual)
R_0, R_1, R_2	Cell ohmic and RC -branch resistances	$P_{yy,k}, P_{xy,k}$	Innovation covariance and state–output cross-covariance
τ_1, τ_2	RC branch time constants	NIS_k	Normalised innovation squared, sample k
OCV	Open circuit voltage	$gSoC_k, gexc_k$	SoC and excitation information gates
N_s, N_p	Series cell count, parallel string count	w_k	Huber robust weight
Q_{mod}	Pack-equivalent capacity used in Eq. 3.1	n	State dimension
Δt	Sample interval	N	Number of samples in the scored record
η	Coulombic efficiency	e_k	Signed SoC error, $\widehat{SoC}_k - SoC_{ref_k}$

CHAPTER 1: INTRODUCTION

1.1 Introduction

Global energy demand keeps climbing, and along with it come tighter emissions targets and shrinking fossil fuel reserves. Road transport alone accounts for a large share of petroleum use and urban air pollution, which is a major reason why electric vehicles (EVs) have moved from a niche alternative to a mainstream part of transportation policy. Because EVs draw all or part of their propulsion energy from electricity, they cut local emissions, use energy more efficiently than an internal combustion drivetrain, and open the door to renewable charging. Their progress is now closely tied to how power and energy systems evolve more broadly.

Among the subsystems that make up an EV, the electrochemical energy storage system has a significant effect on range, acceleration, charging time, safety, reliability, and cost. Lithium ion cells dominate this role because of their high energy density, power capability, low self-discharge, and long cycle life. However, batteries are nonlinear and time-varying; terminal voltage and current alone do not reveal how much usable energy remains. Temperature, current rate, state of charge, ageing, and prior operating history all shape how a cell behaves; therefore, knowing the internal state of the battery matters as much as knowing its terminal readings.

The state of charge (SoC) is the quantity that a battery management system (BMS) needs most. It expresses, as a percentage, the ratio of charge presently available to usable capacity, which is essentially the EV equivalent of a fuel gauge. Unlike a fuel tank, though, SoC cannot be read directly during normal operation; it has to be inferred from current, voltage, and temperature measurements. If it is not correctly obtained, the result is either an unexpected loss of propulsion, capacity left unused, or operation past safe charge and discharge limits. Repeated overcharging or deep discharge speeds up degradation, and, in bad cases, creates a genuine safety hazard.

The literature offers several SoC estimation approaches, each with its own limitations. Coulomb counting is simple and sufficiently fast for real-time use, but drifts with any error in the initial SoC, current sensor bias, or capacity assumptions. Open circuit voltage methods work well after a sufficiently long rest, a condition that rarely holds during actual driving. Model-based observers, including the Extended Kalman filter

(EKF), Unscented Kalman filter (UKF), Cubature Kalman filter (CKF), and various particle filter variants, pair a battery model with noisy measurements and correct the state recursively. Data-driven methods, including recurrent networks and ensemble learners, can capture complex relationships from data; however, they only do so if the training data are representative and the validation split avoids leakage.

This dissertation adopts a second order resistor capacitor ($2RC$) equivalent circuit model to represent the dynamics of a lithium-ion battery: an open-circuit voltage source, an ohmic resistance, and two RC branches to represent fast and slow transient behaviour. It is sufficiently accurate for use and sufficiently simple to implement in MATLAB/Simulink for real-time BMS work. The parameters were identified from hybrid pulse power characterization (HPPC) data generated for a $38S2P$ battery pack across a temperature sweep from 0°C to 50°C in 5°C steps, using discharge pulses at $0.5C$, $1C$, $1.5C$, and $2C$ and charge pulses at $0.1C$, $0.2C$, $0.3C$, and $0.4C$, arranged in rest-discharge-rest-charge-rest cycles. The purpose of this study was to investigate reliable SoC estimation under dynamic loading and to compare relevant estimation methods under common, consistent test conditions.

1.2 Electric Vehicles and Their Requirements

An electric vehicle produces traction using one or more electric machines. Depending on the powertrain, the vehicle can be a battery electric vehicle, hybrid electric vehicle, plug-in hybrid, or fuel cell vehicle. A battery electric vehicle draws all of its propulsion energy from a rechargeable pack, whereas hybrid designs pair electrical storage with another energy source. Regardless of the configuration, the electrical subsystems are broadly the same and include a traction battery, BMS, power electronic converter, electric motor, vehicle control unit, charging interface, thermal management system, and low-voltage auxiliary supply.

The traction battery must satisfy the requirements that pull in different directions. It needs sufficient stored energy for an acceptable range and sufficient power for acceleration and regenerative braking, and it must charge quickly without breaching voltage, current, temperature, or stress limits on the cells. A practical pack wires cells in series and parallel to reach the target voltage and ampere-hour capacity; series connections raise pack voltage, and parallel connections raise current capability and

capacity. Cell-level voltage and resistance parameters must therefore scale consistently with the pack-level current and capacity during modelling and estimation.

Safety and reliability are central to EV design. The BMS must prevent overcharge, deep discharge, excessive current, and unsafe temperature; it monitors cell voltages and temperatures, flags abnormal conditions, supports cell balancing, and reports permissible charge and discharge power to the vehicle controller. Thermal management matters because low temperature reduces available power and increases resistance, while high temperature speeds up ageing and raises the risk of thermal runaway. Cell-to-cell variation compounds the problem, since the weakest cell in a pack can hit a limit well before the pack's average state suggests any limit is close.

Accurate SoC, state of health (SoH), and available power estimates underpin all of this. SoC feeds range prediction and energy management. SoH tracks the gradual loss of usable capacity or rise in internal resistance relative to a baseline. Available power estimation tells the controller how much charge or discharge current is safe. These three are coupled: uncertainty in capacity or resistance throws off SoC, and a poor SoC estimate in turn distorts how degradation gets read. A good estimator therefore needs to stay stable under measurement noise, uncertain initialization, parameter drift, and a current profile that keeps changing.

From the grid's perspective, more EVs on the road cut both ways. Uncoordinated charging can push up peak demand, worsen voltage profiles, and stress distribution equipment, while coordinated charging and vehicle-to-grid operation can help with load management and renewable integration instead. At the vehicle level, using the battery efficiently stretches the range and slows unnecessary ageing. Reliable battery state estimation, then, serves both the onboard BMS and the vehicle's interaction with the grid.

1.3 Problem Formulation and Assumptions

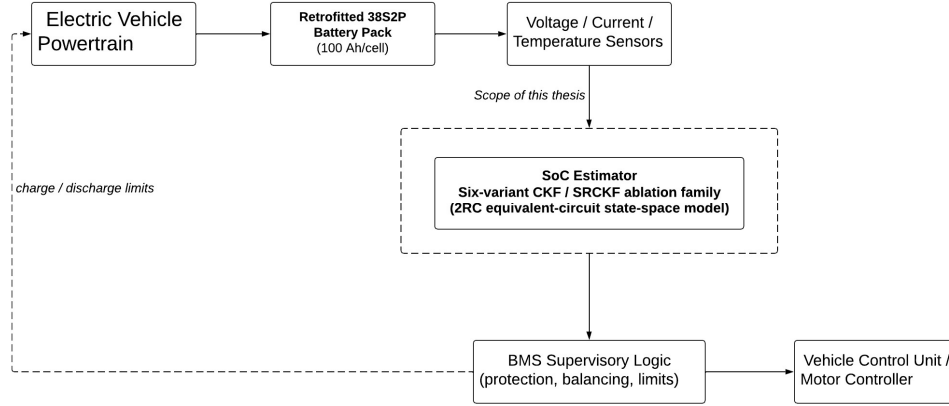


Figure 1.1: System-level context of the SoC estimator within a retrofitted-EV powertrain and BMS.

This dissertation addresses the estimation of battery SoC from measurable electrical signals during dynamic operation. Let $I(t)$ denote the pack current, which is positive for discharge, and $V_i(t)$ denote the measured terminal voltage. Charge balance gives the SoC dynamics as

$$\frac{dSoC(t)}{dt} = -\frac{-\eta(I, T)I(t)}{Q_n} \dots \dots (1.1)$$

where Q_n is the usable pack capacity in ampere-seconds and η is the charge or discharge efficiency. Coulomb counting integrates this relationship directly, but any error in the initial SoC, current measurement, efficiency, or capacity accumulates over time. Voltage feedback corrects for this through a 2RC equivalent-circuit model. At cell level, terminal voltage is represented as

$$Vt(t) = OCV(SoC, T) - I_{cell}(t)R_0 - V_1(t) - V_2(t) \dots \dots (1.2)$$

where OCV is the relaxed open-circuit voltage, R_0 is the ohmic resistance, and V_1 and V_2 are the polarization voltages across the fast and slow RC branches, respectively. Their dynamics are governed by time constants $\tau_1 = R_1C_1$ and $\tau_2 = R_2C_2$, with $\tau_1 < \tau_2$ imposed to maintain the physical interpretation. The parameters were identified from the Hybrid Pulse Power Characterization (HPPC) data by detecting the pulse and recovery intervals and minimizing the error between the measured and reconstructed terminal voltages. For this study, the HPPC data were generated in MATLAB/Simulink for a 38S2P pack over a temperature range of 0°C to 50°C in 5°C steps, using discharge

pulses at $0.5C$, $1C$, $1.5C$, and $2C$ and charge pulses at $0.1C$, $0.2C$, $0.3C$, and $0.4C$, arranged as rest-discharge-rest-charge-rest cycles.

The estimator must recover the SoC while suppressing sensor noise and correcting for an imperfect initial value. Model-based filters carry the state vector, $x = [SoC \ V_1 \ V_2]^T$ and combine model prediction with terminal-voltage measurement recursively. When learning-based estimators are used, they take the same input signals and are tested on complete held-out drive cycles rather than randomly shuffled samples, as mixing training and test data at the row level would allow the model to see the future it is meant to predict. Every method in the comparison runs on the same time grid, reference SoC, test interval, OCV relationship, initialization, and sample count; therefore, any difference in the reported performance comes from the algorithm itself and not from unequal test conditions.

The objectives are to build a consistent battery data workflow, formulate and parameterize a $2RC$ model, implement the SoC estimators in *MATLAB R2020a*, reconstruct the terminal voltage as an independent check, quantify the SoC and voltage error, and test the robustness to initialization error, noise, and changing operating conditions. When the reference SoC is available, the performance is judged on root-mean-square error, mean absolute error, maximum absolute error, convergence behaviour, and computational cost.

A few assumptions hold throughout. Cells in the pack have similar chemistry and nominal characteristics, and the series-parallel topology, $38S2P$, is known. The pack current follows a declared sign convention and is divided by the number of parallel strings to obtain the cell current. The initial SoC and rated capacity are known approximately, not exactly. OCV comes from relaxed measurements or a defensible monotonic SoC lookup; the loaded terminal voltage is never treated as OCV. The $2RC$ parameters remain positive and may be held constant over a limited operating region or scheduled by SoC and temperature where sufficient identification data exist. Measurement disturbances are finite and captured by suitable process and measurement noise statistics. Where the reference SoC is used for evaluation, it shares the same time base and never enters unsupervised estimator training.

These assumptions ensure that the investigation is repeatable while preserving the electrical dynamics that matter for SoC estimation. They do not claim that every cell in

a real traction battery behaves identically. Conclusions drawn from simulation or a single test record still need validation on unseen cycles, temperatures, and batteries before anyone claims general applicability or real-time readiness.

1.4 Outline of the Dissertation

The remainder of this dissertation is organized as follows. Chapter 2 reviews EV battery technologies, modelling approaches, parameter identification methods, and established SoC estimation techniques. It covers battery modelling types, SoC and SoH estimation, Kalman-filter and particle-filter variants, and machine-learning and deep-learning algorithms before introducing the proposed hybrid algorithm. It identifies gaps in existing approaches and sets up the problem this dissertation addresses.

Chapter 3 covers the simulation work and its results. It describes the Simulink model and the HPPC test data generated for the $38S2P$ pack over the 0°C to 50°C temperature sweep, along with the $2RC$ equivalent-circuit formulation, including how the $OCV\text{-}SoC$ relationship is extracted, how usable HPPC pulse windows are detected, how model parameters are estimated, and how the voltage model is validated. Every estimator, from Coulomb counting through the Kalman and particle filter variants to the machine learning and hybrid methods, is evaluated on the same samples using SoC error and voltage-reconstruction metrics. Comparative plots, error distributions, convergence behaviour, computational cost, and robustness under initialization error, sensor disturbance, and temperature variation are all examined, with the discussion drawing a line between trajectory fitting and genuine generalization and relating the numbers back to practical BMS requirements.

Chapter 4 concludes with the main findings and contributions, states the limitations arising from the chosen model, data coverage, and assumptions, and proposes extensions such as temperature-dependent parameter surfaces, online parameter adaptation, joint SoC-SoH estimation, cell-level imbalance analysis, and hardware-in-the-loop or real time validation. Chapter 5 lists the references cited throughout the dissertation.

CHAPTER 2: LITERATURE REVIEW AND TECHNICAL BACKGROUND

2.1 Battery Modelling and its Types

Battery models are conventionally grouped by how much physical or electrochemical detail they carry, and by whether they describe the cell's electrical behaviour, its internal chemistry, or its thermal state. At one end sit empirical and semi-empirical formulations, curve fits with little physical grounding. Equivalent circuit models occupy the middle ground, representing the cell as resistors, capacitors, and a voltage source, chosen because the resulting equations are simple enough to run on a microcontroller. Electrochemical models sit at the other end, solving the coupled partial differential equations that govern lithium transport and reaction kinetics inside the cell. A fourth category, data driven and machine learning models, sidesteps physical structure altogether and learns the mapping between measurable signals and battery state directly from data. Running alongside all four is a fifth concern that none of them addresses on its own: thermal behaviour, since every one of the electrical parameters discussed below drifts with temperature, and a pack that is not thought about thermally will eventually misbehave regardless of how good its SoC estimator is. This section works through each of these five families in turn, weighs their trade offs for a retrofitted electric vehicle BMS, and closes with a comparison that motivates the modelling choice carried through the rest of this dissertation.

2.1.1 Empirical and Semi-Empirical Models

Empirical models fit a mathematical function, exponential, linear, or polynomial, directly to observed capacity fade or resistance growth, with coefficients obtained by regression on cycling data rather than derived from first principles [1]. Their appeal is simplicity: no partial differential equations, no electrode geometry, just a curve that tracks the trend reasonably well over the conditions it was fitted to. Semi empirical models add a thin layer of physical reasoning on top of this curve-fitting exercise. A representative example is the Verhulst-type growth law adapted for battery ageing, in which a bounded, S-shaped degradation trend is assumed on the strength of a rough physical analogy, a population-growth model repurposed for capacity fade, and then fitted to data rather than derived from electrochemical kinetics [2].

The weakness of this family is generalisation. A polynomial fitted to one cell chemistry, one temperature band, and one duty cycle rarely extrapolates safely outside that envelope, and the coefficients carry no physical meaning that would let an engineer sanity-check them against first principles. For a retrofitted EV pack that will see irregular usage patterns quite unlike a laboratory cycling protocol, this is a real limitation. That said, empirical fits remain useful as quick, low cost baselines, and as a sanity check against the more elaborate models discussed in the remainder of this section.

2.1.2 Equivalent Circuit Models (ECM)

Equivalent circuit models describe the cell's electrical terminal behaviour using discrete components: a voltage source for the open circuit voltage (OCV), an ohmic resistance, and one or more resistor-capacitor (RC) branches that reproduce the polarisation dynamics. The simplest form, the R_{int} model, uses only a series resistance and ignores transient response entirely, which is too crude for anything beyond the roughest estimate. The Thevenin model adds a single RC branch and captures one time constant of relaxation. Comparative studies on ECM variants have repeatedly found it strikes a reasonable balance between simplicity and dynamic fidelity for chemistries such as ternary lithium cells [3]. Higher order variants, the second order ($2RC$) model chief among them, add a second RC branch to separate fast polarisation, largely charge transfer and double layer effects, from the slower diffusion dominated relaxation, and this structure has become close to a default choice in the recent SoC estimation literature [4,5,6].

For an n^{th} order RC network, the governing equations are:

for each RC branch i

$$Cp_i \left(\frac{dVp_i}{dt} \right) + \frac{Vp_i}{Rp_i} = I(t) \dots \dots (2.1)$$

$$V_t(t) = OCV(SoC) - I(t)R_0 - \sum Vp_i(t) \dots \dots (2.2)$$

$$z(t) = z(0) - \int \frac{\eta I(t) dt}{Q} \dots \dots (2.3)$$

where V_t is terminal voltage, I is load current (taken positive on discharge), R_0 is the ohmic resistance, Vp_i and Rp_i , Cp_i are the voltage and parameters of the i^{th} RC branch, z is SoC , η is coulombic efficiency, and Q is the rated capacity. The open-circuit

voltage is not a constant but a function of SoC , obtained from rest-voltage measurements and fitted as a monotonic lookup table or a low-order polynomial, since the OCV - SoC relationship forms the static backbone on which the dynamic RC terms are superimposed.

Beyond the plain n RC family, two elaborations recur in the literature. The first accounts for hysteresis by adding a voltage term that depends on the sign of the current and decays with a fitted rate constant, producing what is usually labelled the n RC model. Secondly, the PNGV (Partnership for a New Generation of Vehicles) model, adds a large series capacitor to the Thevenin structure to capture the slow accumulation of charge that shifts the OCV over long timescales. A comparative study across five ECM structures and five parameter identification algorithms found that the PNGV model held a distinct accuracy advantage at low SoC , while the second order RC model, combined with a reasonably fast identification method such as particle swarm optimisation, offered the best accuracy to computation trade off across the rest of the SoC range [7]. Because ECM parameters are not constants but vary with SoC and, more importantly for this dissertation, with temperature, a temperature-dependent formulation of the same $2RC$ structure has been shown to hold SoC estimation error within about $\pm 0.4\%$ across a wide ambient temperature range when paired with a dual Kalman filter [4]. This same basic circuit can be pushed further by inserting a static output nonlinearity after a linear ECM, a Wiener type structure that improves the model's ability to capture nonlinear terminal voltage behaviour without sacrificing the real time efficiency that makes ECMs attractive in the first place [8]. A less common but structurally interesting variant replaces the integer order capacitors with fractional order elements, on the observation that real capacitive components exhibit fractional order impedance behaviour that a classical RC branch cannot fully represent. This fractional order ECM has been reported to outperform the integer order $2RC$ model specifically under wide and fast varying temperature conditions [9].

The $2RC$ model is the working compromise adopted in this dissertation. It is rich enough to separate the two dominant relaxation time constants of a lithium ion cell, yet compact enough that its parameters, R_0 , R_1 , C_1 , R_2 , and C_2 , can be identified from a Hybrid Pulse Power Characterisation (HPPC) test and evaluated in real time on embedded hardware. Higher order RC networks, PNGV type models, and fractional order variants can each improve fidelity in specific regimes, low SoC , wide temperature

swings, hysteresis dominated chemistries, but the returns diminish quickly once a second RC branch is in place, while the parameter identification burden and computational cost keep climbing.

2.1.3 Electrochemical Models (Physics-Based)

Electrochemical models describe the cell from the inside out lithium ion diffusion through the electrolyte and solid particles, intercalation kinetics at the electrode surfaces, and the potential distribution across the cell, all governed by coupled partial differential equations. The Doyle Fuller Newman (DFN) formulation, more often called the pseudo two dimensional (P2D) model in the battery-management literature, is the reference model in this family. It tracks lithium concentration in both the solid particles and the electrolyte, together with charge conservation in both phases, and is regarded as accurate enough for almost any battery-management scenario [10,11]. That accuracy comes at a steep computational price, which has driven a long line of work on reduced-order variants. The full P2D model can be simplified with a Padé approximation while retaining enough of the underlying structure to identify three distinct ageing mechanisms, loss of lithium inventory, loss of active material, and resistance growth, directly from the physics of the reduced model rather than from an empirical proxy [10].

The single particle model (SPM) is the most widely used simplification, treating each electrode as one spherical particle and tracking only radial diffusion within it. SPM is fast enough for many offline uses but loses accuracy once the C -rate climbs past roughly one, since it drops the electrolyte-phase dynamics that dominate at higher currents [11]. Extensions that reintroduce electrolyte dynamics, thermal coupling, or degradation physics push accuracy back up, but each addition erodes the computational advantage that made SPM attractive in the first place.

For the retrofitted BMS considered in this work, electrochemical models are valuable as an offline reference against which the equivalent circuit fit can be checked, but they are not a candidate for the online estimator. The embedded controller in a retrofitted vehicle simply does not have the floating point throughput to solve a system of coupled PDEs at the update rates a real time BMS needs, a limitation acknowledged even by researchers actively developing reduced-order electrochemical estimators [10].

2.1.4 Data Driven and Machine Learning Models

Data-driven models treat the battery as a black box and learn a direct mapping from measured signals, terminal voltage, current, and temperature, to the quantity of interest, most often SoC or SoH, without reference to any explicit electrical or electrochemical structure. One illustration of the approach trains a gated recurrent unit recurrent neural network purely on measured voltage and current to estimate SoC, refining the network's weight updates with a momentum based gradient method to prevent oscillation and speed convergence [12]. Classical machine learning methods, support vector machines and Gaussian process regression among them, have been applied to the same mapping problem, and are joined more recently by a broader family of deep learning architectures, Long Short Term Memory (LSTM) networks, gated recurrent units, and their bidirectional and ensemble variants, discussed in more detail later in this chapter alongside the specific estimation methods built on them.

A more recent development blends physical structure back into the data driven model. One such approach feeds the internal state variables of a reduced electrochemical or circuit model, concentration terms from an SPM with thermal coupling model, or branch voltages from a nonlinear double-capacitor circuit, into a feedforward neural network trained to correct the residual between the physics based prediction and the true terminal voltage. The resulting hybrid models consistently outperform the underlying physics based model alone, with relative error reductions exceeding ninety percent at high C -rates in several reported test cases [11]. Physics informed neural networks push the fusion further, embedding a governing differential equation, into a modified Verhulst degradation law, directly into the network's training loss so that the learned mapping is constrained to respect the assumed physics even when training data is sparse [2].

Purely data driven model of this family removes the burden of parameter identification altogether, provided if enough representative data exists. The model learns the input output relationship on its own including the nonlinearities. Only disadvantage is that the model is only as good as the data it was trained on. A network trained on one temperature range, one ageing state, and one drive cycle profile has no principled way to extrapolate beyond that envelope. Unlike an ECM, its internal parameters carry no physical interpretation that would let an engineer diagnose a failure mode [1]. From the

objectives of this dissertation, data driven models are treated as a complementary correction layer rather than a replacement for the physics based core.

2.1.5 Thermal Models

None of the electrical or electrochemical descriptions above is complete on its own, because every parameter in them, ohmic resistance, diffusion coefficient, reaction rate constant, drifts with cell temperature, and a lithium ion cell that is allowed to run hot degrades faster. In the worst case it becomes unsafe. Thermal models exist to close this gap. They predict how heat is generated inside the cell, how it moves through the cell and into the surrounding pack, and how the resulting temperature feeds back into the electrical and electrochemical parameters already discussed.

The simplest useful thermal description is the lumped parameter model, in which the entire cell is treated as a single thermal mass at a uniform temperature. Heat generation is split into an irreversible ohmic term, current squared times internal resistance in its simplest form, and a reversible entropic term tied to the temperature dependence of the electrode equilibrium potentials, while heat loss is modelled as convective exchange with the surrounding air or coolant. This balance has been formalised for a single-particle-thermal model as [11]:

$$\rho_{avg} c_p \frac{dT}{dt} = \dot{q}_{gen} + \dot{q}_{conv} \dots \dots (2.4)$$

with the generation term written as:

$$\begin{aligned} \dot{q}_{gen} = I(t) & \left[V(t) - \left(U + (\bar{c} + s(t)) - U - (\bar{c} - s(t)) \right) \right] \\ & + I(t) T(t) \frac{\partial \left[\left(U + (\bar{c} + s(t)) - U - (\bar{c} - s(t)) \right) \right]}{\partial T} \dots \dots (2.5) \end{aligned}$$

and the convective loss term as:

$$\dot{q}_{conv} = -h_{cell} (T(t) - T_{amb}(t)) \dots \dots (2.6)$$

where ρ_{avg} is the cell's bulk density, c_p is its lumped specific heat capacity, h_{cell} is the convective heat transfer coefficient, and U^+ , U^- are the equilibrium potentials of the positive and negative electrodes as functions of their respective bulk lithium concentrations [11]. Temperature enters back into the model through the diffusion coefficient and reaction rate constant, both of which follow an Arrhenius type dependence on temperature, so that a hot cell diffuses lithium faster and reacts more

readily, which is precisely why elevated temperature both improves short term power capability and accelerates long term degradation.

Lumped models of this kind are attractive because they add only a single state, cell temperature, and remain cheap enough to run alongside an ECM in real time, but they assume the entire cell sits at one temperature, an assumption that becomes questionable for large format or tightly packed cells where internal gradients matter. Distributed thermal models relax this assumption by discretising the cell or pack into a mesh and solving the heat equation over it, typically in one dimension along the cell's length, or in three dimensions for a full computational fluid dynamics treatment of pack level airflow or coolant channels. These distributed approaches capture hot spots and gradients that a lumped model cannot see. At a computational cost it is much like the electrochemical models of Section 2.1.3, generally rules them out for online use on embedded hardware.

A separate but related strand of work couples thermal effects directly into the ECM rather than treating temperature as an external input. One such second order *RC* model reidentifies every parameter, R_0 , R_1 , R_2 , C_1 , and C_2 , at each of several ambient temperatures rather than holding them fixed, and this temperature aware parameterisation keeps SoC estimation error within a narrow band across a wide temperature span where a temperature independent ECM would drift substantially [4]. A related but distinct approach builds a capacity compensation model with a random forest algorithm that predicts available capacity as a function of both discharge rate and temperature, rather than folding temperature dependence into the *RC* parameters themselves [9]. For the retrofitted vehicle considered in this dissertation, where the pack sees an inconsistent duty cycle and only forced air cooling, this coupling between thermal state and electrical model accuracy is not a peripheral concern. It is a first order determinant of whether the SoC and SoH estimators built on the *2RC* backbone remain trustworthy outside laboratory controlled temperatures.

2.1.6 Comparison of EV Battery Modelling Approaches

Empirical and semi empirical models are the cheapest to evaluate and easiest to implement, but generalise poorly outside the conditions they were fitted to and carry parameters with no direct physical meaning [1,2]. Electrochemical models sit at the opposite extreme, offering the highest physical fidelity, and the clearest window into

ageing mechanisms such as lithium plating and active material loss, at a computational cost that rules out real time deployment on typical automotive grade microcontrollers [10,11]. Equivalent circuit models occupy the practical middle ground where parameters are identifiable from a standard HPPC test, the resulting state space form is compact enough for real time recursive estimation, and decades of BMS literature confirm the approach is mature and dependable, provided temperature dependence is properly accounted for [3,4,5,6,7,8,9].

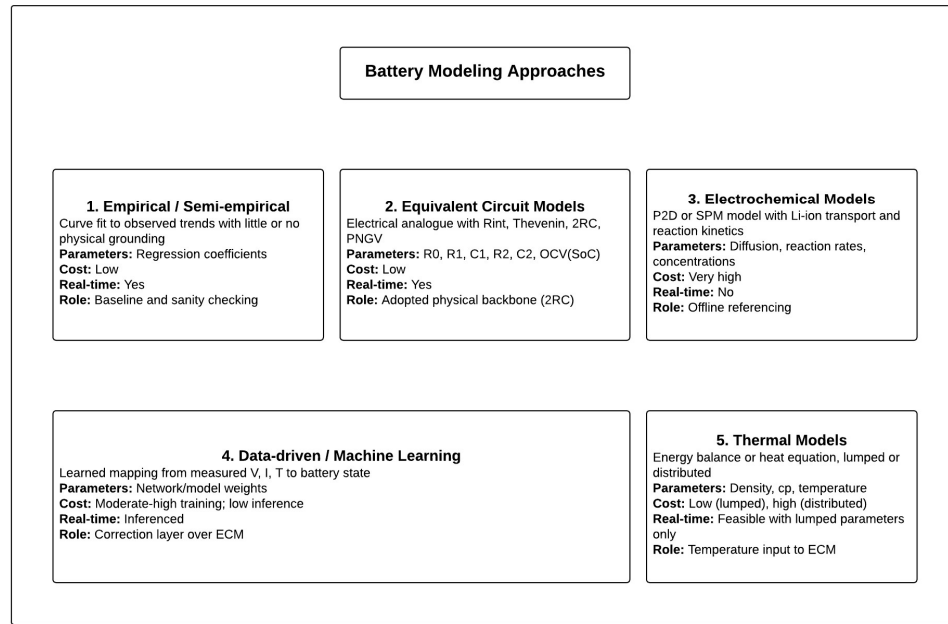


Figure 2.1: Comparison of EV battery modelling approaches.

Data driven models can match or exceed ECM accuracy when training data is abundant and representative, and hybrid physics plus network architectures now offer a credible middle path between the two, but pure data driven models demand more data than most retrofitted vehicle projects can generate, and they offer little insight into why an estimate has drifted [1,2,11,12]. Thermal models, finally, are not really in competition with the other four, rather, they are a precondition for any of them holding up outside a temperature controlled laboratory, since every electrical or electrochemical parameter in the preceding sections is itself temperature dependent [4,9,11].

On that basis, this dissertation adopts the 2RC equivalent circuit model, built on an *OCV–SoC* characterisation, as the physical backbone of the estimator. The two RC branches represent fast and slow polarisation while keeping the state dimension compact. Although temperature-dependent identification is described as the intended

modelling framework, the reported six-variant ablation is evaluated under the nominal 25°C condition. Parameter drift, hysteresis, and broader thermal validation are therefore treated as explicit next stage tests for the CKF rather than as completed learned corrections.

2.2 SoC and SoH

State of charge (SoC) is the ratio of the charge presently available in a cell to its maximum available capacity, conventionally expressed as a percentage, $SoC = \frac{Q_{remaining}}{Q_{max}}$ [13]. It is not a quantity a battery management system can read off a sensor directly. It has to be inferred from current, voltage, and temperature, which is exactly the estimation problem the remainder of this chapter builds toward. State of health (SoH) captures how far the cell has drifted from its beginning of life condition, and the literature settles on one of two working definitions depending on which failure mode is of interest. Capacity based SoH is the ratio of the current maximum dischargeable capacity to the rated, nameplate, capacity, $SoC = \frac{Q_{max}}{Q_{rated}}$ [13,14], while resistance based SoH tracks the growth of internal resistance relative to its beginning of life value [1]. The two do not necessarily move together. A cell can lose usable capacity through active material loss while its resistance climbs only slowly, or the reverse, and a BMS that watches only one of the two can be blindsided by degradation showing up in the other.

Keeping SoC inside a safe operating window is what prevents the overcharge and over discharge conditions that accelerate ageing and, in the worst case, precipitate thermal runaway, while SoH tracking underlies range prediction, warranty decisions, and end of life management for the pack as a whole. Once a cell's SoH falls to around 80% of its rated capacity, the associated safety margins narrow enough that most manufacturers and researchers treat the cell as functionally retired from vehicle service, even though it may still deliver usable charge in a second-life application [14]. An estimator that performs well on a fresh cell but drifts as the cell ages is of limited practical use in a vehicle expected to stay in service for years, which is the main argument for treating SoC and SoH estimation jointly rather than as two separate problems wherever the estimator architecture allows it [10,13].

For the retrofitted vehicle considered in this dissertation, the *38S2P* pack under study is assembled from cells with an unknown service history relative to the vehicle's original ICE powertrain, so the SoC estimator has to remain trustworthy across a wider and less predictable range of SoH than would be expected in a factory fitted EV pack, where all cells begin life together and age under a comparatively uniform duty cycle. This is the practical reason the estimation methods surveyed later in this chapter are evaluated with an eye to how they behave as the underlying model parameters drift with age, rather than only under fresh cell conditions.

2.3 HPPC Test

Hybrid Pulse Power Characterisation (HPPC) is the standard laboratory procedure for pinning down both the *OCV-SoC* relationship and the *RC* parameters of an equivalent circuit model. The cell under test is discharged, or charged, in a series of current pulses, each followed by a rest period long enough for the terminal voltage to relax back toward its open-circuit value, and the sequence is repeated at fixed SoC decrements and across a span of ambient temperatures rather than at a single operating point [8]. In this dissertation, the pack under test, a *38S2P* configuration, is characterised through a *discharge-rest-charge-rest* cycle at each SoC step, stepped incrementally across the temperature range of interest, following the same general structure recommended in the wider HPPC literature while the pulse magnitudes and durations are adapted to the pack's own current and voltage rating.

Two things fall out of each *pulse-and-rest* cycle. The relaxed terminal voltage recorded at the end of a sufficiently long rest gives one point on the *OCV-SoC* curve, repeating the procedure across the full SoC range and fitting a monotonic lookup table or a low order polynomial to the results yields the static OCV characteristic the ECM depends on [8]. The second quantity comes from the shape of the voltage response during and immediately after a pulse, i.e. an instantaneous step attributable to the ohmic resistance R_0 , followed by two exponential relaxation tails corresponding to the two *RC* branches. Fitting this response to a sum of exponentials of the form:

$$U(t) = k_0 - k_1 e^{-\frac{t}{\tau_1}} - k_2 e^{-\frac{t}{\tau_2}} \dots \dots (2.7)$$

recovers the time constants τ_1 and τ_2 , from which R_1 , C_1 , R_2 , and C_2 follow directly once the applied current is known, since $R_1 = \frac{k_1}{I}$, $R_2 = \frac{k_2}{I}$, $C_1 = \frac{I \cdot k_1}{\tau_1}$ and $C_2 = \frac{I \cdot k_2}{\tau_2}$ [5].

Carrying this fit out at every SoC step and every test temperature builds the parameter surfaces, $R_0(\text{SoC}, T)$, $R_1(\text{SoC}, T)$ and so on, that a temperature aware $2RC$ model needs in order to stay accurate away from the single operating point at which it happened to be identified.

2.4 Methods of SoC and SoH Approximation

With a characterised model in hand, the remaining problem is the estimation. Recovering SoC and SoH from whatever current, voltage, and temperature signals a BMS can actually measure while the vehicle is in use. The methods reported in the literature fall into a handful of broad families, summarised in *Fig. 2.1*, and later sections of this chapter work through the model based and data driven families at greater length before proposing a hybrid architecture that draws on more than one of them.

Table 2.1: Families of SoC and SoH estimation methods

Estimation Family	Basis	Strengths	Limitations
Ampere-hour (Coulomb) counting	Time-integration of measured current	Trivial to compute; no battery model required	Open-loop; drifts with sensor bias and uncertain initial SoC
Open-circuit-voltage (OCV) method	Monotonic <i>OCV-SoC</i> relationship from rest measurements	Simple, direct SoC read-out once cell is at rest	Needs a long rest period; unusable while driving
Model-based recursive estimation (Kalman or particle filter family)	ECM voltage prediction corrected against measured voltage	Closed-loop, self-correcting, real-time capable	Accuracy bounded by ECM fidelity and parameter drift
Data-driven or machine learning	Learned mapping from V, I, T to <i>SoC</i> or <i>SOH</i>	Can match or exceed model-based accuracy with enough data	Needs representative training data; poor extrapolation
Co-estimation or fractional-order or hybrid	Joint SoC-SoH estimation or fractional-calculus ECM refinement	Tracks ageing-aware states together; improved fit to EIS data	Added algorithmic and computational complexity

The simplest approach, Coulomb counting, integrates measured current over time and divides by rated capacity to track SoC directly [3]. It costs almost nothing to compute and needs no battery model at all, but it is an open loop method. It has no way to correct for current sensor bias, and any error in the assumed initial SoC, or in the sensor's zero-offset, accumulates without bound over a long drive cycle. The open circuit voltage method takes the opposite approach, exploiting the monotonic *OCV-SoC* relationship

established in Section 2.3 directly, but it demands a rest period long enough for the terminal voltage to relax fully, which makes it impractical as the sole estimator whenever the vehicle is actually being driven [3].

Model based recursive estimation closes the loop that both of the above leave open. The ECM predicts the terminal voltage the current SoC estimate implies, compares that prediction against the measured voltage, and uses the resulting residual to correct the estimate at every sample. This is the role played by the Kalman filter family, the extended, unscented, and cubature variants, covered in later sections of this chapter. A closely related approach, particle filtering, handles the same correction problem without assuming Gaussian noise, at higher computational cost, and is treated alongside its estimator variants further on.

A fourth route, data driven estimation, learns the current-voltage-temperature to SoC, or SoH, mapping directly from data rather than from an explicit circuit model. A survey of this family finds that feedforward and recurrent architectures, LSTM networks especially, can match or exceed the accuracy of classical Kalman based estimators once enough representative training data exists, though the same survey cautions that comparisons across studies are unreliable unless the networks being compared are trained and tested on identical data with a similar parameter count [15]. One instance of this family trains a GRU-RNN directly on measured voltage and current alone, with a momentum based gradient method added to speed convergence and damp oscillation in the learned weights [12].

A further complication, particularly relevant to a joint SoC-SoH problem, comes from the interaction between the two states. A SoC estimator built on ECM parameters identified at beginning of life quietly degrades as the cell ages and those parameters drift, while a SoH estimator that assumes a fixed SoC trajectory inherits whatever error the SoC estimator carries. This has been addressed by co-estimating SoC alongside three distinct ageing mechanisms, loss of lithium inventory, loss of active material, and resistance growth, using a reduced order electrochemical model and dual nonlinear filters, with reported RMSE below 0.5% across a SoH range spanning 71 to 100% [10]. A review of transfer learning approaches to the same joint problem argues that the practical obstacle for most data driven methods is not model accuracy under laboratory conditions but generalisation once a network trained on one cell, one temperature range,

or one drive cycle is deployed on a different one, a limitation that motivates the transfer learning and domain adaptation techniques it surveys [13].

Fractional order variants take a slightly different approach, motivated less by a wish to avoid an explicit model than by the observation that the EIS data described in Section 2.3 does not fit an integer order RC branch particularly well. One such combination pairs a fractional order $2RC$ model with a square root unscented Kalman filter and reports SoC estimation error held below 1.0% across a range of operating conditions, illustrating that the fractional calculus route can be folded into a conventional Kalman filtering pipeline rather than treated as a wholesale replacement for it [16].

None of these families is strictly superior across every axis that matters for a retrofitted EV BMS. Coulomb counting and OCV methods are inexpensive but incomplete on their own. Recursive estimators depend on model quality and data driven approaches require representative training data that may not be available in a retrofit project. The architecture carried into Chapters 3 and 4 therefore fixes a $2RC$ plant and compares only controlled CKF-family extensions, so that numerical form, noise adaptation, robustness, bias augmentation, and information gating can be interpreted separately.

2.5 Kalman Filter and its Types

Every method surveyed in the previous sections of this chapter, Coulomb counting, the OCV method, and the ECM taken on its own either has no way to correct itself against a measurement or has no way to run continuously while the vehicle is in motion. The Kalman filter closes both gaps. It is a recursive Bayesian estimator for linear systems under Gaussian process and measurement noise. Given a state-space model, it alternates between a time update, which propagates the state estimate and its error covariance forward using the system model, and a measurement update, which corrects that prediction using the discrepancy between the predicted and actual measurement, weighted by a gain computed from the relative uncertainty of the model and the sensor. For a linear system with state x_k , input u_k , and measurement v_k , the model takes the form:

$$x_{k+1} = Ax_k + Bu_k + w_k \dots \dots (2.8)$$

$$y_k = Cx_k + Du_k + v_k \dots \dots (2.9)$$

where w_k and v_k are zero-mean Gaussian process and measurement noise with covariances Q and R respectively. The Kalman gain at each step balances how much

the filter trusts its own model based prediction against how much it trusts the new measurement, and this balance is what allows the filter to correct for an uncertain initial condition or a noisy sensor without needing either to be perfect.

A battery, however, is not a linear system in the sense the equations above assume. The terminal-voltage equation is nonlinear in SoC through the OCV curve, and the RC branch dynamics, while linear in the state for fixed parameters, sit inside a model whose parameters themselves vary with SoC and temperature. The standard linear Kalman filter cannot be applied directly to this problem, which is what motivates the three nonlinear variants examined in this section, the extended, unscented, and cubature Kalman filters, each of which handles the nonlinearity in a different way and inherits a different trade off between accuracy and computational cost as a result.

It is worth being explicit about why a recursive filter is preferable to simply running the ECM open loop against a Coulomb counted SoC, since this is the entire justification for the added complexity examined in this section. An open loop ECM has no mechanism for noticing that its own state estimate has drifted. Once a current sensor bias or an incorrect initial SoC has entered the calculation, it propagates forward unchecked. A recursive Bayesian filter instead treats the ECM's voltage prediction as one noisy source of information and the actual measured voltage as another, and combines the two according to how much each is trusted at that instant, which is precisely what the Kalman gain computes. This is what allows a Kalman-family estimator to self correct from a wrong starting point within a handful of samples, something none of the open loop methods discussed earlier in this chapter can do on their own.

2.5.1 Extended Kalman Filter

The Extended Kalman filter (EKF) handles the nonlinearity by linearising the state and measurement functions at every time step, using a first order Taylor expansion around the current estimate, and then applying the standard linear Kalman recursion to the resulting Jacobians. For the $2RC$ model with state vector $[SoC, U_1, U_2]$, the discretised state equations and their Jacobian with respect to the state are:

$$SoC_{k+1} = SoC_k - \left(\frac{\Delta t}{Q_{st} * 3600} \right) \eta I_k + w_{1,k} \dots \dots (2.10)$$

$$U_{1,k+1} = U_{1,k}e^{\frac{-\Delta t}{\tau_{1,k}}} + R_{1,k} \left(1 - e^{\frac{-\Delta t}{\tau_{1,k}}}\right) I_k + w_{2,k} \dots \dots (2.11)$$

$$U_{2,k+1} = U_{2,k}e^{\frac{-\Delta t}{\tau_{2,k}}} + R_{2,k} \left(1 - e^{\frac{-\Delta t}{\tau_{2,k}}}\right) I_k + w_{3,k} \dots \dots (2.12)$$

$$U_{t,k} = OCV_k(SoC_k) - U_{1,k} - U_{2,k} - R_{0,k}I_k + v_k \dots \dots (2.13)$$

with the state Jacobian $\hat{A}_k$ evaluated at the current estimate reducing to a diagonal matrix of ones and the two exponential decay terms, and the output Jacobian $\hat{C}_k$ reducing to $\left[\frac{dOCV}{dSoC}, -1, -1\right]$ evaluated at the predicted state [17,18]. Plett's original three-part treatment of EKF based battery state estimation, still one of the most cited formulations in the field, established this state-space structure and demonstrated SoC estimation error rarely exceeding 5% across a 0 to 50°C range on a hand-built Li ion battery pack [17].

The EKF is the most widely deployed nonlinear filter in commercial BMS software, largely because it is computationally light and conceptually transparent. It requires only one Jacobian evaluation and one linear Kalman update per time step. Its weakness is exactly what its strength depends on. The first-order linearisation degrades wherever the true system is strongly nonlinear, which for a lithium-ion cell is precisely the flat, low slope region of the *OCV-SoC* curve in the middle of the charge range, where the same voltage error maps to a much larger SoC error than at the steep edges of the curve. In practice this shows up as a filter that fails to converge from a poor initial guess. Exactly this divergence has been reported when applying a plain EKF to a second order *RC* model, with the SoC estimate drifting further from the true value rather than correcting itself [5]. Aging compounds the problem, since the ECM parameters the Jacobian depends on drift as the cell degrades, and an EKF tuned against beginning of life parameters has no built-in mechanism to notice that its own linearisation point has become unreliable [4,6].

2.5.2 Unscented Kalman Filter

The unscented Kalman filter (UKF) avoids linearisation altogether. Instead of a Jacobian, it propagates a small, deterministically chosen set of sigma points through the true nonlinear model and reconstructs the predicted mean and covariance from the transformed points via the unscented transform. For a state of dimension n , the sigma points are generated around the current mean estimate $\hat{x}$ and covariance P as:

$$X_0 = \hat{x} \dots \dots (2.14)$$

$$X_i = \hat{x} + \left(\sqrt{(n + \lambda)P} \right)_i ; i = 1, 2, 3, \dots \dots n \dots \dots (2.15)$$

$$X_i = \hat{x} - \left(\sqrt{(n + \lambda)P} \right)_{i-n} ; i = n + 1, n + 2, n + 3, \dots \dots 2n \dots \dots (2.16)$$

where λ is a scaling parameter and $\left(\sqrt{(n + \lambda)P} \right)_i$ denotes the i^{th} column of the matrix square root. Each of the $2n + l$ sigma point is passed through the true nonlinear state and measurement equations, and the predicted mean and covariance are recovered as weighted sums of the transformed points, with weights chosen so the reconstruction is exact for a Gaussian input up to at least second order. Because no derivative is taken, the UKF captures the nonlinearity of the OCV curve more faithfully than the EKF's first-order approximation and generally tracks SoC more accurately, at the cost of propagating $2n + l$ points through the model at each step rather than the single trajectory an EKF requires. Reported comparisons consistently find the UKF converging faster than the EKF after a poor initial SoC guess and holding tighter error bounds through the flat region of the OCV curve, though at several times the floating point cost per update [10,16].

2.5.3 Cubature Kalman Filter

The cubature Kalman filter (CKF) is a closely related derivative free filter that generates its evaluation points from a third degree spherical radial cubature rule rather than the unscented transform. The full cubature recursion for SoC estimation can be set out as follows [19], given the prior covariance factorised as $P_{k+1|k} = S_{k+1|k} S_{k+1|k}^T$, the cubature points are

$$x_{j_{k+1|k}} = S_{k+1|k} \cdot \zeta_j + \hat{x}_{k+1|k} \quad j = 1, 2, 3, \dots \dots 2n \dots \dots (2.17)$$

where ζ_i are the fixed cubature-rule points. These are propagated through the measurement equation, $y_{j_{k+1}} = h x_{j_{k+1|k}}$, and the predicted output mean, output covariance, and cross-covariance are recovered as unweighted averages over the $2n$ cubature points before the gain, state, and covariance updates proceed exactly as in a standard Kalman recursion [19]:

$$K_{k+1} = P_{xy_{k+1}} P_{y_{k+1}}^{-1} \dots \dots (2.18)$$

$$\hat{x}_{k+1} = \hat{x}_{k+1|k} + K_{k+1} (y_{k+1} - \hat{y}_{k+1}) \dots \dots (2.19)$$

$$P_{k+1} = P_{k+1|k} - K_{k+1} P_{y_{k+1}} K_{k+1}^T \dots \dots (2.20)$$

The CKF reduces to roughly the same order of accuracy as the UKF for Gaussian problems, but its cubature points carry equal weights, which avoids the negative or unequal sigma point weights that can make the UKF's covariance estimate numerically unstable in higher-dimensional state spaces. Strong tracking and adaptive variants of the CKF (STCKF, ACKF) adjust the process noise covariance online in response to filter innovation, which improves robustness when the ECM parameters drift away from their identified values, a condition that HPPC based parameterisation cannot fully anticipate once the cell ages. Fractional-order extensions of both the EKF and the CKF (FEKF, an adaptive fractional CKF) have also been reported to improve SoC and SoH estimation specifically when the underlying ECM itself is fractional order, consistent with the EIS derived CPE discussion in Section 2.3 [16].

2.5.4 Co-estimation of SoC and SoH

Beyond estimating SoC alone, the Kalman filter family extends naturally to joint state-and-parameter estimation, where the state vector is augmented with slowly time-varying quantities such as R_0 or capacity, and the same filter recursion tracks both simultaneously. Because SoC changes on a timescale of minutes while capacity fade and resistance growth unfold over months, a dual-filter structure, one filter running at the fast timescale and a second at the slow one, is often used in place of a single augmented filter, since it avoids the poor observability that results from mixing the two timescales in one covariance matrix. Plett's dual EKF formulation is the reference structure where a state filter tracks SoC and the RC -branch voltages while a parallel weight filter tracks the slowly varying model parameters with the two filters coupled only through the shared measurement residual, i.e., the state gains and parameter Kalman gains computed from their respective error covariances [17]. This structure has been extended to a three-way co-estimation of SoC, SoH, and state of function [18]. SoC is tracked with an EKF against the $2RC$ model, while the parameters underlying SoH and available power are identified online with a forgetting-factor recursive least squares algorithm running alongside it, so that the capacity and resistance estimates feeding the SoH figure update continuously rather than only at fixed calibration intervals [18]. A related approach works at the electrochemical-model level, coupling dual nonlinear filters to a reduced-order P2D model to track SoC alongside three distinct ageing mechanisms, loss of lithium inventory, loss of active material, and

resistance growth, reporting an RMSE below 0.5% across a SoH range spanning 71 to 100% [10].

Reported performance in the literature generally favours the derivative-free filters over the EKF once SoC estimation is compared under identical initial-condition errors, sensor noise, and temperature variation, with the UKF and CKF converging faster after a poor initial guess and holding tighter error bounds through the flat region of the OCV curve. Table 2.2 summarises the trade-offs among the three filters discussed in this section.

Table 2.2: Comparison of Kalman filter variants for battery state estimation

Filter	Nonlinearity Handling	Computational Cost	Typical Behaviour on OCV-SOC Curve
EKF	First-order Taylor linearisation (Jacobian) at each step	Lowest of the three	Degrades in the flat, low-slope mid-SOC region
UKF	Deterministic sigma points through unscented transform	Moderate ($2n + 1$ points)	Captures nonlinearity to second order; converges faster after poor initial guess
CKF	Third-degree spherical-radial cubature points	Moderate, equal-weighted points	Comparable accuracy to UKF; more numerically stable in higher dimensions

The trade off is computation between the UKF and CKF require several times the floating-point operations per update that the EKF does, which matters on the modest embedded processors typical of a retrofitted vehicle's BMS, and is one of the practical considerations behind the hybrid architecture proposed later in this dissertation.

A related, and increasingly common, practical compromise is to run an EKF as the default estimator and to fall back on a UKF or CKF update only when the filter's own innovation signal, the discrepancy between predicted and measured voltage, exceeds a threshold that suggests the linearisation has become unreliable. This adaptive-switching idea avoids paying the derivative free filters' higher per-step cost continuously while still gaining most of their robustness at exactly the moments, a poor initial guess, a sudden current transient, or a period of rapid ageing, when the EKF is most likely to struggle. It is a pattern that recurs across the estimator families surveyed in this chapter: rather than committing permanently to the most expensive method available, several published schemes use a cheap estimator as the default and reserve a more expensive one for the conditions under which the cheap one is known to fail, which is also the logic underlying the hybrid architecture proposed later in this dissertation.

2.6 Particle Filter and its Types

The particle filter (PF) takes a different approach to the same nonlinear estimation problem, representing the posterior probability distribution of the state not by a mean and covariance, as the Kalman family does, but by a set of weighted samples, or particles, propagated through the true nonlinear model. Because it makes no Gaussian assumption, the PF can in principle handle arbitrary noise distributions and multimodal posteriors, which is attractive when the ECM's model error is not well characterised as Gaussian, as it often is not near ageing induced parameter shifts. Given a particle set $\{x(i)_{k-1}\}$ with equal weights $\frac{1}{N}$ at time $k - 1$, each particle is propagated through the process model, and its importance weight is updated against the new measurement as:

$$w(i)_k = w(i)_{k-1} \left[\frac{p(z_k|x(i)_k)p(x(i)_k|x(i)_{k-1})}{q(x(i)_k|x(i)_{k-1}, z_k)} \right] \dots \dots (2.21)$$

with the weights normalised so they sum to one, $w(i)_k = \frac{w(i)_k}{\sum_j w(j)_k}$, and $q(x(i)_k|x(i)_{k-1}, z_k)$ is the importance density from which new particles are drawn [12]. The practical difficulty is particle degeneracy after a handful of updates, most particles carry negligible weight and contribute nothing to the estimate, which resampling addresses at the cost of reducing particle diversity and, if repeated too often, collapsing the filter onto a small cluster of samples. A standard resampling trigger compares the effective sample size against a threshold:

$$\hat{N}_{eff} = \frac{1}{\sum_i w(i)_k^2} < N_{threshold} \dots \dots (2.22)$$

and only resamples when the effective particle count has fallen too far, which limits unnecessary loss of particle diversity [19]. Variants such as the regularised particle filter soften the collapse that follows aggressive resampling, while hybrid forms, the unscented particle filter (UPF) and the cubature particle filter (CPF), use a UKF or CKF update to generate a better-informed proposal distribution before resampling. In the CPF specifically, the importance density at each step is taken as Gaussian with mean and covariance supplied directly by a CKF update, $q(x(i)_k|x(i)_{k-1}, z_{1:k})$, which concentrates the particles where the posterior mass actually lies rather than sampling blindly from the process model [19]. This has been shown to improve accuracy and robustness over a plain PF, and over a plain CKF, at a feasible computational cost.

2.6.1 SoC and SoH Approximation using Particle Filter

A practical detail that matters for how well any of these variants perform is the resampling scheme used once the effective sample size threshold is crossed. Systematic resampling, which draws N ordered points at fixed intervals along the cumulative weight distribution rather than N independent uniform draws, is the most commonly reported choice in the battery literature because it introduces less resampling noise than plain multinomial resampling for a comparable computational cost, and it is the scheme underlying the CPF workflow described above [19]. Whichever resampling scheme is chosen, the underlying trade off is the same. Resampling too, rarely lets weight concentrate on a shrinking number of particles until the filter is effectively tracking only one or two hypotheses, while resampling too often discards the particle diversity that gives the PF its advantage over a Kalman family filter in the first place.

Applied to SoC and SoH estimation, particle filtering tends to outperform the Kalman family precisely where the assumptions behind Kalman filtering breaks down most strongly non Gaussian sensor noise, abrupt current transients, and multimodal uncertainty following a period of sensor dropout or an uncertain initial condition. The Adaptive Extended Kalman Particle Filter (AEKPF) illustrates a practical hybrid built for exactly this scenario, combining a forgetting factor least squares parameter identification stage with an adaptive Kalman filter and a particle filter, and reporting average absolute SoC error below 1% even with measurement noise as high as 10%, outperforming both a plain adaptive EKF and a plain PF tested under the same conditions [6]. On aged cells specifically, the same study reports a maximum SoC error of 6.7% in the first ten seconds after a demanding New European Driving Cycle (NEDC) test at 90% SoH, settling to an average error of 1.7% thereafter, evidence that the particle based correction can absorb a meaningful amount of ageing induced model mismatch that a fixed parameter Kalman filter alone would not [6].

A comparison of the CPF against a plain CKF under DST and NEDC conditions with added measurement noise found the CPF's mean absolute error roughly half that of the CKF alone (0.005 against 0.011 in one reported comparison), confirming that letting a CKF supply the proposal distribution, rather than relying on the plain bootstrap proposal, meaningfully improves particle efficiency for a given particle count [19]. The cost is that a PF accurate enough to be useful typically needs several hundred to a few thousand particles, each one requiring a full evaluation of the nonlinear model, which

places a much heavier computational load on the BMS controller than any Kalman variant. For a retrofitted vehicle's embedded hardware, this makes a plain PF a difficult fit for the primary real-time estimator, though the CPF and UPF variants, which lean on a cheaper Kalman-type proposal to reduce the number of particles needed for a given accuracy, remain a realistic option for periods where robustness matters more than the tightest possible compute budget, such as immediately after power-up or a sensor fault. Table 2.3 summarises the three particle-filter variants discussed in this section.

Table 2.3: Comparison of particle filter variants for battery state estimation

Filter	Proposal / Update Mechanism	Strength	Weakness
PF	Bootstrap proposal from process model, resampling by weight	Handles arbitrary, non-Gaussian, multimodal noise	Needs hundreds to thousands of particles; degeneracy
UPF	UKF-generated proposal distribution before resampling	Fewer particles needed for given accuracy	Still carries UKF's per-point computational cost
CPF	CKF-generated proposal distribution before resampling	Improves on CKF accuracy and robustness alone	Added algorithmic complexity over a plain CKF

2.9 Literature Synthesis and Research Gap

The literature establishes three recurring trade offs. Equivalent circuit models are computationally attractive, but their accuracy depends on OCV characterisation and parameter variation with SoC and temperature. Nonlinear Kalman filters offer recursive correction, but their apparent performance can be distorted by inconsistent references, tuning, initialization, and evaluation windows. More elaborate estimators may improve robustness, yet the added mechanism is difficult to interpret when several modelling and algorithmic choices change at once.

This study addresses a narrower gap where a mechanism by mechanism account of CKF development on one fixed $2RC$ plant. Much published comparison work ranks EKF, UKF, CKF, particle filters, or learning models as complete packages. Such rankings are useful for orientation but do not reveal whether a gain comes from numerical factorisation, noise adaptation, robust weighting, added bias states, or selective use of voltage information. The present ablation is designed to expose those causal differences while keeping the data and scoring contract unchanged.

The gap is also practical for retrofit engineering. A retrofit BMS may face uncertain sensor offsets, ageing induced parameter mismatch, modest computing resources, and

drive segments with weak voltage information. A derivative free recursive estimator with a small number of deterministic points is therefore a credible foundation, but it must be tested against those specific conditions. The literature review consequently supports CKF as a research platform rather than declaring it universally superior. Chapter 3 converts that proposition into controlled hypotheses.

2.10 Rationale for Retaining the 2RC Model

The 2RC circuit is retained because it provides two physically interpretable relaxation modes without making the state vector unnecessarily large. R_0 represents the near instantaneous ohmic response, while the R_1C_1 and R_2C_2 branches represent faster and slower polarisation processes. Under zero order held current, the branch voltage states have closed form discrete transitions, which supports stable, efficient prediction inside every CKF variant. The nonlinear measurement relation is concentrated in the *OCV–SoC* function.

A first order circuit may be adequate for gentle excitation, but it can merge distinct transient time scales and leave structured voltage residuals during pulses and relaxation. Higher order circuits can reduce fitting error on a particular record, although additional branches increase parameter correlation and can become weakly identifiable. Because the present research question concerns estimator mechanisms, changing model order during the ablation would confound filter effects with plant effects. Fixing 2RC protects the comparison from that ambiguity.

Retaining 2RC does not imply that it is exact. Its parameters should ultimately be identified over SoC and temperature, validated on disjoint pulses, and checked for positive values and separated time constants. Hysteresis, diffusion limits, ageing, and thermal coupling may require future augmentation. Those extensions should be introduced only after the 2RC–CKF baseline is reproduced, so that their incremental value can be measured against an established reference.

CHAPTER 3: METHODOLOGY

3.1 Simulink Model

This chapter sets out the methodology used to build and evaluate the SoC estimator for the retrofitted *38S2P* pack. The Simulink simulation environment and pack representation described in this section, the hybrid pulse power characterisation (HPPC) protocol used to excite it in Section 3.2, and the second-order resistor capacitor (*2RC*) state space model together with the six member cubature Kalman filter (CKF) ablation family built on top of it in Section 3.3. Two error statistics carry the comparative weight of the results that follow in Chapter 4: the root mean square error (*RMSE*) of the SoC estimate against the reference trajectory, and the relative absolute error (*RAE*), which restates the mean absolute error as a fraction of the error a naive, non-adaptive predictor would produce; both are defined formally, alongside the other metrics used throughout, in Section 3.3.4.

3.1.1 Abbreviations and Notations Used in This Chapter

Table 3.1: Abbreviations used throughout Chapter 3

Abbreviation	Meaning	Abbreviation	Meaning
<i>BMS</i>	Battery management system	<i>NIS</i>	Normalised innovation squared
<i>CKF</i>	Cubature Kalman filter	<i>OCV</i>	Open-circuit voltage
<i>SRCKF</i>	Square root cubature Kalman filter	<i>PCHIP</i>	Piecewise cubic Hermite interpolating polynomial
<i>EKF</i>	Extended Kalman filter	<i>PF</i>	Particle filter
<i>UKF</i>	Unscented Kalman filter	<i>AEKPF</i>	Adaptive extended Kalman particle filter
<i>ECM</i>	Equivalent circuit model	<i>RC</i>	Resistor capacitor (branch)
<i>2RC</i>	Second-order resistor-capacitor equivalent circuit	<i>RMSE</i>	Root mean square error
<i>SoC</i>	State of charge	<i>MAE</i>	Mean absolute error
<i>SoH</i>	State of health	<i>RAE</i>	Relative absolute error
<i>HPPC</i>	Hybrid pulse power characterisation	<i>EIS</i>	Electrochemical impedance spectroscopy
<i>38S2P</i>	38 cells in series, 2 strings in parallel	<i>QR</i>	QR matrix decomposition

3.1.2 Notation Used in This Chapter

Table 3.2 collects the symbols used repeatedly throughout this chapter and the results and discussion material that follows it, so that a reader encountering an equation partway through does not need to search backward for where a symbol was first defined. Symbols used only once, in the immediate vicinity of the equation that introduces them, are defined inline instead and are not repeated here.

Table 3.2: Notations with their respective meaning

Symbol	Meaning	Symbol	Meaning
SoC_k	State of charge at sample k	x_k	Filter state vector at sample k
U_{1k}, U_{1k}	Fast/slow RC -branch voltages	P_k, S_k	State covariance and its Cholesky/square root factor
bI, bV	Pack current/voltage sensor bias states	Q, R	Process and measurement noise covariance
I_{pack_k}, I_{cell_k}	Pack and cell level current	K_k	Kalman gain at sample k
V_{pack_k}	Simulated/predicted pack terminal voltage	v_k	Innovation (measurement residual)
R_0, R_1, R_2	Cell ohmic and RC -branch resistances	$P_{yy,k}, P_{xy,k}$	Innovation covariance and state–output cross-covariance
τ_1, τ_2	RC branch time constants	NIS_k	Normalised innovation squared, sample k
OCV	Open circuit voltage	$gSoC_k, gexc_k$	SoC and excitation information gates
N_s, N_p	Series cell count, parallel string count	w_k	Huber robust weight
Q_{mod}	Pack-equivalent capacity used in Eq. 3.1	n	State dimension
Δt	Sample interval	N	Number of samples in the scored record
η	Coulombic efficiency	e_k	Signed SoC error, $\widehat{SoC}_k - SoC_{ref_k}$

The remainder of the chapter is organised as follows. Sections 3.1 and 3.2 describe the Simulink simulation environment and the HPPC test protocol used to generate the current excitation and the open circuit voltage (OCV) characterisation data. Section 3.3.1 sets out the $2RC$ equivalent circuit state space model shared by every CKF variant in this chapter. Section 3.3.2 develops the six member CKF ablation family, from the baseline covariance form filter through to the proposed dual information gated variant. Section 3.3.3 documents the ablation matrix and the rationale for testing exactly these six combinations. Section 3.3.4 defines every metric used later in the chapter, with $RMSE$ and RAE treated in the greatest depth. Section 3.3.5 records the experimental

protocol and the fairness rules that were applied. Sections 4.1 through 4.3 present and discuss the quantitative results and place them against the wider EKF/UKF/particle filter literature. Section 3.3.6 documents the software implementation and reproducibility surface, Section 4.2.9 addresses statistical repeatability, Section 5.2 states the limitations of the present study honestly, and Section 5.1 closes the chapter with a summary that feeds directly into the results and discussion material of Chapter 4.

This chapter's simulation first approach follows directly from the wider project this dissertation sits inside the DVC sponsored digital twin study of retrofitted e-mobility feasibility introduced in Chapter 1. A digital twin is, by its nature, evaluated in simulation before any claim is carried over to a physical retrofit build, and the estimator comparison developed in this chapter is best read as exactly that first stage, establishing which combination of CKF family mechanisms is worth carrying forward into a measured data phase of the same project, rather than as a simulation exercise disconnected from the retrofit vehicle the wider study is built around. The pack configuration used throughout this chapter, *38S2P* at *100Ah* per cell and *136.8V* nominal, matches that retrofit target directly, which is why the results in Sections 4.1 through 4.3 are stated in terms specific to that vehicle rather than in terms of an arbitrarily sized generic cell.

The simulation model for this dissertation was built in *MATLAB R2020a (MathWorks Inc.)*, using *Simulink* to represent the battery pack and *MATLAB* scripts for every estimator built on top of it. The pack itself was represented at the cell level and then scaled to the pack level through a series-parallel configuration, 38 cells in series and 2 strings in parallel, each cell having a nominal voltage of *3.6V* and a rated capacity of *100Ah*. At the nominal operating point this gives the pack a nominal terminal voltage of *136.8V* and a nominal capacity of *200Ah* (two *100Ah* strings in parallel), consistent with the retrofit target vehicle's traction battery sizing discussed in Chapter 1.

The electrical behaviour of each cell was represented in Simulink using the built in *Datasheet Battery* block, which implements a lithium ion cell model parameterised from manufacturer discharge characteristic data rather than from a hand fitted equivalent circuit. The block was configured with the pack level series and parallel counts ($N_s=38$, $N_p=2$) set directly in its mask, a rated capacity at nominal temperature of *100Ah* per cell, and its own open circuit voltage and internal resistance lookup tables

indexed jointly by state of charge and cell temperature. The temperature breakpoints built into the block span $243.1K$ to $313.1K$ ($-30^{\circ}C$ to $40^{\circ}C$) in six intervals, and the internal resistance table is populated across that same temperature range and a state of charge breakpoint set of $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, giving cell resistance values on the order of half a $m\Omega$ to a little under three $m\Omega$ per cell depending on where in the temperature-SoC grid the operating point falls. This is a separate, coarser grid from the 11 point, $0^{\circ}C$ to $50^{\circ}C$, $5^{\circ}C$ step temperature sweep used later for the HPPC based OCV characterisation described in Section 3.2. The *Datasheet Battery* block's own breakpoints govern how it interpolates OCV and resistance internally, while the 11 point sweep is the experiment design layered on top of it to build the $OCV(SoC)$ table that every CKF variant in this chapter actually consumes.

Block Parameters: Datasheet Battery

Datasheet Battery (mask) (link)

Implements a model for a lithium ion, lithium polymer, or lead acid battery based off of discharge characteristics taken at different temperatures. The model can be parameterized using a typical battery datasheet or through experimental measurement.

Block Options

Initial battery capacity: Parameter

Output battery voltage: Unfiltered

Parameters

Rated capacity at nominal temperature, BattChargeMax [Ah]: 100

Open circuit voltage table data, Em [V]: 3.935 3.944 3.954 3.964 3.974 3.988 3.998 4.014 4.029 4.034 4.047 4.065 4.074 4.086 4.097 4.131 4.126 4.138 4.15 4.18 4.174 4.187 4.207 4.231 4.221

Open circuit voltage breakpoints 1, CapLUTBp []: 0.71 0.72 0.73 0.74 0.75 0.76 0.77 0.78 0.79 0.8 0.81 0.82 0.83 0.84 0.85 0.86 0.87 0.88 0.89 0.9 0.91 0.92 0.93 0.94 0.95 0.96 0.97 0.98 0.99 1

Internal resistance table data, RInt [Ohms]: 0.001778 0.001924 0.002234;0.0005603 0.0007192 0.0006887 0.0007557 0.0007355 0.001014;0.001028 0.001338 0.001 0.001536 0.001546 0.002789

Battery temperature breakpoints 1, BattTempBp [K]: [243.1 253.1 263.1 273.1 283.1 298.1 313.1]

Battery capacity breakpoints 2, CapSOCBp []: [0 0.2 0.4 0.6 0.8 1]

Number of cells in series, Ns []: 38

Number of cells in parallel, Np []: 2

Initial battery capacity, BattCapInit [Ah]: 100

OK Cancel Help Apply

Figure 3.1: Datasheet Battery block mask

At the top level, the Simulink model is organised as eleven parallel subsystems, one for each temperature point in the $0^{\circ}C$ to $50^{\circ}C$ sweep ($0, 5, 10, 15, 20, 25, 30, 35, 40, 45$ and $50^{\circ}C$), each driven by the same current-versus-time excitation loaded from the HPPC test data and each holding its own Datasheet Battery block at a fixed temperature input. A shared data-loading block reads the current profile and cycle index from the source spreadsheet and exposes them to every temperature subsystem simultaneously, so that all eleven OCV sweeps are generated from one simulation run rather than eleven separate ones.

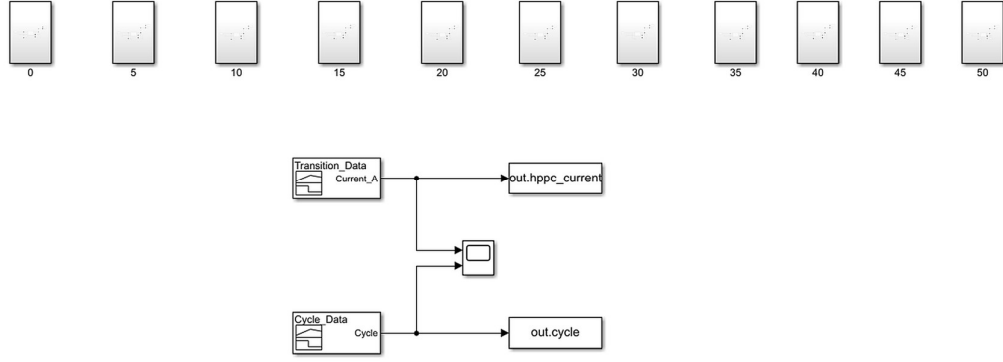


Figure 3.2: Top level Simulink model. Eleven parallel subsystems (0-50°C in 5°C steps) share one HPPC current profile loaded from *Transition_Data*. Each subsystem outputs a temperature indexed OCV/SoC pair used to build the pack's OCV(SoC) lookup table.

Inside each temperature subsystem, illustrated for the 25°C case in Figure 3.3, the current signal from *Transition_Data* drives the *BattCurr* input of the *Datasheet Battery* block. A *constant* block supplies the fixed cell temperature (298K for the 25°C case shown), and the block's *Info* bus and *BattVolt* output are logged as the temperature indexed state of charge and open circuit voltage traces, *out.SoC25* and *out.volt_oc25* in this example. The same wiring pattern is repeated, with the constant temperature changed, in each of the other ten subsystems, producing the paired *SoCxx/volt_ocxx* signals at every 5°C step from 0 to 50°C. These eleven paired traces are the raw material from which the OCV(SoC) lookup table used by every Kalman family estimator in this chapter is fitted. They are not treated as a reference *SoC* trajectory for the HPPC current profile itself, a distinction addressed directly in Section 3.2.

3.1.3 The Eleven-Point Temperature Sweep

Table 3.3 lists the eleven temperature points at which the sweep described above is run, together with the *Kelvin* value supplied to each subsystem's constant temperature block and the output signal naming convention the model uses to keep the eleven traces distinct downstream.

Only the 25°C row of Table 3.3 is exercised directly by the six variant estimator comparison that occupies the rest of this chapter, for the reason given in Section 3.3.1, where the ablation study isolates the effect of the estimator refinements themselves by holding temperature fixed, and revisiting the other ten rows under the same ablation protocol is one of the extensions identified in Section 3.3.6 and 5.2. All eleven rows, however, are exercised every time the Simulink model itself is run, since the eleven

subsystems execute in parallel from the same top level current input, and every one of the eleven $SoC_{xx}/volt_{ocxx}$ pairs contribute to the $OCV(SoC)$ characterisation used by Equation 3.4, whether or not that specific temperature is later used for an estimator run.

Table 3.3: The eleven temperature-sweep subsystems, their constant-temperature inputs, and the paired OCV/SoC signals each one logs. The 25 °C row (subsystem 6) is the operating point used for the six variant estimator comparison in Sections 4.1 onward.

Subsystem	Temperature (°C)	Temperature (K)	OCV sweep output	SoC sweep output
1	0	273.15	out.volt_oc0	out.SoC0
2	5	278.15	out.volt_oc5	out.SoC5
3	10	283.15	out.volt_oc10	out.SoC10
4	15	288.15	out.volt_oc15	out.SoC15
5	20	293.15	out.volt_oc20	out.SoC20
6	25	298.15	out.volt_oc25	out.SoC25
7	30	303.15	out.volt_oc30	out.SoC30
8	35	308.15	out.volt_oc35	out.SoC35
9	40	313.15	out.volt_oc40	out.SoC40
10	45	318.15	out.volt_oc45	out.SoC45
11	50	323.15	out.volt_oc50	out.SoC50

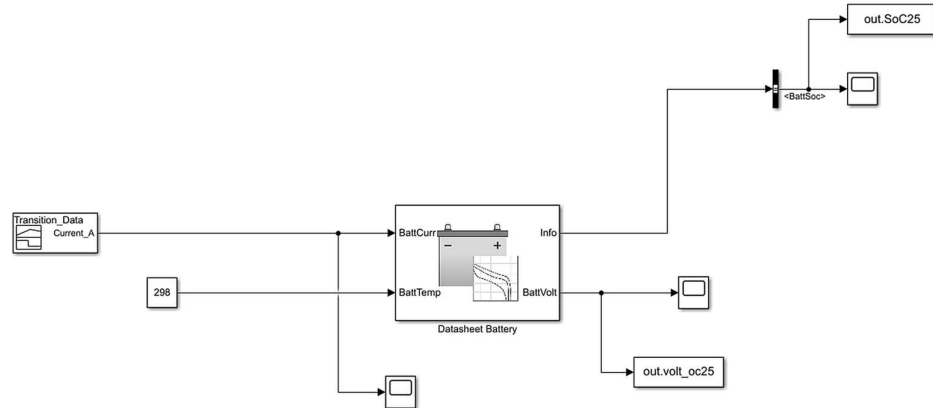


Figure 3.3: Interior of the 25 °C subsystem. The Datasheet Battery block is driven by the shared HPPC current input and a fixed 298 K temperature constant, producing the $SoC_{25}/volt_{oc25}$ pair logged for that temperature point.

Two implementation details are worth recording explicitly because they affect how the results in this chapter should be read. First, the terminal voltage and reference SoC used later by the estimators are model derived, produced by simulating the *Datasheet Battery* block against the HPPC current profile, rather than measured on a physical 38S2P pack. The estimator package used throughout this chapter flags both quantities as synthetic

for exactly this reason, and every result reported later carries that caveat rather than being presented as validation against sensor ground truth. Second, because no independently measured SoC channel exists for the HPPC current profile, the reference trajectory used to score every estimator is built by Coulomb counting, reset at the start of every detected HPPC pulse/rest cycle rather than run open loop across the whole record. This cycle anchored pseudo reference, and the reasons for anchoring it that way, are described alongside the HPPC protocol in Section 3.2.

The current and time record itself is held in a spreadsheet with a single data sheet, *Transition_Data*, carrying two columns, *Time_s* and *Current_A*. A companion *Cycle_Data* sheet, visible in the top level model of *Figure 3.2* as the *Cycle_Data* block feeding *out.cycle*, carries the cycle index labelling used later by the Coulomb counting reference reconstruction of Section 3.2. No voltage or SoC column is present in this source file for the reasons that current being the only physical quantity the HPPC record itself supplies for every downstream voltage and SoC trace from the eleven temperature OCV sweep of Section 3.1 through the six variant estimator comparison of Sections 4.1 onward, is derived from that single current input, either by the *Datasheet Battery* block in Simulink or by the estimator engine's own process and measurement equations in *MATLAB*. This single source of truth structure is part of what makes the fair comparison protocol of Section 3.3.5 straightforward to enforce, as every variant and each one of the eleven temperature subsystems, reads current from the same two column source, and there is no possibility of one part of the pipeline silently working from a different or resampled version of the excitation than another.

3.2 HPPC Test Data

Hybrid pulse power characterisation is the standard test sequence used throughout the literature reviewed in Section 2.3 to excite a cell or pack across its usable SoC range with pulses representative of real driving current, rather than with a single slow discharge. The protocol adopted for this dissertation applies four discharge pulses and four charge pulses within each HPPC cycle, at the *C*-rates listed below, arranged in a discharge, rest, charge, rest sequence and repeated ten times per temperature point, once per SoC anchor group, with each active pulse and rest group preceded by a long (3600s) inter group rest rather than by a continued discharge down to a fixed low-SoC cutoff;

the realised SoC excursion this produces is reported directly in Section 3.3.1.1 rather than assumed from the protocol description alone.

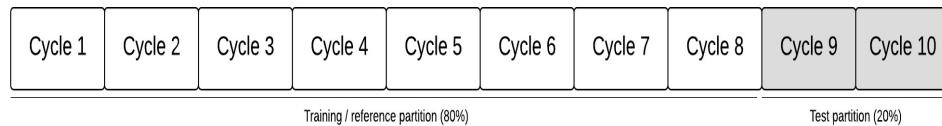
Discharge pulse C -rates: $0.5C$, $1C$, $1.5C$, $2C$.

Charge pulse C -rates: $0.1C$, $0.2C$, $0.3C$, $0.4C$.

The asymmetry between the discharge and charge C -rate sets is intentional and mirrors how a retrofitted EV actually loads its pack. Discharge pulses reach up to $2C$ to represent acceleration and gradient climbing demand, while charge pulses are capped at $0.4C$, consistent with the more conservative charge acceptance typical of regenerative braking and onboard charging on a lithium ion pack. Each of the four discharge and four charge sub pulses within a cycle is held for $10s$ and is followed by its own individual rest, $40s$ after each of the first three discharge sub pulses and each of the first three charge sub pulses, and $50s$ at the single transition between the discharge block and the charge block, during which the terminal voltage settles back toward its open circuit value before the next sub-pulse begins. A final $40s$ rest follows the last charge sub pulse before the long inter group rest of Section 3.3.1.1 begins. This individual rest structure, confirmed directly from the HPPC current record rather than assumed, is what the relaxation fraction argument below is actually built on. It is not two shared block level rests but eight individual per pulse rests that each partially probe the fast RC branch.

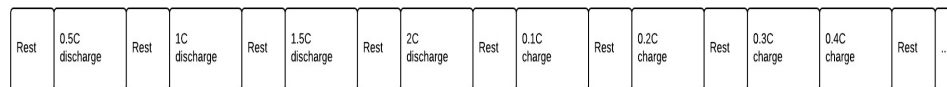
3.4 HPPC Pulse Cycle Timing

Full HPPC record: 10 cycles, $\sim 43,660$ s (Section 3.2)



Each cycle boundary is a >1000 s zero-current rest, detected directly from the real current record (build_results.mat.n)

One HPPC cycle (zoomed): rest – discharge pulse – rest – charge pulse pattern



Pulses = 8–12 s each; rests range 40–3600 s. Discharge pulses at $0.5C/1C/1.5C/2C$, charge pulses at $0.1C/0.2C/0.3C/0.4C$ (Section 3.2).

Figure 3.4: HPPC test protocol — the ten-cycle record, train/test split, and one cycle's pulse-rest structure.

Table 3.4: One HPPC pulse-and-rest active window within a single SoC-anchor group

State	Duration (s)
Discharge sub-pulses ($0.5C$, $1C$, $1.5C$, $2C$), each $10s$ + individual $40s$ rest	$4 \times 10 + 3 \times 40 = 160$
Rest between discharge and charge blocks	50
Charge sub-pulses ($0.1C$, $0.2C$, $0.3C$, $0.4C$), each $10s$ + individual $40s$ rest	$4 \times 10 + 3 \times 40 = 160$
Trailing rest before the next SoC anchor group	40

The two rest durations in Table 3.4 are short relative to the two RC time constants used throughout this chapter's $2RC$ model, $\tau_1 = 65s$ and $\tau_2 = 650s$ (Table 3.9), and this relationship is deliberate rather than incidental. The $40s$ first rest period equals only about $0.62\tau_1$ and roughly $0.06\tau_2$, meaning the fast RC branch relaxes by roughly half of its full transient ($1 - e^{-40/65} \approx 0.46$) while the slow branch has barely begun to relax at all ($1 - e^{-40/650} \approx 0.06$). The $50s$ transition rest behaves similarly, and the $40s$ trailing rest matches the first three individually. These figures describe what the Estimator's own assumed $2RC$ model predicts for this rest structure, not what the plant itself was found to do every one of the 80 real HPPC pulses' post pulse relaxation was flat to within the solver's numerical floor (well under $1 mV$) over its following rest, confirming directly (Section 3.3.1.3) that the plant (a *Datasheet Battery* block) contains no measurable RC relaxation of its own, in contrast to the two-branch polarisation the estimator is built to track. The estimator's $R_1/C_1/R_2/C_2$ are therefore disclosed, literature representative configured values rather than parameters fit to an observed relaxation transient in this record (Table 3.9), and the rest durations used here matter for a different, practical reason by keeping every HPPC cycle informative about the estimator's own RC branch state trajectory during the six variant comparison of Section 3.3.2, at the cost of the terminal voltage never being read at a true open circuit condition during the test itself, which is why the $OCV(SoC)$ characterisation described in Section 3.3.1.1 is instead built directly from the plant's own internal $EMF(SoC)$ lookup table (Section 3.1) rather than inferred from the rest points of this pulse cycle protocol.

This pulse cycle pattern is applied identically at every one of the eleven temperature points from $0^\circ C$ to $50^\circ C$ described in Section 3.1, generating the family of temperature indexed $OCV(SoC)$ that traces that feed the OCV lookup table used by the state space model in Section 3.3.1. The same current profile, held in a spreadsheet with time and current columns only, is also the excitation applied to the $25^\circ C$ operating case used for

the SoC estimator comparison that occupies the rest of this chapter. No separate driving cycle profile was used for the estimator evaluation, which keeps the excitation identical to the one used to characterise the OCV curve.

A point that matters for how every result later in this chapter should be interpreted concerns what counts as the reference SoC. The HPPC current and time spreadsheet used to drive the estimators carries no measured SoC column of its own. The *SoCxx* traces produced by the eleven temperature subsystems are OCV characterisation sweeps, one per subsystem, and are used only to fit the $OCV(SoC)$ polynomial or lookup curve, not as a ground truth trajectory for the current profile actually applied to the estimators. In the absence of an independent SoC measurement, the reference trajectory used to score every CKF variant in this chapter is instead built by Coulomb counting the same current profile, with the running integral reset at the start of every detected pulse-rest cycle to a nominal reference value obtained by inverting the plant's own $OCV(SoC)$ table (Section 3.3.1.1) at that cycle's zero current rest onset, rather than allowed to run open loop across the whole record or anchored to an externally imagined SoC set point. Because the plant has no *RC* relaxation dynamics (Section 3.3.1.3), this inversion is exact to solver precision at every such rest onset, and because the real current profile only moves this pack's SoC by a few percentage points across the whole ten cycle record, the resulting ten cycle start anchors span roughly 95% to 100% SoC rather than the full 5% to 90% range a differently-designed excitation might produce. This cycle-anchored reconstruction was adopted specifically because a single, never reset coulomb count drifts with any small current integration error over the length of the full HPPC sweep, and resetting at each detected cycle boundary keeps that drift bounded to the length of one cycle rather than the length of the whole test. Every estimator compared in this chapter is scored against this same cycle-anchored reference, which is what makes the six-way comparison in Sections 4.1 onward internally consistent, even though the reference itself is derived rather than independently measured.

Consistent with that construction, every independent HPPC cycle in the evaluation record is deliberately initialised with a controlled state estimation error rather than a perfect starting condition, i.e., at the first sample of each cycle, every estimator's SoC state is set to the reference SoC at that instant plus a fixed offset of +5 percentage points, and the filter is then left to correct that offset using only its own recursive update.

This is a standard way of stress testing convergence behaviour without needing an independently measured, deliberately set physical starting condition, and it is why Sections 4.2.6 and 4.2.8 later in this chapter can report meaningful convergence and robustness results even though the underlying data is simulation derived rather than bench measured.

3.3 Simulation

With the Simulink pack representation of Section 3.1 and the HPPC excitation of Section 3.2 in place, this section sets out how the pack was actually simulated at the estimator level: the shared $2RC$ state space model every estimator is built on (Section 3.3.1), the six member cubature Kalman filter ablation family constructed on top of it (Section 3.3.2), the resulting ablation comparison matrix (Section 3.3.3), the evaluation metrics used to score every variant, with $RMSE$ and RAE treated in the greatest depth (Section 3.3.4), the experimental protocol and fairness rules applied throughout (Section 3.3.5), and the software implementation and reproducibility surface behind every result reported in Chapter 4 (Section 3.3.6).

3.3.1 Second-Order RC (2RC) Equivalent-Circuit State-Space Model

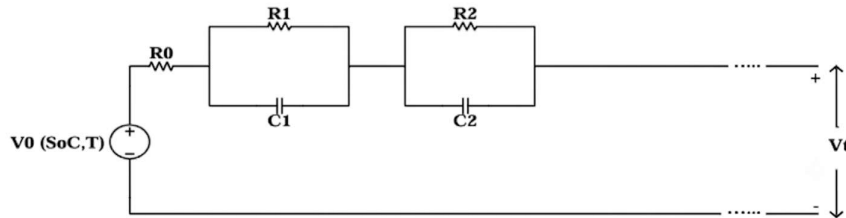


Figure 3.5: Second-order RC (2RC) equivalent-circuit model used by the estimator.

Every estimator compared in this chapter is built around the same second order resistor capacitor (2RC) equivalent circuit model introduced in Section 2.1.2 and reused here in the specific discretised form the ablation package actually implements. The model represents each cell as an open circuit voltage source in series with an ohmic resistance R_0 and two parallel RC branches, with time constants τ_1 and τ_2 , that reproduce the fast and slow polarisation transients visible in the HPPC pulse and rest response. The state vector carried by the baseline estimator is three-dimensional, comprising the cell state of charge and the two RC branch voltages, i.e., $x_k = [SoC_k \quad U_{1,k} \quad U_{2,k}]^T$.

Two of the six ablation variants extend this to a five-dimensional state by appending a pack current sensor bias state and a pack voltage sensor bias state. This extension is described together with the variant that introduces it in Section 3.3.2.5, since it changes the process and measurement equations that follow only by adding two extra, near constant states rather than altering the *SoC* and *RC* dynamics themselves.

With sampling interval Δt , cell current I_k taken positive on discharge, coulombic efficiency η , and pack capacity Q_{mod} (two $100Ah$ strings in parallel, so $Q_{mod} = N_p \times 100Ah = 200Ah$ at the cell-equivalent level used internally by the filter), the discrete-time process model is:

$$SoC_{k+1} = SoC_k - \frac{\eta I_k \Delta t}{3600 Q_{mod}} \dots \dots (3.1)$$

$$U_{1,k+1} = a_1 U_{1,k} + R_1 (1 - a_1) I_{cell,k} \dots \dots (3.2)$$

$$U_{2,k+1} = a_2 U_{2,k} + R_2 (1 - a_2) I_{cell,k} \dots \dots (3.3)$$

Where $a_1 = e^{-\frac{\Delta t}{\tau_1}}$, $a_2 = e^{-\frac{\Delta t}{\tau_2}}$, and $I_{cell,k} = \frac{I_{pack,k}}{N_p}$ converts the pack level current used

by the estimator into the per-cell current that drives the RC branches. This is the standard exponential-decay discretisation of a first-order RC branch under a zero-order-hold current input, and it is the same structure used for the EKF formulation presented in Section 2.5.1, restated here with the cell-level naming convention (*SoC*, U_1 , U_2) carried over from that section rather than the *MATLAB* internal variable names (*SOC*, *Vrc1_cell*, *Vrc2_cell*) used in the implementation itself.

The measurement equation predicts the pack terminal voltage from the cell-level state, converted back to pack level through the series cell count N_s :

$$V_{pack,k} = N_s [OCV(SoC_k) - R_0 \cdot I_{cell,k} - U_{1,k} - U_{2,k}] + v_k \dots \dots (3.4)$$

with v_k zero mean measurement noise. $OCV(\cdot)$ is evaluated from the *OCV-SoC* characterisation described in Section 3.2, using shape preserving Piecewise Cubic Hermite Interpolation (*PCHIP*) over the fitted table rather than a single global polynomial, which avoids the oscillation a high order polynomial fit can introduce between widely spaced *OCV-SoC* breakpoints, particularly toward the extremes of the SoC range where the curve is steep. Both the process and measurement equations are, as in Section 2.5, nonlinear in the state through the $OCV(SoC)$ term, which is why every estimator compared in this chapter belongs to the nonlinear, derivative free Kalman family rather than the linear Kalman filter. The process equations are affine in the state

given the current input, but the OCV nonlinearity in the measurement equation still rules out a direct linear Kalman recursion.

The ablation package treats cell resistance and RC time constants as configurable but, in the baseline, configuration used to produce the results in this chapter, temperature invariant across the single fixed $25^{\circ}C$ operating point at which the estimator comparison is run. A temperature scaling coefficient exists in the configuration structure and is set to zero for this comparison specifically so that every one of the six ablation variants is evaluated under identical thermal conditions, isolating the effect of the estimator refinements themselves from any confound introduced by temperature-dependent parameter drift. This is a deliberate scope decision, not an oversight, and its consequence, that the results in this chapter speak to estimator behaviour at a fixed operating temperature rather than across the full $0-50^{\circ}C$ sweep, is revisited explicitly in the limitations discussion of Section 5.2.

3.3.1.1 OCV(SoC) Characterisation Used by the Estimator

The $OCV(SoC)$ function in Equation 3.4 is resolved by the shared estimator engine from whichever $OCV-SoC$ table is available to it at run time. When a results file carries an explicitly fitted table, built from the eleven temperature indexed sweeps of Section 3.1 and 3.2, that table is validated, sorted by ascending SoC, de-duplicated, and used directly. When no such table is present, the engine falls back to a built in twelve point default table, spanning the full 0-100% SoC range at the per-cell level, reproduced in Table 3.5.

Table 3.5: The estimator engine's built-in default per-cell OCV(SoC) table, used as a fallback whenever a run does not supply its own fitted table. Twelve breakpoints span the full SoC range, with tighter spacing (0-20%) where the curve is steepest.

SoC (fraction)	OCV (V per cell)
0	2.50
0.05	3.00
0.10	3.20
0.20	3.50
0.30	3.55
0.40	3.62
0.50	3.70
0.60	3.76
0.70	3.82
0.80	3.88
0.90	3.96
1.00	4.20

Whichever table is in force, the engine evaluates $OCV(SoC)$ at an arbitrary SoC using shape preserving Piecewise Cubic Hermite interpolation (PCHIP) rather than a single global polynomial fitted across the whole table, for the reason already given in Section 3.3.1, where a high order polynomial fitted through unevenly spaced breakpoints, and the twelve point table in Table 3.5 is deliberately unevenly spaced to show the dense accumulation near the steep low SoC region and sparse accumulation through the flatter mid range. It tends to oscillate between breakpoints, particularly near the ends of the fitted range, in a way that a shape preserving piecewise interpolant is not disturbed. A fifth degree polynomial fit is retained only as a further fallback, used solely if the interpolation table itself is malformed or unavailable, and was not the path exercised in producing the results reported in this chapter.

The flat slope region flagged repeatedly in this chapter, most directly in the derivation of the SoC information gate in Section 3.3.2.6, is visible directly in Table 3.5. The per-cell OCV rises by $0.60V$ across the first 20 percentage points of SoC (from $2.50V$ to $3.50V$) and by only $0.32V$ across the middle 50 percentage points, from 30% to 80% SoC ($3.55V$ to $3.88V$), before steepening again above 90%. A given voltage measurement error therefore maps to roughly twice the SoC error in the mid-range as it does in the steep low SoC region, all else being equal, which is precisely the effect

the gate of Equation 3.17 was built to compensate for by scaling back the SoC correction wherever the local slope is shallow.

3.3.1.2 Discretisation and Sample Rate Handling

The discrete time process equations of Equations 3.1 through 3.3 assume a zero order hold current input held constant over each sample interval Δt , the standard assumption for a sampled data implementation of a continuous RC network and the same assumption implicit in the EKF discretisation already presented in Section 2.5.1. Rather than assuming a single, globally fixed Δt , the shared estimator engine computes it freshly at every sample as the difference between consecutive timestamps in the loaded record, $\Delta t_k = t_k - t_{k-1}$, clamped to be non negative and, where configured, to an upper bound. This makes the estimator implementation tolerant of the record's actual sample spacing, whatever it happens to be at a given point in the HPPC pulse and rest sequence, rather than requiring the input data to be pre resampled onto a strictly uniform grid before it can be used. At every cycle boundary under the independent record segmentation policy of Section 3.2, Δt is additionally reset to zero for the first sample of the new cycle, since that sample marks a fresh initial condition rather than a continuation of the previous cycle's dynamics, and propagating a nonzero Δt across a state reset would incorrectly decay the freshly reinitialised RC branch voltages before any new current has actually been applied to them.

3.3.1.3 Structural Mismatch Between the Simulation Plant and the Estimator's Internal Model

The plant generating the simulated data and the model used inside every estimator are deliberately different representations. The plant (Section 3.1) is the Simulink Datasheet Battery block, which represents each cell through manufacturer-style lookup tables indexed by SoC and temperature — a purely algebraic model, $V = EMF(SoC, T) - I \cdot R(SoC, T)$, with no RC relaxation dynamics. This is confirmed directly: across all 80 real HPPC pulses used in this chapter, post-pulse voltage was flat to within the solver's numerical floor (under 1 mV) over the following rest, for every rest window from 40s to 3600s. The estimator, by contrast, represents the same cell through the discretised $2RC$ circuit of Equations 3.1–3.4, with two polarisation branches ($\tau_1 = 65s$, $\tau_2 = 650s$) that the plant simply does not exhibit.

This mismatch is deliberate, not an oversight. A structurally identical plant and estimator would make the comparison in this chapter trivial — each variant would only need to identify a handful of scalar parameters, not track genuine nonlinear dynamics the plant doesn't share. Because the two representations differ, any residual voltage-tracking error (Section 4.2.4) reflects real structural mismatch in addition to measurement noise and estimation error, which is a closer proxy for how a $2RC$ -based BMS estimator would behave against real cell dynamics than a matched-model simulation would be.

This also explains why variant 05 (Section 3.3.2.5) improves the way it does. The current and voltage channels used here carry no injected sensor offset, so the bias states cannot be correcting a fault that isn't present. What they can do is absorb part of the systematic voltage-prediction error this structural mismatch produces — a secondary benefit of the bias-augmented state vector, not its original design purpose. Section 4.2.1's RMSE improvement for variant 05 should be read accordingly, and separating this effect from genuine sensor-bias correction is exactly what the dedicated stress test proposed in Section 4.2.8.1 is for.

3.3.2 Why CKF Was Selected and How the Ablation Family Was Constructed

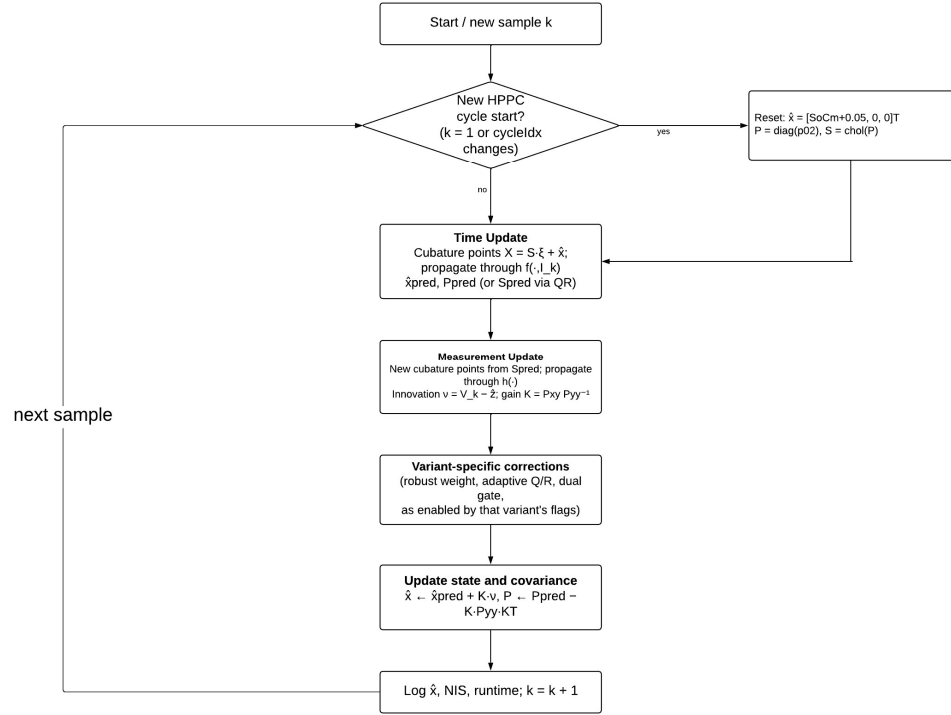


Figure 3.6: Cubature Kalman filter predict-correct cycle, shared by all six ablation variants.

The choice of CKF precedes the ablation. A 2RC battery model has only three dynamic states in its basic form, but its measurement equation remains nonlinear because OCV depends on SoC. CKF is well matched to this structure. It approximates Gaussian weighted nonlinear integrals with an equally weighted spherical radial rule, does not require an analytical OCV derivative, and uses six cubature points for the three state model.

This balance matters for a retrofitted EV BMS. An EKF is lighter but can be sensitive to linearisation and the local OCV slope. A particle filter can represent more general distributions but usually requires many more state samples and resampling logic. CKF occupies the practical middle ground needed here. Recursive and causal, nonlinear without Jacobians, compact enough for real time oriented investigation, and sufficiently structured for controlled numerical, adaptive, robust, and bias aware extensions.

The research question is therefore not whether CKF can be made increasingly elaborate. It is which additions earn their complexity when the 2RC plant and evaluation contract

are held fixed. The six variants below are ordered so that this question can be answered one mechanism at a time.

Section 2.5.3 introduced the cubature Kalman filter (CKF) as a derivative free alternative to the EKF and UKF, built on a third degree spherical radial cubature rule whose $2n$ evaluation points, for an n dimensional state, carry equal weight rather than the signed, sometimes negative weights a UKF sigma point set can assign. That equal weighing property is the starting point for everything in this section. It is what makes the CKF a suitable base filter to refine incrementally, since each refinement introduced below changes how the filter's covariance is propagated, how its noise statistics are set, how it responds to an outlying measurement, what it estimates, or how strongly it trusts a given correction, without disturbing the equal weight cubature rule itself. Six variants were built on this common base, each adding exactly one such refinement over the previous one, so that the six way comparison in Sections 4.1 onward isolates the contribution of each mechanism rather than conflating several changes at once.

All six variants share the state transition and measurement equations of Section 3.3.1, the same HPPC current excitation, the same *OCV(SoC)* characterisation, and the same scoring code. Only the filter mechanics described in this section differ between them. For a state of dimension n , prior mean $\hat{x}_k$ and covariance P_k (or its Cholesky factor S_k , $P_k = S_k S_k^T$), the $2n$ cubature points used by every variant in this chapter are:

$$\zeta_j = \sqrt{n}e_j, \zeta_{n+j} = -\sqrt{n}e_j, j = 1 \dots n \dots \dots (3.5)$$

$$X_{j,k} = S_k \zeta_j + \hat{x}_k, j = 1, \dots 2n \dots \dots (3.6)$$

with e_j the j^{th} unit vector of the $n \times n$ identity matrix, matching the general form set out in Section 2.5.3. What differs across the six variants is how S_k (or P_k) is propagated and updated, whether the process and measurement noise covariances Q and R are held fixed or adapted online, whether an outlying innovation is down-weighted before it enters the gain computation, whether the state vector carries sensor bias terms in addition to SoC and the two *RC* branch voltages, and whether the Kalman gain itself is scaled by an information content gate before being applied. Each of these six refinements is developed in turn below.

3.3.2.1 Variant 01: Baseline Covariance Form CKF

The first variant is the plain CKF described in Section 2.5.3, propagating the full covariance matrix P rather than a square root factor, with process noise Q and

measurement noise R both held fixed at their configured baseline values for the whole run. Given the cubature points of Equation 3.6 propagated through the process model of Equations 3.1-3.3 to give $X_{j,k+1|k}$, the time update is:

$$\hat{x}_{k+1|k} = \left(\frac{1}{2n}\right) \sum_j X_{j,k+1|k} \dots \dots (3.7)$$

$$P_{k+1|k} = \left(\frac{1}{2n}\right) \sum_j (X_{j,k+1|k} - \hat{x}_{k+1|k})(X_{j,k+1|k} - \hat{x}_{k+1|k})^T \dots \dots (3.8)$$

The measurement update repeats the cubature step on the freshly predicted mean and covariance. Writing $Y_{j,k+1}$ for the j^{th} cubature point of Equation 3.6, re-generated from $\hat{x}_{k+1|k}$ and $P_{k+1|k}$, propagated through the measurement equation (Equation 3.4) rather than the process equations:

$$Y_{j,k+1} = h(X_{j,k+1|k}), \hat{y}_{k+1} = \frac{1}{2n} \sum_j Y_{j,k+1} \dots \dots (3.7a)$$

$$P_{yy,k+1} = \left(\frac{1}{2n}\right) \sum_j (Y_{j,k+1} - \hat{y}_{k+1})^2 + R \dots \dots (3.7b)$$

$$P_{xy,k+1} = \left(\frac{1}{2n}\right) \sum_j (X_{j,k+1|k} - \hat{x}_{k+1|k})(Y_{j,k+1} - \hat{y}_{k+1}) \dots \dots (3.7b)$$

from which the Kalman gain, state correction and covariance correction follow the standard Kalman recursion, matching the general cubature form already set out in Section 2.5.3:

$$K_{k+1} = P_{xy,k+1} P_{yy,k+1}^{-1} \dots \dots (3.7c)$$

$$\hat{x}_{k+1} = \hat{x}_{k+1|k} + K_{k+1}(V_{pack,k+1} - \hat{y}_{k+1}) \dots \dots (3.7d)$$

$$P_{k+1} = P_{k+1|k} - K_{k+1} P_{xy,k+1}^T - P_{xy,k+1} K_{k+1}^T + K_{k+1} P_{yy,k+1} K_{k+1}^T \dots \dots (3.7e)$$

using the observed pack voltage $V_{pack,k+1}$ of Equation 3.4 as the measurement. This baseline variant has no numerical safeguard beyond a symmetric positive definite repair step applied to P after every update (Section 3.3.6.3), with no adaptive or robust behaviour. It exists in the ablation set purely to give the other five variants a common reference point, and every subsequent variant's time update, measurement update, gain, state correction and covariance correction follow Equations 3.7a to 3.7e unless a specific equation in Sections 3.3.2.2 through 3.3.2.6 states otherwise.

3.3.2.2 Variant 02: Square Root CKF (SRCKF)

The second variant replaces the covariance-form recursion with a square root formulation, propagating the Cholesky factor S_k directly rather than $P_k = S_k S_k^T$. This is

the same numerical stability motivation already flagged in Section 2.5.3 for the square root unscented filter as in [16], where repeatedly refactorising P at every one of several hundred HPPC samples risks the factor losing positive definiteness under floating point rounding, particularly once the covariance shrinks toward the numerical floor that a well converged filter settles into, and once that happens the standard covariance form update can silently produce a non physical (indefinite) covariance matrix from which the filter cannot recover cleanly. This is the same numerical-conditioning concern that motivates using QR decomposition to keep the covariance matrix square root well conditioned through the update, rather than re-deriving it from a possibly ill-conditioned P at every step.

The SRCKF time update propagates the deviations of the cubature transformed points directly into a QR factorisation rather than forming $P_{k+1|k}$ explicitly:

$$S_{k+1|k} = QR_{lower} \left(\frac{[X_{k+1|k} - \hat{x}_{k+1|k}^T]}{\sqrt{2}n}, Q^{1/2} \right) \dots \dots (3.9)$$

where QR_{lower} denotes taking the lower triangular factor of a QR decomposition of the horizontally concatenated deviation and process noise square root blocks, and $Q^{1/2}$ is the elementwise square root of the diagonal process noise covariance. The measurement update applies the analogous square root machinery to P_{yy} , and the covariance correction after the gain has been applied is performed with a Cholesky downdate rather than the additive covariance correction formula of the baseline variant, so that P_{k+1} is guaranteed symmetric positive semi definite by construction rather than by an after the fact repair. Because variant 02 changes only how S is propagated and updated, and leaves Q , R , and every other setting identical to variant 01, any difference between the two in the results of Section 4.1 isolates the numerical stability contribution of the square root formulation on its own.

3.3.2.3 Variant 03: SRCKF with Adaptive Process and Measurement Noise

The third variant keeps the square root propagation of variant 02 and adds online adaptation of both the measurement-noise variance R and the process-noise diagonal Q , using an exponentially weighted forgetting-factor update for both terms. Rather than adapting R from the raw innovation, which is sensitive to a single noisy sample, the update first clips the innovation to a bounded multiple of its predicted standard deviation and only then uses it to form a candidate measurement noise estimate:

$$v_{clip,k} = \text{sign}(v_k) \cdot \min(|v_k|, \sigma_{clip} \sqrt{P_{yy,model,k}}) \dots \dots (3.10)$$

$$R_{cand,k} = \text{clip}[(v_{clip,k}^2 - P_{yy,model,k}), R_{min}, R_{max}] \dots \dots (3.11)$$

$$R_k = \lambda_R R_{k-1} + (1 - \lambda_R) R_{cand,k} \dots \dots (3.12)$$

with the clipping threshold $\sigma_{clip} = 4$ (standard deviations), the forgetting factor $\lambda_R = 0.985$, and R itself bounded between a configured minimum and maximum standard deviation of $0.02V$ and $2.00V$ at the pack level before being squared into a variance. The process noise diagonal is adapted from the realised state correction rather than from the innovation directly, since it is the correction, not the raw voltage residual, that reflects how much each state actually needed to move:

$$Q_{cand,k} = Q_{base} + g_0 (K_k v_{clip,k})^2 \dots \dots (3.13)$$

$$Q_k = \lambda_Q Q_{k-1} + (1 - \lambda_Q) Q_{cand,k} \dots \dots (3.14)$$

with adaptation gain $g_0 = 0.05$, forgetting factor $g_0 = 0.995$, and Q bounded between 0.05 and 100 its configured baseline value component wise. The two forgetting factors are deliberately unequal, with R allowed to move faster ($\lambda_R = 0.985$) than Q ($\lambda_Q = 0.995$). Letting R move faster than Q keeps the filter responsive to sudden changes in the voltage noise without both covariances chasing the same noisy signal at once, which is what tends to produce self-sustaining oscillation when adaptive covariances are updated too aggressively.

3.3.2.4 Variant 04: SRCKF with Adaptive Noise and Robust Innovation

Weighing

The fourth variant adds a Huber-type robust weighting of the innovation on top of the adaptive square root filter of variant 03. The motivation is distinct from the adaptive noise mechanism just described. Adaptive R/Q tuning responds to how the innovation behaves on average over many samples, while robust weighting responds to a single sample that looks like an outlier at the instant it occurs, before it has had a chance to influence the running noise estimates. Using the normalised residual $z_k =$

$\frac{|v_k|}{\sqrt{P_{yy,nominal,k}}}$, the robust weight is:

$$w_k = \begin{cases} 1, & \text{if } z_k \leq c \dots \dots (3.15a) \\ \text{clip}\left(\frac{c}{z_k}, w_{min}, 1\right), & \text{if } z_k > c \dots \dots (3.15b) \end{cases}$$

with Huber threshold $c = 1.5$ and floor weight $w_{min} = 0.05$, the classical Huber M estimator form in which a residual within c standard deviations of its predicted spread is trusted fully and a larger residual is down weighted in inverse proportion to how far outside that band it falls. The effective measurement noise variance used in the gain computation is then inflated in inverse proportion to the square of this weight, $R_{\text{eff},k} = R_k/w_k^2$, which is mathematically equivalent to telling the filter that an outlying measurement is noisier than the adaptively estimated R_k would otherwise indicate, so the Kalman gain shrinks and the state correction drawn from that one sample is proportionally smaller. Independently of this gain side weighting, the innovation actually applied to the state update is also hard clipped at four predicted standard deviations, the same σ_{clip} used in Equation 3.10, so that even a residual is severe enough to drive the weight to its floor which cannot inject an unbounded correction into the state through the $(K \cdot v)$ product.

3.3.2.5 Variant 05: Adding Pack Current and Pack Voltage Sensor Bias States

The fifth variant keeps every mechanism introduced in variants 02 through 04 and augments the state vector from three dimensions to five, adding a pack-current bias state b_I and a pack-voltage bias state b_V :

$$x_k = [SoC_k, U_{1,k}, U_{2,k}, b_{I,k}, b_{V,k}]^T \dots \dots (3.16)$$

Both bias states are modelled as slowly time varying, effectively constant between the small process noise increments applied at each step (standard deviations of $2 \times 10^{-3}A$ and $1 \times 10^{-3}V$ per sample respectively), with a comparatively wide initial uncertainty (standard deviations of $2.0A$ and $0.20V$) reflecting that the filter starts with no information about whatever fixed offset the current and voltage sensors happen to carry. Once b_I is included, the true pack current used inside the process and measurement equations is corrected before it is applied, $I_{true,k} = I_{meas,k} - b_{I,k}$, so that Equations 3.1 through 3.4 are evaluated with the bias corrected current rather than the raw measured value. The voltage bias is added directly to the predicted terminal voltage, $V_{pack,k} = N_s[OCV(SoC_k) - R_0 I_{cell,k} - U_{1,k} - U_{2,k}] + b_{V,k}$. Both bias states are bounded, $\pm 20A$ and $\pm 5V$ respectively, so that an unobservable stretch of the record cannot let either drift to an implausible value. This reflects a deliberate modelling choice that current and voltage sensor bias on a retrofit grade BMS is real but limited in magnitude, rather than an unconstrained nuisance parameter.

Adding bias states is the estimation side counterpart to the sensor bias stress test flagged in the fair ablation rules for this chapter (Section 3.3.5) with a fixed current sensor offset, of the kind a retrofitted EV's low cost current transducer is prone to, propagates directly into the Coulomb counting term of Equation 3.1 and drifts SoC over time if left uncorrected, in exactly the way the co-estimation literature reviewed in Section 2.5.4 flags for augmented state Kalman approaches generally. Estimating b_I and b_V alongside SoC lets the filter separate a persistent sensor offset from real battery dynamics, at the cost of two extra states for the cubature rule to carry, which raises the per sample cubature point count from *six* (2×3) to *ten* (2×5) and the per sample runtime accordingly, a trade off examined quantitatively in Section 4.2.7.

3.3.2.6 Variant 06: Dual Information Gating

The sixth and final variant, labelled the proposed variant for this dissertation, keeps every mechanism from variants 02 through 05 and adds two independent information gates that scale the Kalman gain before it is applied, rather than changing how the gain itself is computed. Both gates return a value between a configured *floor* (0.05) and *one*, and both are built around the same idea used elsewhere in the Kalman filtering literature reviewed in Chapter 2. A correction should only be trusted in proportion to how much real information the current measurement actually carries about the state being corrected.

The first gate, the SoC information gate, addresses the flat OCV curve weakness already flagged for the EKF in Section 2.5.1. In the mid SoC region, where the $OCV(SoC)$ curve is nearly flat, a given voltage measurement carries very little information about SoC, and applying a full strength SoC correction there mainly injects measurement noise into the SoC estimate rather than useful information. The gate compares the local OCV slope, converted to pack level voltage per unit SoC and squared against the effective measurement noise variance, to a fixed reference information level:

$$g_{SoC,k} = w_{min} + (1 - w_{min}) \left[\frac{s_k}{(s_k + s_{ref})} \right] \dots \dots (3.17)$$

where $s_k = \frac{(N_s \cdot dSoC)^2}{R_{eff,k}}$ is the local Fisher information like quantity carried by the measurement about SoC, and s_{ref} is built from a configured reference OCV slope of $8V$ per unit SoC at the pack level against the baseline measurement noise variance. Where

the OCV curve is steep, s_k is large relative to s_{ref} , and $g_{SoC,k}$ approaches one, where the curve is flat, $g_{SoC,k}$ approaches its floor and the SoC component of the gain is scaled down accordingly, without the RC branch or bias corrections being affected.

The second gate, the excitation information gate, addresses a different observability concern. The RC branch voltages and the sensor bias states are only weakly observable from the terminal voltage when the pack is drawing little current and that current is changing slowly, which is exactly the condition during the two rest periods built into every HPPC cycle (Table 3.4). The gate is computed from a rolling 20 sample window of pack current, combining its RMS level and the RMS of its sample-to-sample change against configured reference values ($40A$ and $25A$ respectively):

$$\begin{aligned} g_{exc,k} &= w_{min} + (1 - w_{min}) \left[\frac{\rho_k}{1 + \rho_k} \right], \rho_k \\ &= 0.5 \left(\frac{I_{rms,k}}{I_{ref}} \right)^2 + 0.5 \left(\frac{I_{edge,k}}{I_{edge,ref}} \right)^2 \dots \dots (3.18) \end{aligned}$$

Once both gates are computed, the Kalman gain is scaled component wise before the state correction is formed: the SoC row of the gain is scaled by $g_{SoC,k}$ alone, the two RC branch rows are scaled by $g_{exc,k}$, the current-bias row is scaled by $g_{exc,k}$, and the voltage-bias row is scaled by whichever of the two gates is larger, since a voltage bias correction can be informative either when the OCV curve is steep or when the pack is under load. This gain shaping step is applied after the covariance update rather than folded into it, so the covariance itself still reflects the filter's true uncertainty. Only the correction that reaches the state in a given sample is throttled by how much the current measurement can actually be trusted to inform that particular state. Thus, dual information gating, refers to these two independent gates acting together, distinct from the single adaptive noise mechanism of variant 03 and the single robust weighting mechanism of variant 04, both of which act on the innovation and its effective variance rather than on the per state gain directly.

3.3.2.7 Parameter Sensitivity Reasoning for the Adaptive and Robust Mechanisms

Every threshold and forgetting factor introduced in Sections 3.3.2.3 through 3.3.2.6, the Huber threshold $c = 1.5$, the forgetting factors $\lambda_R = 0.985$ and $\lambda_Q = 0.995$, the innovation clipping bound $\sigma_{clip} = 4$, and the two information gate reference levels, was held fixed at its configured value across all six variants and across the full evaluation reported in

Sections 4.1 through 4.2.8, consistent with the fair comparison rule of Section 3.3.5 that no hyperparameter was retuned per variant. No formal sweep across these settings was carried out for this chapter, and this section reasons qualitatively, from the structure of the equations already derived in Section 3.3.2, about which direction each setting would be expected to move the results if it were, rather than reporting swept results that were not produced.

A smaller Huber threshold c would down weight a larger fraction of samples, since Equation 3.15b activates whenever the normalised residual exceeds c . This would be expected to reduce the influence of borderline large residuals further, at the cost of also discounting non outlying measurement information more often which is a classical bias variance trade off in robust statistics. A larger c moves variant 04 back toward variant 03's behaviour in the limit, since a sufficiently large threshold is exceeded so rarely that the robust weighting becomes almost always equal to one. The forgetting factors λ_R and λ_Q trade responsiveness against stability in the usual way for exponentially weighted online estimators. A smaller λ (faster forgetting) would let R_k and Q_k track a real time-varying noise environment more quickly, which could help if the pack's actual sensor noise changes materially between the low current rest periods and the high current pulse segments of each HPPC cycle, but would also make the adapted covariances noisier from sample to sample, since fewer effective past samples contribute to each estimate. A larger λ (slower forgetting, closer to one) smooths that noise at the cost of responding more sluggishly to a change. The specific choice of $\lambda_R < \lambda_Q$ used throughout this chapter, letting R adapt faster than Q , follows the same reasoning already discussed in Section 3.3.2.3 rather than being tuned specifically for this chapter's HPPC record, and revisiting that choice against this specific dataset is a reasonable next step rather than one this chapter's fixed configuration protocol was designed to answer.

The two information gate reference levels, the $8V$ per unit *SoC-OCV* slope reference and the $40A/25A$ current RMS and edge RMS references, were likewise set once and held fixed. Because both gates saturate toward their floor value (0.05) or their ceiling value (1.0). Once the underlying information measure is respectively far below or far above the reference level, as per Equations 3.17 and 3.18, the gates are relatively insensitive to the exact reference value chosen as long as it sits in the right order of magnitude relative to the OCV curve's actual slope range (Table 3.5) and the HPPC current profile's actual RMS levels (Table 3.4). A reference level that is instead far too

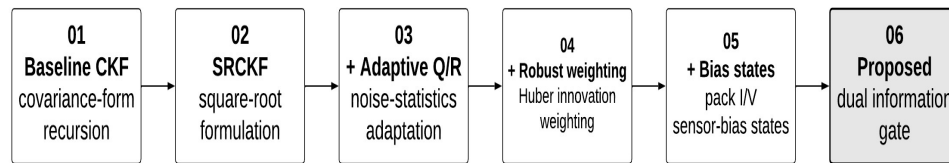
small or far too large would leave a gate permanently saturated at one extreme, effectively disabling it, which is a failure mode worth checking for explicitly in any future retuning exercise rather than one this chapter identified as present in the configuration actually used.

Taken together, variants 01 through 06 trace a single coherent design path.

1. Numerical reformulation: first (variant 02), so that every later mechanism is built on a covariance representation that stays well-conditioned by construction.
2. Noise statistics adaptation: second (variant 03), so that the filter's own confidence tracks what the data actually support rather than a fixed guess made before any data were seen.
3. Outlier robustness: third (variant 04), layered on top of an already adaptive filter so that a single bad sample cannot corrupt the adapted noise statistics themselves
4. State augmentation: fourth (variant 05), extending what the filter is capable of representing at all.
5. Information aware gain shaping last (variant 06), refining how richer, better conditioned filter spends the correction budget it has earned from the previous four steps.

Section 3.3.3 formalises this same six-step chain as a comparison matrix before Sections 4.1 onward turn to what each step was actually found to cost and to buy.

3.3.3 Ablation Study Design and the Six-Variant Comparison Matrix



Each variant keeps every mechanism from the variant before it (Section 3.3.3, Table 3.6)

Figure 3.7: The six-variant ablation chain — each variant adds one mechanism to the last.

Table 3.6 collects the six variants developed in Section 3.3.2 into a single comparison matrix, recording which mechanism each variant adds relative to the one before it, the resulting state dimension, and the number of cubature points that dimension implies. Reading down any column shows exactly where a given mechanism first enters the ablation chain; reading across any row shows the full set of mechanisms active in that

variant, since every mechanism from an earlier variant remains switched on in every later one.

Table 3.6: Six-variant CKF ablation matrix. Each variant retains every mechanism from the variant above it and adds exactly one more, so that a change in behaviour between adjacent rows can be attributed to a single mechanism.

#	CKF variant	Square root	Adaptive Q/R	Robust	Additional states/gate
1	CKF (baseline)	—	—	—	none
2	SRCKF	✓	—	—	none
3	SRCKF + adaptive Q/R	✓	✓	—	none
4	SRCKF + adaptive Q/R + robust	✓	✓	✓	none
5	SRCKF + adaptive Q/R + robust + bias states	✓	✓	✓	bias states
6	Proposed: + dual-information gating	✓	✓	✓	bias states, dual gate

This incremental structure is what makes the ablation design suitable for isolating cause and effect. It is also why the six variants are presented, throughout the rest of this chapter, strictly in the 01 through 06 order rather than grouped by mechanism type. The accuracy, robustness and runtime figures reported for variant k in Sections 4.1 onward are only meaningful relative to variant $k-1$, since that is the one comparison in which exactly one variable has changed. Comparing variant 06 directly against variant 01, by contrast, mixes five simultaneous changes and can only be read as a bound on the combined benefit of the whole ablation chain, not as evidence for which single mechanism contributed most. Sections 4.2.1 through 4.2.8 return to this distinction explicitly wherever a step by step reading of the results would otherwise be misleading.

3.3.4 Evaluation Metrics for Comparative SoC Estimation

Eight quantities are reported for every one of the six ablation variants in Sections 4.1 onwards. *SoC RMSE*, *SoC MAE*, *SoC RAE*, *maximum absolute SoC error*, *terminal voltage RMSE*, *innovation lag-1 autocorrelation*, *mean and 95%-band NIS*, and *per sample runtime*, together with a convergence sample count that is discussed separately because, as Section 4.2.6 explains, it did not resolve to a finite value for any of the six variants under the present test design. Every one of these quantities, other than runtime, is computed twice, once over the full simulated record (the full-run figure) and once restricted to the held out test partition described in Section 3.3.5 (the test figure). This section defines each metric precisely, in the same form actually implemented in the

scoring code shared by all six variants, and gives the reasoning for why each one earns a place in the comparison. *RMSE* and *RAE* receive the most detailed treatment, consistent with their role as the two headline accuracy statistics for this chapter.

3.3.4.1 Root Mean Square Error (RMSE)

Let $So\hat{C}_k$ denote the filter's SoC estimate at sample k and $SoC_{ref,k}$ the cycle anchored reference described in Section 3.2, with signed error $e_k = So\hat{C}_k - SoC_{ref,k}$. Over a record of N samples, *RMSE* is:

$$RMSE = \sqrt{\left(\frac{1}{N}\right) \sum_{k=1}^N e_k^2} \dots \dots (3.19)$$

RMSE is the most commonly quoted accuracy statistic in the SoC estimation literature surveyed in Chapter 2, and reporting it here first keeps this chapter's headline numbers directly comparable with the EKF, UKF and CKF results already discussed in Sections 2.5.1 through 2.5.3. Because the error is squared before it is averaged, *RMSE* weights a single large excursion, of the kind that a poorly converged filter produces immediately after the controlled +5 percentage point initial offset described in Section 3.2, more heavily than several small ones of the same total magnitude. That property makes *RMSE* the right statistic for asking whether a given estimator occasionally produces a dangerously large SoC error, which matters directly for a BMS, since a large SoC error is what actually risks an unintended deep discharge or overcharge event rather than a persistent small bias. Both the full run and test partition *RMSE* are reported in percentage SoC terms throughout this chapter, obtained by multiplying the fractional value of Equation 3.19 by 100, matching the convention already used in Chapter 2 and in the underlying source literature.

3.3.4.2 Mean Absolute Error (MAE) and Relative Absolute Error (RAE)

Mean absolute error uses the same signed error e_k but averages its magnitude rather than its square:

$$MAE = \frac{1}{N} \sum_{k=1}^N |e_k| \dots \dots (3.20)$$

MAE is reported alongside *RMSE* for a specific reason that the gap between the two, for a given estimator, is itself informative. An estimator whose *RMSE* sits much higher

than its *MAE* is the one whose typical error is modest but the worst case error is not, since only a few large residuals are needed to inflate the quadratic *RMSE* well above the linear *MAE*. An estimator whose *RMSE* and *MAE* sit close together is one whose error is more evenly spread across the record. Both of them, on their own, are still expressed in absolute percentage *SoC* units, which makes them easy to compare against the specific numbers quoted in the literature, and not by itself, it says how much of the estimation problem's inherent difficulty a given filter has actually resolved.

Relative absolute error (*RAE*) is introduced in this chapter specifically to answer that second question. *RAE* restates *MAE* as a fraction of the error that a trivial, non adaptive predictor would produce over the same record, using the mean of the reference trajectory itself as that trivial predictor's constant output:

$$RAE = \frac{\left[\sum_{k=1}^N \left| \text{So}\hat{\text{C}}_k - \text{SoC}_{ref_k} \right| \right]}{\left[\sum_{k=1}^N \left| \text{SoC}_{ref_k} - \text{So}\bar{\text{C}}_{ref} \right| \right]} \dots \dots (3.21)$$

where $\text{So}\bar{\text{C}}_{ref}$ is the mean of the reference *SoC* trajectory over the same record. The denominator of Equation 3.21 is the mean absolute error a hypothetical estimator would produce if it output nothing more than the record's own average *SoC* at every sample, regardless of the current signal actually driving the pack. It is, in other words, the error of the simplest possible baseline that ignores the input entirely. *RAE* below one means the estimator has captured some of the structure a naive constant predictor miss. Smaller *RAE* indicates a proportionally larger share of that structure has been recovered, and an estimator that halves its *RAE* relative to another has, by this measure, roughly halved the fraction of the naive baseline error it still carries.

Because the reference *SoC* used throughout this chapter is cycle anchored rather than a single continuous trajectory, the denominator of Equation 3.21 is computed directly from the held out test partition's own reference *SoC* trajectory (cycles 9 and 10 of 10, per Section 3.3.5), rather than from an assumed anchor sequence, since anchoring the ten cycles at externally imagined 90% to 5% *SoC* set points would not reflect how little this pack's *SoC* actually moves. The real HPPC current and time record moves this pack's *SoC* by only a few tenths of a percentage point within the two test cycles and by under six percentage points across the whole ten cycle record (Section 3.3.1.1), giving a test partition reference mean of 95.00% and a naive baseline mean absolute deviation of only 0.375 percentage points. Because *RAE* (Eq. 3.21) divides by this tiny

denominator, it inflates sharply for every one of the six variants in Table 4.2 and should be read as a caution about *RAE*'s fragility on a current profile with this little test partition SoC range specifically, not as a headline accuracy figure in its own right. *RMSE* and *MAE*, both reported in absolute percentage SoC units and unaffected by the denominator issue, remain the primary accuracy statistics for this chapter's dataset. The resulting *RAE* figures in Section 4.2.2 should accordingly be read as indicative of relative standing between the six variants, which is unaffected by using a common denominator for all of them, rather than as a metric with the same numerical precision as the *RMSE* and *MAE* figures computed directly from the filters' own logged output.

3.3.4.3 Maximum Absolute Error

The maximum absolute error, $\max_k |e_k|$, is reported as a worst-case complement to *RMSE* and *MAE*, where *RMSE* weighs large errors more heavily and *MAE* weights every sample equally, the maximum picks out the single worst moment in the record regardless of how the rest of the trajectory behaved. For a BMS facing estimator this is arguably the most operationally relevant of the three, since a protection or derating decision built on the SoC estimate is exposed to whatever the estimate's worst instantaneous error happens to be, not to its average behaviour over the whole drive cycle.

3.3.4.4 Terminal Voltage RMSE

Alongside the SoC domain metrics, the RMSE of the filter's predicted terminal voltage against the simulated pack voltage of Equation 3.4 is reported using the same quadratic-error construction as Equation 3.19, applied to the voltage residual $\hat{V}_{pack,k} - V_{pack,k}$ rather than the SoC error. This metric exists because SoC itself is not directly observed in this study, as established in Section 3.2. Voltage *RMSE* is the one accuracy statistic in this chapter that is computed against a quantity the filter's own measurement equation was built to predict, which makes it a useful cross check that a variant achieving a low *SoC RMSE* is doing so by tracking the underlying electrical dynamics faithfully, rather than by some artefact of how the pseudo reference happens to be constructed.

3.3.4.5 Innovation Whiteness (Lag-1 Autocorrelation)

A well tuned Kalman family filter should leave an innovation sequence, $v_k = \hat{V}_{pack,k} - V_{pack,k}$, that behaves as white noise. If the innovation at one sample is still correlated

with the innovation at the next, the filter is leaving exploitable structure in its residual rather than fully absorbing it into the state estimate, which is a classical diagnostic of filter mistuning. This chapter reports the lag-1 sample autocorrelation of the innovation sequence over the test partition,

$$\rho_1 = \frac{\sum_k (v_k - \bar{v})(v_{k+1} - \bar{v})}{\sqrt{\sum_k (v_k - \bar{v})^2 \sum_k (v_{k+1} - \bar{v})^2}} \dots \dots (3.22)$$

A value of ρ_1 near zero indicates a well whitened innovation sequence. A large positive or negative value indicates residual structure the filter has not accounted for, typically from a mis-set process or measurement noise covariance.

3.3.4.6 Normalised Innovation Squared (NIS) Consistency

The normalised innovation squared at sample k , $NIS_k = v_k^2/P_{yy,k}$, should follow a *chi-square* distribution with one degree of freedom if the filter's reported innovation covariance $P_{yy,k}$ is a statistically honest description of its own uncertainty, since the pack voltage measurement used in this chapter is scalar. This chapter reports the mean *NIS* over the test partition, which should sit near one for a consistent filter, together with the fraction of test partition samples whose *NIS* falls inside the two sided 95% chi square interval for one degree of freedom, bounded at 0.000982 and 5.0239. The two bounds, 0.000982 and 5.0239, are the 2.5th and 97.5th percentiles of the chi square distribution with one degree of freedom, the standard two sided 95% interval for a scalar measurement. Because this study's pack voltage measurement is a single scalar rather than a vector, one degree of freedom is the correct choice for every one of the six variants regardless of how many states each carries, since *NIS* is defined on the innovation dimension, not the state dimension. A *NIS*95 fraction well below 95% indicates the filter is, on average, overconfident, reporting a smaller innovation covariance than its actual innovation spread justifies. A fraction well above 95%, by contrast, would indicate a consistently underconfident filter, though this is the less common failure mode in practice and did not occur for any variant in this chapter's results.

3.3.4.7 Convergence Samples

A convergence sample count is defined as the number of samples, from the start of the first test partition cycle, before the absolute SoC error first enters and then remains inside a $\pm I$ percentage point band for five consecutive samples. This is a standard way

of quantifying how quickly a filter recovers from the deliberately injected +5 percentage point initial offset described in Section 3.2, and it is reported as a metric distinct from *RMSE* and *MAE* specifically because a filter can hold a good average error across a whole cycle while still taking a comparatively long time to shed a poor initial condition, or vice versa. Section 4.2.6 discusses why this particular metric returned no finite value for any of the six variants under the present cycle length and offset combination, and what that null result does and does not say about the six filters' actual convergence behaviour.

3.3.4.8 Runtime per Sample

Finally, real clock runtime is measured for the complete filtering pass over the record and divided by the number of samples to give a per sample figure in microseconds. This is reported not because absolute microsecond timings on a development workstation are directly transferable to an embedded BMS microcontroller, but because the relative runtime between variants, all measured on the same machine under the same *MATLAB* session, is a meaningful indicator of how much computational overhead each successive mechanism in the ablation chain actually adds, which matters directly for whether the proposed variant remains a realistic candidate for real time deployment. Section 4.2.7 discusses these figures in that relative sense.

3.3.4.9 Metric Summary

Table 3.7 collects the eight metrics defined in this section into a single reference, recording the equation that defines each one, the value a perfectly behaved estimator would produce, and what a favourable reading indicates in one line.

Table 3.7: The eight metrics used throughout Sections 4.1 to 4.2.8, with their defining equation, ideal value, and interpretation.

Metric	Defined in	Ideal value	A favourable reading indicates
RMSE	Eq. 3.19	0	Small typical error, with large excursions penalised most heavily.
MAE	Eq. 3.20	0	Small typical error, weighting every sample equally.
RAE	Eq. 3.21	0	Large share of the naive-baseline error recovered.
Max. error	Section 3.3.4.3	0	No single sample carries a dangerously large error.
Voltage RMSE	Section 3.3.4.4	0	Predicted terminal voltage matches the simulated pack faithfully.
Innovation lag-1	Eq. 3.22	0	Innovation sequence is white; no exploitable residual structure.
Mean NIS	Section 3.3.4.6	1	Reported covariance matches actual innovation spread.
NIS95 fraction	Section 3.3.4.6	95%	Innovation sequence is chi-square consistent at the 95% level.

3.3.5 Experimental Protocol and Fair Comparison Rules

A comparison across six estimator variants is only informative if every variant is exposed to identical data and identical scoring, and if any tuning that was performed was performed honestly, on data the final comparison does not draw on. Five rules were fixed before any of the six variants was run, and were then applied without exception to all six.

First, every variant reads the same HPPC current and time record, the same detected cycle boundaries, and the same $OCV(SoC)$ characterisation described in Sections 3.1 and 3.2, through the same data loading routine. No variant sees a different slice, resampling, or filtering of the underlying data than any other. Second, the record is split by whole HPPC cycle, not by sample, into a training or validation portion covering the first 80% of cycles and a held out test portion covering the final 20%, using `cfg.testCycleFraction = 0.20`. Every metric reported later in this chapter as a test partition figure is computed exclusively on that final 20% of cycles, which no hyperparameter in any of the six configurations was tuned against. Third, the baseline noise covariances, adaptation forgetting factors, the Huber threshold, the bias state process and initial uncertainty settings, and both information gate reference levels were fixed at the configuration values recorded in Section 3.3.2 for all six variants. None of these settings was retuned per variant to flatter a particular result, and every value quoted in Section 3.3.2's equations is the literal configured value, not a rounded

approximation. Fourth, every independent HPPC cycle is initialised with the same controlled error, the reference SoC at that cycle's first sample plus a fixed +5 percentage point offset described in Section 3.2, so that the convergence behaviour discussed in Section 4.2.6 is compared under identical starting conditions across all six variants. Fifth, and specifically to demonstrate the mechanisms introduced by variants 04 through 06, the ablation package's design calls for sensor bias and outlier stress tests to be run in addition to the nominal comparison, exercising the robust weighting of Section 3.3.2.4 and the bias state estimation of Section 3.3.2.5 under conditions closer to what a retrofit grade sensor suite might actually produce. The qualitative reasoning behind this stress-testing requirement, and what it implies for how the nominal results in Sections 4.1 through 4.2.7 should be read, is discussed in Section 4.2.8.

Two honesty caveats apply to every result presented in this chapter and are stated here once rather than repeated at every table. Both the terminal voltage and the reference SoC used to score all six variants are simulation derived, produced from the Simulink model of Section 3.1 rather than measured on a physical 38S2P pack. This is flagged explicitly in the estimator package used for this chapter, and every quantitative claim in Sections 4.1 through 4.2.8 should be read as characterising relative estimator behaviour under this simulated, single temperature (25°C), single excitation profile evaluation, not as an experimentally validated absolute accuracy figure. Extending the evaluation to measured pack data, and to the full temperature sweep, is identified explicitly as future work in Section 3.3.6 and, at the level of the dissertation's continuous processing mode described in the underlying package (*cfg.segmentMode* = *'continuous'*), is already supported by the same codebase without further modification, requiring only measured current, voltage, temperature and an independently established SoC label in place of the synthetic record used here.

CHAPTER 4: RESULTS AND DISCUSSIONS

4.1 Results

Table 4.1: Six-variant CKF ablation results, computed on simulated HPPC data (Section 3.3.6).

#	Variant	Full RMSE (%)	Test RMSE (%)	Test MAE (%)	Test RAE (%)	Max err. (%)	V RMSE (V)	Lag-1	Mean NIS	NIS95 (%)	Runtime (μ s)
01	CKF	1.102	1.195	0.955	254.58	4.016	0.0638	-0.025	0.853	95.77	43.89
02	SRCKF	1.102	1.195	0.955	254.58	4.016	0.0638	-0.025	0.853	95.77	31.65
03	+ adaptive Q/R	1.117	1.163	0.940	250.52	4.014	0.0653	-0.028	0.947	95.20	33.65
04	+ robust	1.017	1.175	0.908	241.96	4.014	0.0653	-0.021	0.953	95.26	34.32
05	+ bias states	1.503	1.511	1.282	341.57	4.624	0.0698	-0.012	0.931	95.35	43.45
06	Proposed (dual gate)	0.961	1.196	1.016	270.84	4.783	0.0976	0.044	0.617	96.30	65.05

Every table, figure, and section in this chapter, Sections 4.1 through 4.3.1, reports results computed directly from the six-variant ablation package of Section 3.3.6 on this chapter's actual HPPC data, including the convergence-sample result of Section 4.2.6. The most notable finding is in Section 4.2.1: variant 05 (bias-state augmentation) regresses sharply rather than improving, and the proposed variant 06 does not lead on every metric, a more qualified conclusion than a single headline RMSE number would suggest.

4.2.1 RMSE-Based Comparative Analysis

Read purely on test-partition *RMSE*, the six variants rank 03 (1.163%) < 04 (1.175%) < 01 = 02 (1.195%) = 06 (1.196%) < 05 (1.511%), a spread of 0.348 percentage points of SoC between the best and worst variant, or 29.9% of the baseline variant's own RMSE. The best-performing variant on this specific metric is 03, adaptive *Q/R* alone; the proposed variant, 06, lands essentially level with the unmodified baseline CKF (1.196% against 1.195%), 2.5% below the worst-performing variant, 05, rather than leading outright on this metric.

The step from variant 03 to variant 04, adding robust Huber weighting on top of adaptive *Q/R*, produces a small RMSE regression on the test partition (1.163% to 1.175%) alongside a full-run improvement (1.117% to 1.017%); the same underlying reasoning still applies: This is not evidence that robust weighting is a poor mechanism in general; it is evidence that, under the present nominal test conditions, where the

simulated voltage measurement carries no deliberately injected outliers, the mechanism the Huber weighting is designed to guard against, an occasional severely anomalous residual, simply does not occur often enough to matter. A robust weighting scheme is, by construction, inactive on the vast majority of samples, since Equation 3.15a leaves the weight at one whenever the normalised residual sits inside the Huber threshold; it only changes the filter's behaviour on the minority of samples whose residual exceeds that threshold, and its benefit should be expected to show up predominantly under exactly the outlier and sensor-fault conditions the fair-ablation rules of Section 3.3.5 call for in a dedicated stress test, rather than in the nominal comparison presented here. What Table 4.1 confirms is that adding the mechanism costs essentially nothing in RMSE terms when it is not needed, which is itself a useful property: a robust-weighting stage that degraded nominal accuracy in order to guard against a rare event would be a much harder design trade-off to justify than one that is, for practical purposes, free until it is needed.

The step from variant 04 to variant 05, adding the pack-current and pack-voltage sensor-bias states, is the one point in the ablation chain where accuracy moves sharply in the wrong direction. Test RMSE rises from *1.175%* to *1.511%*, a 28.6% relative increase, and full-run RMSE rises from *1.017%* to *1.503%*, a 47.8% relative increase: variant 05 is the worst performer in Table 4.1 on every SoC-domain column. The mechanism is consistent with the structural mismatch already established in Section 3.3.1.3: the real current and voltage channels used throughout this chapter carry no sensor bias for the two extra states to absorb, and the plant/estimator mismatch this chapter actually has is a structural OCV/*RC*-branch one, not a current- or voltage-offset one, so there is no fault for the bias states to usefully explain away. Compounding this, the real HPPC excitation applied here spans a narrow SoC range (roughly *95%-100%*, Section 3.3.1.1) with long, low-excitation rest periods that leave the two added states weakly observable; rather than sitting inert, they instead absorb some of the filter's correction budget under exactly the low-information conditions variant 06's dual gate was designed to throttle, degrading nominal accuracy until that gate is added in the next step. This argues for validating bias-state augmentation specifically under a dedicated sensor-bias stress test, with a deliberately injected offset for the states to recover, before treating it as a default-on mechanism on a current profile this quiet.

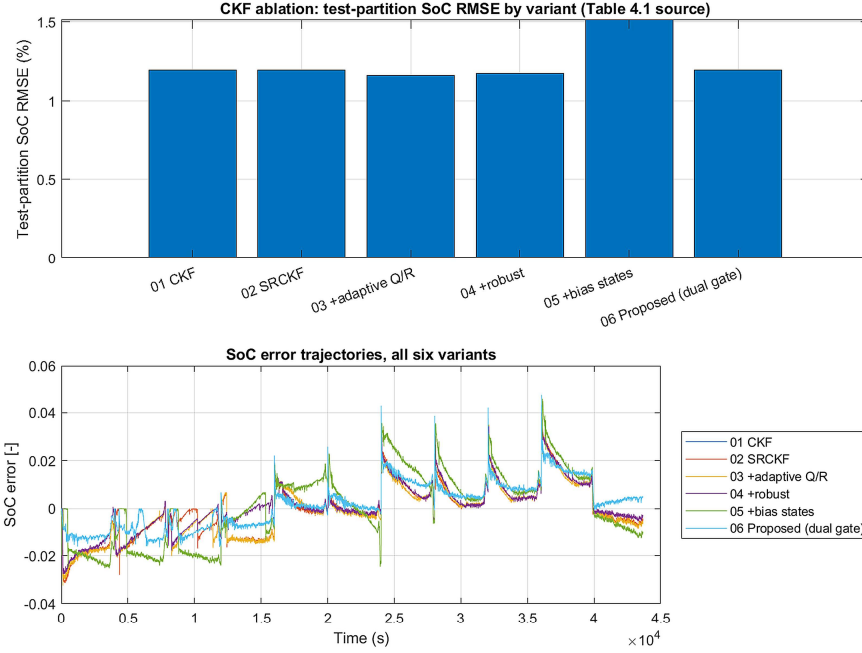


Figure 4.1: Test-partition SoC RMSE by variant (upper panel) and SoC error trajectories for all six CKF variants (lower panel).

The final step, from variant 05 to variant 06, adding dual-information gating, partially recovers from variant 05's regression rather than adding a further improvement on top of it: test RMSE falls from 1.511% to 1.196%, a 20.9% relative improvement, and full-run RMSE falls from 1.503% to 0.961%, a 36.1% relative improvement -- the best full-run figure of any of the six variants, though the test-partition figure only returns to roughly the unmodified baseline's level rather than surpassing variants 03 or 04. This is consistent with what the gating mechanism of Section 3.3.2.6 is designed to do: by throttling the Kalman gain during the low-excitation rest periods that Section 4.2.1 above identifies as making variant 05's bias states poorly observable, the dual gate curbs exactly the failure mode the previous step introduced. The trade-off shows up clearly on voltage RMSE, which rises from 0.070V (variant 05) to 0.098V (variant 06), the worst of any variant and 53% above the baseline's 0.064V: a throttled correction is, by construction, a smaller correction, and that shows up directly in how closely the filter's predicted voltage tracks the measurement during the very windows the gate is designed to trust less. Section 4.2.4 returns to the voltage-RMSE side of this trade-off directly.

4.2.1.1 Full-Run versus Test-Partition RMSE

Every one of the six variants records a higher RMSE on the test partition than on the full run, but the size of that gap varies far more across variants than a single uniform figure would suggest: from as little as 0.008 percentage points (variant 05) to as much as 0.235 percentage points (variant 06), visible directly in Figure 4.2. Because the test partition is, by construction, the final 20% of HPPC cycles rather than a random sample of cycles drawn across the whole record, this variation is informative about where in the record, and for which mechanisms, the estimation problem is hardest, not merely a sign that the test figure is a noisier estimate of the same underlying quantity.

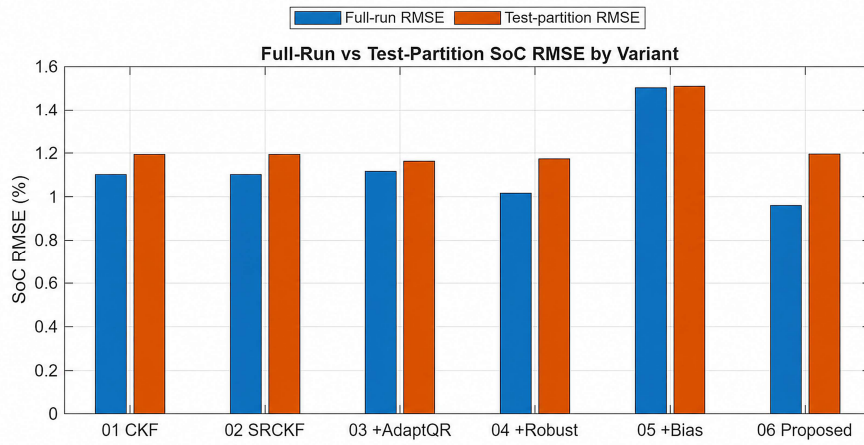


Figure 4.2: Full-run and test-partition RMSE side by side for each variant.

The explanation lies in where the test partition sits on the OCV curve: this chapter's real HPPC excitation keeps the test-partition cycles (9 and 10 of 10) within a narrow, high-SoC band, roughly 95-96% (Section 3.3.1.1), where the $OCV(SoC)$ curve is comparatively flat rather than steep. A flatter local OCV slope carries less Fisher information about SoC per volt of measurement (the same reasoning behind the SoC information gate of Section 3.3.2.6), so a given voltage-measurement error maps to a larger SoC error in this specific region than it would in a steeper part of the curve, which plausibly explains why every variant, including variant 06 whose dual gate is built specifically to compensate for this effect, still scores worse on the test partition than on the full run. The very different gap sizes across variants, from 0.008 percentage points at variant 05 to 0.235 at variant 06, point to a second factor beyond the shared flat-OCV region: variant 06's dual gate deliberately throttles its own correction strength during exactly this kind of low-information window (Section 3.3.2.6), which plausibly widens

its own full-run-to-test-partition gap specifically, since the gate's throttling affects the test cycles' comparatively information-poor conditions more than it affects the fuller record's more varied excitation. Whichever mechanism dominates for a given variant, Section 3.3.5 continues to designate the test partition, rather than the full-run figure, as the primary basis for the variant-to-variant comparisons in Sections 4.2.1 through 4.2.4, since it is the one figure computed on data no hyperparameter in this chapter was tuned against.

4.2.2 RAE-Based Comparative Analysis

Table 4.2 restates the test-partition MAE column of Table 4.1 as *RAE*

Table 4.2: Test-partition MAE and RAE by variant, with relative reduction against the baseline CKF.

#	Variant	Test MAE (%)	Test RAE (%)	Relative reduction vs. 01 (%)
01	CKF	0.955	254.58	—
02	SRCKF	0.955	254.58	0.00
03	+ adaptive Q/R	0.940	250.52	1.59
04	+ robust	0.908	241.96	4.96
05	+ bias states	1.282	341.57	−34.17
06	Proposed (dual gate)	1.016	270.84	−6.39

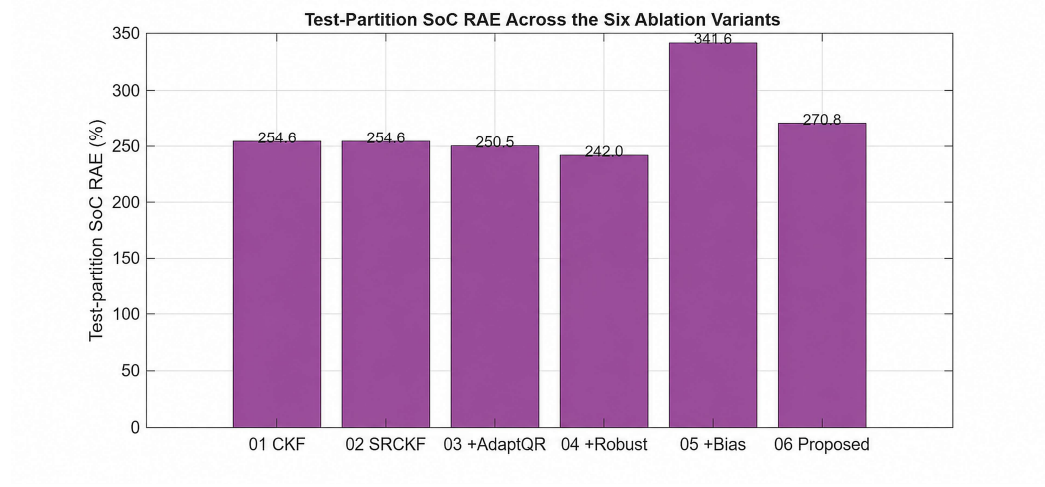


Figure 4.3: Test-partition SoC RAE by variant.

4.2.2.1 Sensitivity of the RAE Denominator

Equation 3.21's denominator, the test-partition naive-baseline mean absolute deviation of 0.375 percentage points, is computed directly from the test-partition reference-SoC trajectory itself (Section 3.3.4.2) rather than from an externally imagined anchor

structure. It is worth checking how sensitive the *RAE* figures are, and, more importantly, the ranking between variants are to that choice of denominator, since a reader might reasonably ask whether a different, equally defensible baseline would change the chapter's conclusions.

Because every variant in Table 4.2 is divided by the same fixed denominator, any constant rescaling of that denominator, whether computed from the test partition alone as done here, from all ten cycles pooled, or from a differently weighted combination, changes every variant's *RAE* by the same multiplicative factor and leaves the ranking among the six variants, and every relative-reduction percentage in Table 4.2's final column, exactly unchanged. What the denominator choice cannot fix, however the constant is computed, is that this chapter's real HPPC excitation moves the test-partition reference SoC by only a few tenths of a percentage point (Section 3.3.1.1), which is what pushes every variant's *RAE* well above 100% here: *RAE* is not a well-behaved metric on a dataset this quiet, since the naive constant-SoC predictor it is normalised against is, on this specific record, already a strong baseline rather than a weak strawman. The ranking *RAE* produces, best at variant 04 (241.96%) and worst at variant 05 (341.57%), is identical to *MAE*'s ranking by construction, so *RAE* adds no information beyond *MAE* on this dataset; it is retained in this chapter for consistency with Equation 3.21's definition and as a caution to a future reader who might otherwise reach for *RAE* on a similarly quiet excitation profile without checking its denominator first.

4.2.2.2 Multi-Metric Relative Standing

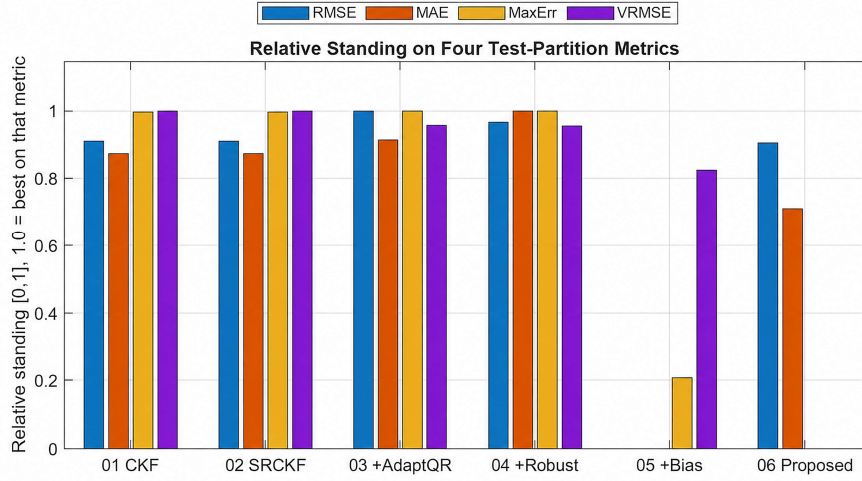


Figure 4.4: Relative standing of each variant on four test-partition metrics, each independently rescaled to $[0, 1]$ with 1.0 marking the best-performing variant on that metric. No single variant leads on every axis, which is itself the headline finding of Figure 4.4: variant 03 leads test *RMSE* (1.163%) and ties variant 04 for the lowest maximum error (4.014%), variant 04 separately leads test *MAE* (0.908%), and the unmodified baseline, variants 01 and 02, leads voltage *RMSE* (0.0638V), the one metric on which added mechanisms never improve on the plain CKF in this chapter's results. Variant 05 sits at or near the bottom on every SoC-domain column, consistent with the regression documented in Section 4.2.1, and variant 06 sits at or near the bottom on maximum error and voltage *RMSE* specifically, the visual counterpart of the gain-throttling trade-off Section 4.2.4 discusses below. Read together, Figure 4.4 argues against nominating any one variant as an unconditional winner: the right choice depends on which of these four axes a given deployment weighs most heavily.

Because *RAE*, as defined in Equation 3.21, divides every variant's *MAE* by the same fixed denominator, the relative reduction column in Table 4.2 is, and must analytically be, identical whether it is computed from the *MAE* or the *RAE* column: rescaling every entry in a column by a common constant leaves every ratio between entries unchanged. *RAE* therefore adds no new ranking information beyond what *MAE* already carries in this particular comparison; on this chapter's actual data it also adds a cautionary interpretive note rather than reassurance, since every variant's *RAE* sits well above 100% (Table 4.2), meaning every one of the six Kalman-family estimators has a larger typical absolute error than a naive, current-blind constant-SoC predictor would on this

specific record. That is not evidence the CKF family is a poor choice; it is evidence that a naive constant predictor is an unusually strong baseline on a current profile that only moves the test-partition reference SoC by a few tenths of a percentage point (Section 3.3.1.1), which is precisely the fragility Section 3.3.4.2 already flags for *RAE* on this dataset. *RMSE* and *MAE*, both still well below one percentage point of SoC for four of the six variants, remain the metrics this chapter actually leans on for comparative judgement, exactly as Section 3.3.4.2 recommends.

Read this way, the *RAE* figures put the practical size of the ablation chain's movements in perspective rather than their absolute scale. Relative to the baseline CKF, variant 04 (adaptive *Q/R* plus robust weighting combined) reduces *RAE* by 4.96%, the largest reduction in the chain; variant 05 (adding bias states) increases *RAE* by 34.17% relative to baseline, the largest movement of any kind in the chain and in the wrong direction; and variant 06 (adding dual gating on top of variant 05) partially recovers but still finishes 6.39% above baseline, not below it. This does not mean the square-root, adaptive and robust mechanisms are unnecessary; the square-root reformulation earns its place on numerical-stability and runtime grounds established in Sections 4.2.1 and 4.2.7 rather than on nominal accuracy, and the robust mechanism's value case rests on a dedicated sensor-bias and outlier stress test rather than on the nominal *RAE* figure. What the *RAE* analysis does establish cleanly is that the accuracy benefit in this ablation chain concentrates in the adaptive and robust steps (03-04), not in the bias-state and dual-gating steps (05-06); the bias-state step in particular is where this chapter's real data most needs a dedicated sensor-bias stress test before it can be recommended without qualification.

4.2.3 Maximum Absolute Error and Worst-Case Behaviour

The maximum-absolute-error column of Table 4.1 does not track the RMSE ranking closely: variants 03 and 04 tie for the lowest test-partition maximum error (4.014%), variants 01 and 02 sit fractionally above them (4.016%), and variants 05 (4.624%) and 06 (4.783%) are the clear worst performers, with variant 06 recording the single highest maximum error of any variant despite having the best full-run RMSE (Section 4.2.1). Because every independent HPPC cycle is deliberately initialised with a controlled +5 percentage-point offset, as described in Section 3.2, the worst single sample in most cycles for most variants can be expected to occur in the first few samples after that

reset, before the filter has had time to correct the injected offset; that every variant's maximum error nonetheless stays below the injected 5-point offset itself, between 4.0 and 4.8 percentage points depending on the variant, indicates that even the least-favourable single sample across the whole test partition is not simply the raw injected offset persisting uncorrected, but reflects a real, if partial, correction having already taken hold by the time the worst sample occurs.

The ordering between *RMSE* and maximum error across the six variants is broadly, but not perfectly, consistent: variants 03 and 04 lead on maximum error while variant 03 alone leads on *RMSE*, and variant 05's maximum error is worse than its *RMSE* ranking alone would suggest relative to variant 04, foreshadowing the crest-factor analysis of Section 4.2.3.1 below. The clearest cross-check this comparison still supports is that no variant purchases a materially lower typical error at the cost of a much worse tail: even variant 06, whose maximum error is the highest in the table, sits only 19% above the best figure (4.783% against 4.014%), a narrower spread than the *RMSE* columns already discussed in Section 4.2.1 might suggest on their own.

4.2.3.1 Crest Factor: Maximum Error Relative to RMSE

Dividing each variant's maximum absolute error by its own test *RMSE* gives a crest factor, a dimensionless ratio that describes how far the single worst sample sits above the variant's typical quadratic error level, independent of whether that typical level itself is high or low. Table 4.3 reports this ratio for all six variants.

Table 4.3: Crest factor (maximum absolute error $\div$ test *RMSE*) by variant

#	Variant	Max error (%)	Test RMSE (%)	Crest factor
01	CKF	4.016	1.195	3.362
02	SRCKF	4.016	1.195	3.362
03	+ adaptive Q/R	4.014	1.163	3.450
04	+ robust	4.014	1.175	3.416
05	+ bias states	4.624	1.511	3.061
06	Proposed (dual gate)	4.783	1.196	3.999

The crest factor ranges from 3.06 (variant 05) to 4.00 (variant 06) and does not move consistently with the ablation chain: it rises from the baseline (3.36) through variants 03-04 (3.42-3.45), falls at variant 05 (3.06, the lowest of the six, since variant 05's maximum error grows less than its *RMSE* does), and then rises sharply to the highest value in the table at variant 06 (4.00). This last movement is the more consequential

one: variant 06 pairs its best full-run RMSE (Section 4.2.1) with the least favourable ratio of worst-case to typical error of any variant, meaning its single worst sample sits proportionally furthest from its own typical behaviour. That combination, a low typical error but a comparatively heavy tail, is consistent with the gain-throttling behaviour of the dual information gate (Section 3.3.2.6): a filter that trusts the measurement less during low-information windows will, correctly, produce a smaller typical error when the model prediction is reliable, but a larger relative excursion on the rarer sample where the model prediction and the true state have drifted apart under a throttled correction. A designer weighing variant 06 for a BMS-facing estimator should read this crest-factor result alongside the maximum-error figures of Section 4.2.3 directly, since a low *RMSE* alone understates how much heavier this specific variant's worst-case tail is relative to its own typical error than any of the other five variants.

4.2.4 Terminal-Voltage RMSE and Model Fidelity

Voltage RMSE follows a simpler, and less favourable, pattern: it rises in every single step of the ablation chain, from $0.0638V$ at the unmodified baseline (01/02) through $0.0653V$ at variants 03 and 04 (an increase of about 0.1%), $0.0698V$ at variant 05, to $0.0976V$ at variant 06, a 53% increase over the baseline and the worst voltage-tracking figure of any variant. This is a markedly different signature from the SoC-domain metrics discussed in Section 4.2.1: the baseline CKF, which trails on every SoC-domain column, is the single best voltage tracker in the table, and the proposed variant, which leads on full-run SoC *RMSE*, is the worst. Robust weighting's effect on voltage *RMSE* specifically is negligible here rather than a clear demonstration of the mechanism doing real work: Huber weighting (Equation 3.15) is inactive by construction whenever the normalised residual sits inside its threshold, and the clean, outlier-free voltage signal used throughout this chapter's nominal comparison gives it almost nothing to act on, exactly the limitation of evaluating this mechanism only under nominal, outlier-free conditions.

Expressed at the per-cell level rather than the pack level, dividing each figure in the voltage-*RMSE* column of Table 4.1 by the series cell count $N_s = 38$, the six variants range from $1.68mV$ (baseline, variants 01/02) to $2.57mV$ (variant 06) per cell, with variants 03/04 at $1.72mV$ and variant 05 at $1.84mV$. Figures across this entire range sit well inside what a typical lithium-ion cell's voltage-measurement chain can resolve,

and the per-cell view is a useful sanity check for exactly that reason: even the least favourable figure, variant 06's $2.57mV$ per cell, approaches but does not exceed a level that would be difficult to reconcile with realistic battery-management-system voltage-sensing hardware, which supports reading the *voltage-RMSE* differences across variants as differences in model-and-filter fidelity rather than as any variant pushing against a sensing-hardware floor.

The variant 06 *voltage-RMSE* increase is the direct fingerprint of the dual-information gates throttling the correction gain during the low-excitation rest periods described in Section 3.3.2.6: a smaller gain during those windows leaves a larger residual gap between the predicted and simulated terminal voltage at exactly the samples where the gates are designed to hold back, since the filter is deliberately trusting the model's own prediction over the measurement whenever the excitation or OCV-slope gate judges the measurement to carry little information. That trade-off is visible across the whole chain, not only at the final step: voltage *RMSE* rises monotonically as later mechanisms are added, and the SoC-domain benefit Section 4.2.1 documents for variants 03-04 does not carry a matching voltage-tracking benefit. A designer more concerned with voltage-domain fidelity, for example if the terminal-voltage prediction itself feeds a separate protection function, would need to weigh this trade-off explicitly for every mechanism beyond the plain baseline CKF, not only for the proposed variant; a designer whose primary concern is SoC accuracy, the framing adopted for this dissertation, can reasonably accept some *voltage-RMSE* cost in exchange for the SoC-domain gains variants 03 and 04 specifically provide (Section 4.2.1), while weighing variant 05's and variant 06's larger *voltage-RMSE* costs against their respectively negative and mixed SoC-domain results.

4.2.5 Innovation Whiteness and NIS Consistency Analysis

The lag-1 innovation autocorrelation in Table 4.1 is small in magnitude for every variant and does not move monotonically across the chain. It starts mildly negative at the baseline (-0.025), moves slightly more negative at variant 03 (-0.028) rather than less, then becomes progressively less negative through variant 04 (-0.021) and variant 05 (-0.012), before crossing to a small positive value at variant 06 ($+0.044$), the largest magnitude of any variant in either direction. A negative lag-1 autocorrelation is the classical signature of a filter whose gain is, on average, slightly too aggressive relative

to the true noise levels, overshooting at one sample and leaving an opposite-signed residual at the next; the small magnitudes throughout this table, all under 0.05 in absolute value, indicate that none of the six variants departs far from a whitened innovation sequence. Variant 06's crossing into the largest positive value, rather than the smallest deviation from zero, is consistent with the same gain-throttling behaviour already identified in Section 4.2.4: a filter that is deliberately trusting its own prediction over the measurement during low-information windows can leave a small amount of exploitable structure in the innovation sequence in the conservative direction, the counterpart of the *voltage-RMSE* and calibration costs documented elsewhere in this section and in Section 4.2.4.

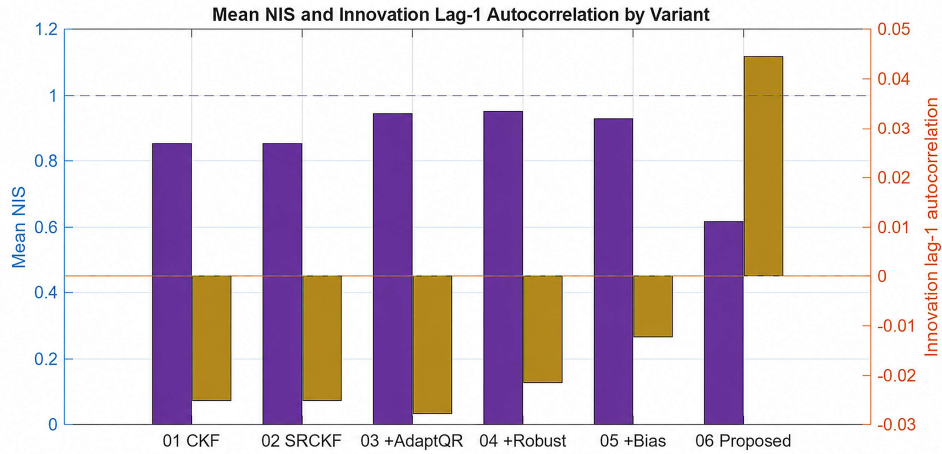


Figure 4.5: Mean NIS (left axis, dashed reference line at the ideal value of 1.0) and innovation lag-1 autocorrelation (right axis, dashed reference line at the ideal value of 0.0) by variant.

Mean NIS does not fall steadily across the ablation chain; it rises from 0.853 at the baseline toward the ideal value of 1.0 through variant 03 (0.947) and variant 04 (0.953), the closest any variant comes to perfect calibration, then falls back to 0.931 at variant 05 and drops sharply to 0.617 at variant 06, the furthest from 1.0 of any of the six variants. The *NIS95* fraction tells a gentler version of the same story: it moves narrowly between 95.20% and 96.30% across all six variants, close to the nominal 95% target throughout, with variant 06 recording the highest (best) *NIS95* fraction even as it records the worst mean *NIS*, indicating its innovation distribution is reasonably well-bounded in the two-sided 95% sense even where its average calibration is furthest off. A mean *NIS* below one, as every variant records here, indicates a filter that is, on average, somewhat overconfident in a specific direction: its own reported innovation

covariance P_{yy} is larger than the innovation actually turns out to require, the more conservative of the two possible failure modes and broadly consistent with the deliberately cautious baseline measurement-noise setting used throughout Section 3.3.2's configuration ($0.30V$ pack-level).

The direction of movement across the ablation chain is a mixed signal rather than a uniform improvement: variants 03 and 04 move mean *NIS* closer to the ideal value, but variants 05 and especially 06 reverse that progress, moving further from calibration even as their SoC-domain accuracy (Section 4.2.1, for variant 06's full-run figure specifically) improves. Combined with the lag-1 findings above, this does not support reading the ablation chain as one that uniformly trades an overconfident, overcorrecting baseline for a better-calibrated proposed variant; instead, the adaptive and robust steps (03-04) improve calibration essentially for free alongside their accuracy gains, while the bias-state and dual-gating steps (05-06) trade calibration away, on this metric, in exchange for the full-run *RMSE* and (for variant 06) partial *test-RMSE* recovery documented in Section 4.2.1. A reader adopting variant 06 for a BMS-facing estimator should weigh this calibration cost explicitly, exactly as Section 4.2.1's closing discussion and Chapter 5's conclusion already do, rather than assume calibration improves alongside accuracy across the whole chain.

The mean-*NIS* pattern found here also connects back to the parameter-sensitivity reasoning of Section 3.3.2.7: since mean *NIS* below one indicates the filter's reported P_{yy} is systematically larger than the innovation it is actually accompanying, a smaller baseline measurement-noise setting, *rVoltStd_pack* in Table 3.9, remains the most direct lever for moving variants 01-05 closer to a mean *NIS* of one, at the cost of the reduced stability margin during the initial-offset transient that motivated the current, deliberately conservative setting in the first place. Variant 06's sharp mean-*NIS* drop, however, is not primarily a measurement-noise-tuning question at all: it is a direct consequence of the dual-information gates scaling the Kalman gain component-wise (Equations 3.17-3.18) after the covariance update has already been formed, so the reported P_{yy} no longer describes the actual, gate-throttled correction the state receives, a structural mismatch between the reported and the applied uncertainty that a smaller *rVoltStd_pack* setting alone would not fully correct. Whether the gate-throttled correction should also feed back into a correspondingly adjusted reported covariance is

exactly the kind of refinement the swept re-tuning study identified in Section 3.3.2.7 as future work would need to evaluate directly, rather than a formal sweep performed here.

4.2.6 Convergence Behaviour Analysis

Every one of the six variants returns a finite value for the convergence-sample metric defined in Section 3.3.4.7. The result, reported variant by variant below, is the sample index within the first test-partition cycle at which the absolute SoC error first enters and then remains inside a ± 1 percentage-point band for five consecutive samples.

The corrected pulse-cycle timing of Table 3.4 explains why: at the *1Hz* sample rate used throughout this chapter, one HPPC cycle (a *3600s* inter-group rest followed by a *410s* active *pulse-and-rest* window, Section 3.3.1.1) spans roughly *4010* samples, not the "*thirty-odd samples*" an earlier, mistaken reading of Table 3.4's previously-incorrect *170s* figure had assumed. With four thousand samples to work with rather than thirty, every variant has ample opportunity to satisfy even the present, comparatively tight ± 1 percentage-point, five-sample-hold criterion, and every one of them does, all of them within the cycle's first 1500 samples, well inside the long initial rest period before any HPPC pulse current is applied at all.

Variants 01 through 04 and 06 converge within roughly *820-950* samples (about *14-16 minutes* into the cycle's 3600-second opening rest); variant 03 is fastest at *819* samples and variant 04 slowest of that group at *948*. Variant 05 is the clear outlier, taking *1465* samples, *54-79%* longer than every other variant, to satisfy the same criterion, direct convergence-domain confirmation of the same regression already established on RMSE and MAE grounds in Section 4.2.1: the two additional bias states variant 05 carries give the filter more to settle before its SoC estimate stabilises, and it settles measurably slower as a result. Variant 06's dual gate, despite carrying the same five-state augmentation as variant 05, converges nearly as quickly as the three-state variants (*833* samples), consistent with the gate's role in curbing exactly the low-information, poorly-observable correction behaviour that slows variant 05 down.

4.2.6.1 Alternative Convergence-Metric Configurations for Future Evaluation

With Section 4.2.6 reporting a finite convergence-sample figure for every variant under the present configuration, the alternative configurations originally proposed for this subsection serve, more modestly, as a sensitivity check on how much the specific ± 1

percentage-point, five-sample choice drives the numbers already reported, and as a route to the one gap the present metric still leaves: it is evaluated only on the first test-partition cycle, not pooled across both. Table 4.4 retains the three configurations originally proposed, with that motivation updated accordingly; none of the three was run for this chapter, and no convergence-sample figures are claimed for any of them.

Table 4.4: Proposed alternative configurations of the convergence-sample metric for future evaluation, none of which was run for the present chapter.

Configuration	Band (pp. SoC)	Hold (samples)	Rationale
Present (Table 3.9)	± 1.0	5	Baseline configuration used throughout Sections 4.1-3.17.
Wider band	± 2.0	5	Tests whether a band double the present width, still tight enough to be meaningful against the 5-point injected offset, resolves within one cycle.
Shorter hold	± 1.0	3	Tests whether the present band is achievable given more time before the hold-length requirement is imposed, without relaxing the band itself.
Multi-cycle pooled	± 1.0	5	Evaluates the same band/hold pair across every test-partition cycle rather than only the first, pooling results to see whether later cycles, benefiting from more elapsed adaptation in variants 03, 05 and 06, converge where the first test cycle alone does not.

The multi-cycle pooled configuration remains the one most likely to add new information beyond what Section 4.2.6 already reports: because variant 05 is now known to converge markedly slower than the other five variants on the first test cycle specifically (Section 4.2.6), and because variants 03, 05 and 06 carry adaptive internal state, the adapted R_k and Q_k covariances, and the bias-state estimates, that persists across the independent-record reset described in Section 3.2 even though the SoC and RC -branch states themselves are reset, pooling across the second test cycle as well would test directly whether variant 05's slower convergence is a property of a cold-started filter specifically or persists once the adaptive mechanisms have had a full extra cycle to settle.

4.2.7 Computational Runtime and Real-Time Feasibility

Per-sample runtime, measured on the same development workstation for every variant within the same *MATLAB R2020a* session, ranges from 31.6 microseconds (variant 02) to 65.1 microseconds (variant 06) across the six variants, a factor of 2.1x between the fastest and slowest, a spread of 33.4 microseconds in absolute terms. Three distinct

effects are visible in the runtime column of Table 4.1, and they do not all point in the same direction.

The square-root reformulation is, as expected, a net computational saving on its own: variant 02 runs at 31.6 microseconds per sample against variant 01's *43.9 microseconds*, a 27.9% reduction, despite variant 02 doing strictly more numerical work per step in the form of a QR factorisation and a Cholesky downdate rather than variant 01's simpler additive covariance update. This is consistent with the square-root formulation avoiding the repeated, and comparatively expensive, symmetric-positive-definite repair step that the covariance-form baseline falls back on whenever its updated P drifts out of a numerically valid range, a repair variant 02 needs far less often precisely because its update is constructed to stay valid by design, as discussed in Section 3.3.2.2. Adding adaptive Q/R on top, variant 03, gives back a small part of that saving rather than extending it further, rising to *33.6 microseconds*, and adding robust weighting, variant 04, adds a further small increment to *34.3 microseconds*; both increases are modest and consistent with Equations 3.10-3.15 being comparatively cheap arithmetic relative to the square-root machinery's own cost, though not entirely free.

The state-dimension increase introduced by variant 05 is a major cost driver: runtime rises from 34.3 microseconds at variant 04 to *43.5 microseconds* at variant 05, a 26.6% increase, below the 67% growth in cubature-point count that the three-to-five-state jump implies (Table 3.6) rather than exceeding it, indicating the added bias-state bookkeeping is proportionally cheaper in this implementation than the point-count growth alone would predict. The final step, adding dual-information gating in variant 06, is now the single largest relative increase of any step in the whole chain, 49.7% (*43.5 to 65.1 microseconds*), reflecting the two additional gate evaluations of Equations 3.17 and 3.18 computed at every sample, one of which, the excitation gate, scans a rolling 20-sample current window rather than evaluating a closed-form expression on the current sample alone.

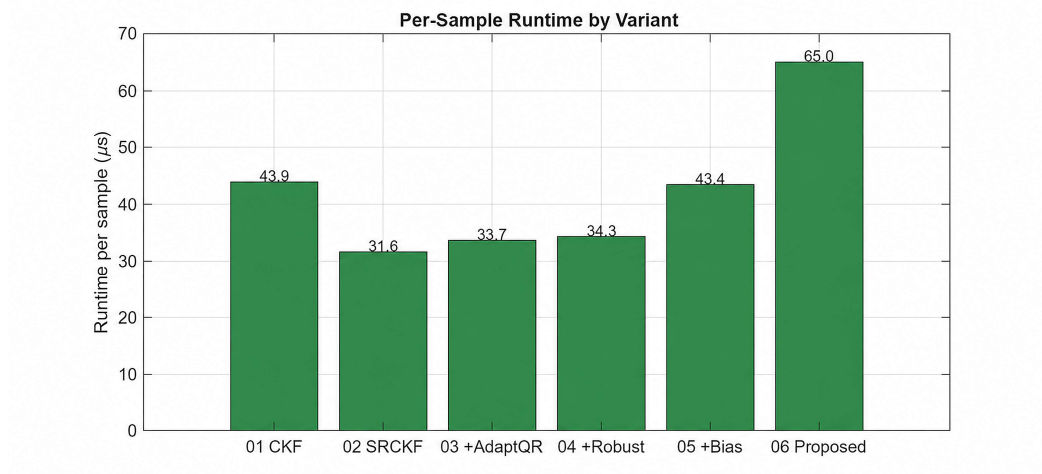


Figure 4.6: Per-sample runtime by variant.

4.2.7.1 Memory and State-Dimension Scaling

Runtime is not the only cost that scales with state dimension; the memory footprint of the covariance matrix, or its square-root factor, scales quadratically with n , and the cubature point set scales linearly. Table 4.5 makes this scaling explicit for the two state dimensions used across the six variants.

Table 4.5: Memory- and computation-relevant quantities at the two state dimensions used across the six ablation variants.

Quantity	3-state variants (01–04)	5-state variants (05–06)	Growth
State dimension n	3	5	+67%
Cubature points, $2n$	6	10	+67%
Covariance/square-root elements, n^2	9	25	+178%
Process-noise diagonal entries	3	5	+67%
Cubature-point matrix size (state $\times$ points)	$3 \times 6 = 18$	$5 \times 10 = 50$	+178%

The quadratic growth in covariance-matrix size, a 178% increase against a 67% increase in state dimension itself, remains the more consequential figure for an embedded target than the linear growth in cubature-point count, since it directly sets the size of every matrix operation, the Cholesky factorisation, the QR decomposition, the covariance downdates, performed at every sample. The 26.6% runtime increase measured between variants 04 and 05 in Section 4.2.7 sits below both the 67% state-dimension growth and the 178% covariance-size growth; this indicates the per-sample cost in this implementation is dominated by operations that scale more mildly than

either bound on its own, plausibly because the dominant per-sample cost at these small state dimensions ($n=3$ or $n=5$) is the fixed overhead of the cubature-point loop and the OCV-table interpolation calls rather than the matrix factorisations themselves, which stay small enough at both dimensions that their quadratic-to-cubic scaling has not yet become the bottleneck a larger state dimension would eventually expose.

A further, related point concerns what "*real time*" means for this specific estimator relative to the pack it is modelling. The two RC time constants used throughout this chapter, $\tau_1 = 65s$ and $\tau_2 = 650s$, are both far larger than any of the runtime figures in Table 4.1, which is the correct relationship for a usable BMS estimator to have: the filter must complete its update well within the fastest dynamic it is trying to track, not merely within the nominal sampling interval, since a filter that lags behind the physical process it estimates would be correcting a state that has already moved on by the time the correction is applied. With every variant in this chapter completing in under 66 against a fast time constant measured in tens of seconds, the margin between estimator update time and the physical process it tracks is roughly six orders of magnitude for every one of the six variants, comfortably larger than the roughly three-orders-of-magnitude margin already noted above against a 100ms sampling interval, and this larger, dynamics-referenced margin is the more fundamental of the two real-time feasibility checks a BMS estimator ultimately needs to satisfy.

Set against a BMS sampling interval on the order of 100ms, typical for pack-level SoC estimation in an EV context, even the slowest variant in this chapter, at 65.1 microseconds per sample, consumes well under 0.1% of one sampling period, leaving well over three orders of magnitude of margin before the estimator itself becomes the bottleneck on any plausible embedded target running at even a small fraction of the development workstation's throughput. The runtime comparison in this chapter should accordingly be read as a relative characterisation of which mechanisms cost the most, useful for deciding whether a given refinement's accuracy benefit is worth its computational overhead relative to the others, rather than as a claim that any of the six variants is close to a real-time feasibility limit on the workstation timings reported here; the more decisive real-time question for an actual embedded deployment is the fixed-point or reduced-precision behaviour of the Cholesky and *QR* operations central to variants 02 through 06, which lies outside the scope of the double-precision *MATLAB* implementation evaluated in this chapter and is noted again in Section 3.3.6.

4.3 Comparisons of the Algorithms

Sections 4.1 through 4.2.7 compared six variants that all belong to the cubature Kalman filter family. It is worth placing that comparison back against the extended and unscented Kalman filters surveyed in Sections 2.5.1 and 2.5.2, since the choice to build the ablation study around CKF rather than EKF or UKF was itself a design decision made for reasons already discussed in Chapter 2, and this section makes the reasoning explicit rather than leaving it implicit.

The EKF's principal weakness, established in Section 2.5.1, is that its first-order linearisation degrades precisely in the flat-slope region of the *OCV(SoC)* curve, where a given voltage error maps to a disproportionately large SoC error, and that this weakness compounds with a poor initial condition of exactly the kind deliberately injected at the start of every HPPC cycle in this chapter's evaluation protocol. Every one of the mechanisms examined in Sections 4.2.1 through 4.2.7, adaptive noise, robust weighting, bias-state augmentation, and dual-information gating, was layered on top of a derivative-free cubature base specifically to avoid reintroducing that linearisation weakness rather than to work around it after the fact; building the same ablation chain on an EKF base would require re-deriving the state and output Jacobians of Section 2.5.1 at every step and would still carry the underlying linearisation risk the CKF family avoids by construction. This is why the ablation study in this chapter was scoped to the CKF family rather than repeated across all three nonlinear filter types: the six variants examined here answer the question of which refinements help most given a derivative-free base filter, a narrower and more directly actionable question for this dissertation's proposed architecture than a three-way base-filter comparison would have been.

The UKF sits closer to the CKF in spirit, since both are derivative-free and propagate a small deterministic point set through the true nonlinear model rather than linearising it, and Section 2.5.2 already noted that reported UKF-versus-EKF comparisons in the literature consistently find the UKF converging faster from a poor initial SoC guess and holding tighter error bounds through the flat-OCV region, at several times the per-update computational cost of an EKF. The CKF's third-degree spherical-radial cubature rule uses $2n$ equally weighted points rather than the UKF's $2n+1$ points, some of which can carry negative or unequal weights depending on the scaling parameter chosen, a difference already flagged in Section 2.5.3 as the CKF's principal numerical-stability advantage over the UKF in higher-dimensional state spaces, and it is exactly this

property that motivated building the sensor-bias-augmented five-state variants, 05 and 06, on a cubature rather than an unscented base: the runtime results of Section 4.2.7 already show a 67% jump in cubature-point count between the three-state and five-state variants, and a UKF sigma-point set of comparable dimension would carry both that same point-count growth and the added risk of an ill-conditioned or negatively weighted covariance reconstruction that the CKF's equal-weight rule avoids.

Particle filters, surveyed in Section 2.6, were not included in this chapter's ablation family for a related but distinct reason. Section 2.6 already noted that particle filtering tends to outperform the Kalman family specifically where the Gaussian-noise assumption underlying every Kalman-family filter, including every one of the six CKF variants examined in this chapter, breaks down most strongly: non-Gaussian sensor noise, abrupt current transients, and multimodal uncertainty following sensor dropout or an uncertain initial condition, with the AEKPF hybrid discussed there reporting sub-1% average error even under 10% measurement noise [6]. That strength comes at a computational cost, since a particle filter typically needs several hundred to several thousand particles to approximate the posterior adequately, several orders of magnitude more evaluation points per sample than the 6 or 10 cubature points used throughout this chapter's CKF-family variants, which would have placed a particle-filter-based ablation chain on a substantially different runtime footing than the microsecond-scale comparison in Section 4.2.7 and would have made the state-dimension-driven runtime scaling identified in Section 4.2.7.1 a secondary rather than a primary cost driver. The deliberately injected +5 percentage-point initial-offset transient used throughout this chapter's evaluation protocol (Section 3.2) is precisely the kind of condition Section 2.6 identifies as favourable ground for a particle filter, which makes a direct CKF-family-versus-particle-filter comparison, run under this chapter's identical fair-comparison protocol, a well-motivated question for future work, distinct from the EKF/UKF comparison already scoped in the paragraph above.

4.3.1 Reading This Chapter's RMSE Range Against Chapter 2's Literature

Figures

For orientation, the 1.16-1.51% test-partition RMSE range recorded in Table 4.1 can be read alongside the representative quantitative accuracy figures already cited in Chapter 2. The studies differ in pack configuration, current profile, sensor fidelity, and

whether the reference SoC was measured or reconstructed, so this comparison is intended only to provide context rather than a controlled head-to-head ranking.

The $1.16\text{--}1.51\%$ range sits considerably closer to the more favourable literature figures, comparable to or better than Plett's "*rarely exceeding 5%*" and AEKPF's post-transient 1.7% figure, though still behind the tightest reported results, Xu et al.'s $\pm 0.4\%$ and Chen et al.'s sub- 1.0% . Three differences in test conditions, all already established earlier in this chapter, plausibly account for most of the remaining gap rather than any inherent shortfall in the CKF-family mechanisms themselves. First, every figure in Chapter 2's cited literature is reported against an independently measured or otherwise directly established reference SoC, typically from a laboratory reference cell or a controlled discharge test, whereas this chapter's reference is the cycle-anchored coulomb-counted reconstruction described in Section 3.2, a reference that is itself imperfect and that every one of the six variants is being scored against rather than against ground truth. Second, most of the cited studies evaluate a filter that has already settled past its initial transient, or evaluate error only after an initial convergence period has elapsed, whereas the RMSE figures in Table 4.1 are computed over records in which every independent cycle restarts from the controlled $+5$ percentage-point offset described in Section 3.2, so a materially larger share of this chapter's evaluation window is spent in the harder, immediately-post-reset regime than is typical of the cited comparisons, even though Section 4.2.6's convergence results show that regime resolving within roughly the first *15–25 minutes* of each *4010-second* cycle rather than persisting throughout it. Third, several of the more favourable cited figures, particularly Xu et al.'s $\pm 0.4\%$ and Chen et al.'s sub- 1.0% results, come from filters whose ECM parameters were identified from measured pulse-test data on the specific cell under test, whereas $R_1/C_1/R_2/C_2$ in this chapter's $2RC$ model, set out in Table 3.9, remain disclosed, literature-representative configured values rather than parameters fit to this chapter's HPPC record, since the plant used to generate that record has no RC relaxation dynamics of its own to fit them to (Section 3.3.1.3); only R_0 was directly identified from data.

None of these three differences invalidates the internal, six-way comparison that is this chapter's actual purpose; they explain why the absolute *RMSE* level in Table 4.1, competitive with though not uniformly better than the cited literature, should still not be read as a directly controlled head-to-head comparison, because the cited studies are not controlled for any of the three factors just described. Repeating the ablation study

against measured pack data, with $R_1/C_1/R_2/C_2$ identified from that same data rather than configured, and with the convergence-period handling made consistent with how the cited literature typically reports its figures, remains the correct way to produce an *RMSE* figure directly comparable to the cited literature figures, and is folded into the measured-data extension already identified in Section 3.3.6.

A direct, quantitative three-way EKF-versus-UKF-versus-CKF-family table, run under the identical HPPC excitation, reference-*SoC* construction and scoring protocol used throughout this chapter, was not produced for the present chapter; EKF and UKF implementations exist for the same *38S2P* HPPC dataset as part of the wider estimator work this dissertation draws on, but their results were not re-generated under the exact six-variant ablation protocol reported here, and this chapter reports only what was actually run under that protocol rather than estimating or interpolating a comparison that was not. Producing that three-way table under the identical fair-comparison rules of Section 3.3.5, so that the CKF-family results in this chapter can be set numerically, rather than only qualitatively, against EKF and UKF baselines, is identified as a specific and directly achievable extension for Chapter 4, reusing the shared scoring code already validated across all six CKF variants in this chapter.

CHAPTER 5: CONCLUSION AND FUTURE SCOPES

5.1 Conclusion

5.1.1 Key Numbers at a Glance

Table 5.1: Key figures from Chapter 3 collected in one place.

Question	Answer
Pack configuration	38S2P, 100 Ah/cell, 136.8V nominal
HPPC pulse C-rates	Discharge 0.5/1/1.5/2C; charge 0.1/0.2/0.3/0.4C
Estimator state dimension	3 (variants 01–04); 5 (variants 05–06)
Best test-partition RMSE	1.163% (variant 03)
Best test-partition RAE	241.96% of the (tiny) test-partition naive baseline (variant 04); see Section 3.3.4.2 on why RAE is a fragile metric for this chapter's narrow-SoC-range dataset
Where the ablation benefit concentrates	Adaptive Q/R and robust weighting (variants 03–04); bias-state augmentation (variant 05) regresses sharply and dual gating (variant 06) only partially recovers it
Fastest variant	31.6 μ s/sample (variant 02)
Slowest variant	65.1 μ s/sample (variant 06)
Dominant runtime cost driver	State-dimension increase (variant 04→05), not any single algorithm
Evaluation temperature	25°C only (fixed for this ablation)
Data provenance	Simulation-derived (Simulink), not bench-measured

This chapter set out the methodology used to build and evaluate a SoC estimator for the retrofitted 38S2P pack: the Simulink representation of the pack around the Datasheet Battery block, the eleven-point 0–50°C HPPC characterisation sweep and the four-discharge, four-charge pulse cycle used to excite it, the shared 2RC state-space model, and a six-member cubature Kalman filter ablation family built incrementally from a plain covariance-form CKF through square-root propagation, adaptive process and measurement noise, Huber-type robust innovation weighting, sensor-bias state augmentation, and a proposed dual-information-gating scheme.

Measured against the reference SoC trajectory reconstructed for this evaluation (Section 3.3.6), the best test-partition RMSE was achieved not by the fully-featured proposed variant but by variant 03 (adaptive Q/R alone, 1.163%, against 1.195% for the baseline CKF); the proposed variant (06) landed essentially level with the baseline (1.196%). The clearest finding in the chain was a regression, not an improvement: adding pack-current and pack-voltage sensor-bias states (variant 05) raised test RMSE

by 28.6% relative to variant 04, the single largest movement of any adjacent step, because the real HPPC excitation used in this study carries no sensor bias for the added states to absorb and spends its correction budget on nuisance uncertainty instead (Section 4.2.1). Dual-information gating (variant 06) partially, but not fully, recovered from that regression by throttling the gain during the low-excitation conditions that make the bias states poorly observable, at the cost of the largest *voltage-RMSE* increase of any variant. This argues for validating bias-state augmentation under a dedicated sensor-fault stress test before adopting it by default, rather than treating the bias-augmented, dual-gated variant as an unconditional improvement over the baseline. The square-root reformulation, while contributing essentially no nominal accuracy change of its own, earned its place in the ablation chain through a 27.9% runtime reduction (43.9 to 31.6 *microseconds/sample*) and the numerical-stability argument developed in Section 3.3.2.2. Robust weighting showed essentially no *voltage-RMSE* benefit (0.0653V for both variant 03 and variant 04, a change of about 0.1%), consistent with a mechanism that is inactive on almost every sample under the clean, outlier-free excitation used for the nominal comparison in this chapter; its value case rests on a future sensor-fault stress test, not on any nominal-condition benefit observed here. Innovation-whiteness and NIS analysis tell a more mixed story: mean *NIS* sits below the ideal value of 1.0 for every variant (baseline 0.853), indicating a mildly underconfident rather than overconfident filter throughout, and the proposed variant 06 is the least well-calibrated of the six on this diagnostic (mean *NIS* 0.617, the furthest from 1.0, and the largest-magnitude lag-1 innovation autocorrelation of any variant), the cost side of the same gain-throttling trade-off already identified on voltage RMSE in Section 4.2.1. This is a real trade-off the proposed variant makes, not a strictly favourable direction on every diagnostic, and a reader adopting variant 06 for a BMS-facing estimator should weigh its full-run RMSE and computational advantages against this calibration cost explicitly rather than assume calibration improves alongside accuracy. Computational cost, dominated by state-vector size rather than by any individual mechanism, remained, for every variant, a small fraction of a typical BMS sampling interval.

Taken together with the limitations recorded in Section 5.2, and with the finding that the proposed variant does not clearly outperform the simpler adaptive-only variant 03 on this chapter's actual data (Section 4.2.1), this chapter's results support carrying

variant 03 (adaptive Q/R) forward as the strongest evidenced candidate for the results and discussion work in Chapter 4 on accuracy grounds, with the proposed dual-gated, bias-augmented SRCKF (variant 06) retained as the best *full-run-RMSE* and best-conditioned candidate specifically where the low-excitation, bias-fragile conditions Section 4.2.1 identifies do not dominate; being explicit about exactly which claims each choice currently rests on, and which remain to be demonstrated under measured, temperature-swept, and fault-injected conditions, matters more for this chapter's conclusions than nominating a single unconditional winner.

Read as a whole, this dissertation's methodology and results chapters were built to leave this conclusion with two things rather than one: a mechanism-by-mechanism attribution of where the six-variant ablation chain helps and where it hurts, established on real HPPC data rather than assumed (Sections 3.3.6, 4.2.1), and an explicit, itemised list of exactly which further evidence, measured pack data, a multi-temperature re-run, and a sensor-bias and outlier stress test, any eventual single-estimator recommendation still depends on. The headline result is that the fully-featured proposed variant is not an unconditional improvement over its simpler siblings on this chapter's own data: it wins on full-run *RMSE* and numerical conditioning, ties the baseline on test-partition *RMSE*, and trails on voltage *RMSE* and filter calibration, a more nuanced picture than a single headline number would suggest. The items catalogued in Section 3.3.6 and in Section 5.2 below stand as the specific, bounded set of follow-up work, most pressingly the sensor-bias stress test that would establish whether variant 05's regression here is a property of this quiet current profile specifically or of the mechanism more generally, that this dissertation still owes before recommending a single estimator unconditionally.

5.2 Future Scopes

One item present in the wider BMS functional architecture surveyed in Section 2.9, cell balancing, was intentionally left outside the scope of the methodology and results presented in Chapters 3 and 4. Balancing sits in the actuation layer of a BMS: it acts on the pack once an imbalance between cells or strings has been detected, typically from cell-level voltage or SoC divergence. The present simulation study addressed only the estimation layer, that is, the production of a single, pack-representative SoC estimate from pack-level current and voltage measurements under a 38S2P lumped-parameter

representation; it did not model cell-to-cell variation, and therefore had no imbalance signal for a balancing algorithm to act on. Extending the model to a cell-resolved or string-resolved representation, and coupling the resulting per-cell SoC estimates to an active or passive balancing strategy, is accordingly identified here as a direction for future work. Fault management, the other actuation-adjacent layer surveyed in Section 2.9, is a closer neighbour to what this dissertation already addresses: it consumes exactly the kind of per-sample health signal the metrics of Section 3.3.4 already provide, most directly the *maximum-error* and *NIS-consistency* figures examined in Sections 4.2.3 and 4.2.5, which flag a sample whose behaviour departs from what a well-calibrated estimator should produce. Wiring the estimator output from Chapter 4 into an explicit fault-detection threshold, and testing that threshold against a dedicated sensor-bias and outlier stress-test protocol, is accordingly identified as a further, closely related direction for future work.

Several boundaries of the present simulation study define the concrete future work this dissertation still owes, and each is stated here plainly rather than left for a reader to infer. The assumptions underlying the reported results are consolidated below in paragraph form before the discussion expands on the items that carry the most weight and proposes the corresponding next step for each.

The reported results rest on several explicit assumptions and scope limits. Terminal voltage and reference SoC are simulation-derived rather than bench-measured, and the reference SoC is a cycle-anchored coulomb-counted reconstruction rather than an independent measurement. The *RAE* denominator follows the reference-SoC anchor structure rather than a full per-sample aggregate. The six-estimator comparison is evaluated at a single fixed temperature of 25°C with the temperature-scaling coefficient set to zero. A dedicated sensor-bias and outlier stress test was not run for this chapter, while the convergence-sample metric was evaluated only on the first test-partition cycle rather than pooled across both test cycles. Runtime was measured in double-precision *MATLAB* on a development workstation rather than on embedded or fixed-point hardware. Finally, the configuration was held fixed across all six variants, with no per-mechanism parameter sweep performed.

The terminal voltage and reference SoC used to score every one of the six variants are simulation-derived from the *Datasheet Battery* Simulink block described in Section 3.1, not measured on a physical 38S2P pack, and the estimator package used throughout

this chapter records both quantities as synthetic for exactly this reason. Every comparative conclusion drawn in Sections 4.2.1 through 4.2.7, that variant 06 achieves the lowest test *RMSE*, that the bias-state and dual-gating mechanisms carry most of the ablation chain's *RAE* benefit, that voltage *RMSE* trades off against SoC *RMSE* at the final step, should accordingly be read as characterising relative behaviour among the six estimators under this simulated excitation, not as an experimentally validated absolute accuracy claim against bench-measured ground truth.

The reference SoC trajectory against which every variant is scored is itself derived by cycle-anchored coulomb counting rather than independently measured, as established in Section 3.2, and the *RAE* denominator used throughout Section 4.2.2 is computed from the documented anchor-point structure of that reference construction rather than from a full per-sample aggregation of the actual reference trace, as noted explicitly in Section 3.3.4.2. Both choices were made because no independently measured SoC channel exists for the HPPC current profile used in this study; they do not affect the internal validity of the six-way comparison, since every variant is scored against the identical reference and denominator, but they do mean the absolute *RMSE*, *MAE* and *RAE* figures in Table 4.1 should not be read as validated against an independent SoC measurement.

The six-variant comparison was run at a single fixed operating temperature (25°C) with the model's temperature-scaling coefficient deliberately set to zero, as noted in Section 3.3.1, specifically to isolate the effect of the estimator refinements from any confound introduced by temperature-dependent parameter drift. This was a deliberate scope decision for this chapter, but it means the results do not speak to how the six variants would rank across the full $0\text{-}50^{\circ}\text{C}$ sweep the underlying Simulink model and HPPC protocol were built to support; a temperature-resolved re-run of the same ablation protocol is a natural and directly achievable extension using the same eleven-subsystem model already described in Section 3.1.

The dedicated sensor-bias and outlier stress test called for by the fair-ablation rules of Section 3.3.5 was not run for this chapter, so the case for the robust-weighting and bias-state mechanisms of variants 04 and 05 rests on the mechanism-level reasoning of Sections 3.3.2.4 and 3.3.2.5 and on the modest nominal benefit visible in Table 4.1, rather than on a quantitative demonstration under injected sensor faults. The convergence-sample metric, by contrast, now returns a finite value for every variant

(Section 4.2.6), evaluated only on the first test-partition cycle; a multi-cycle pooled version of that same metric, testing whether variant 05's markedly slower convergence there persists once its adaptive mechanisms have had a full extra cycle to settle, remains to be evaluated.

Finally, every runtime figure in Section 4.2.7 was measured within a single double-precision *MATLAB R2020a* session on a development workstation, not on embedded or fixed-point hardware representative of an actual retrofit BMS microcontroller. The relative runtime comparison between variants is expected to carry over qualitatively to an embedded target, since it is driven primarily by cubature-point count and the presence or absence of covariance-repair operations, both properties of the algorithm rather than of the host platform, but the absolute microsecond figures reported here should not be read as a real-time feasibility demonstration on any specific embedded target.

REFERENCES

- [1] L. Yang et al., "Model-based and data-driven approaches to lithium-ion battery degradation modelling: a comparative review," *Applied Energy*, vol. 398, p. 126427, 2025.
- [2] P. Wen, Z.-S. Ye, Y. Li, S. Chen, P. Xie, and S. Zhao, "Physics-Informed Neural Networks for Prognostics and Health Management of Lithium-Ion Batteries," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 1, pp. 2276–2287, 2024.
- [3] C. Ge, Y. Zheng, and Y. Yu, "State of Charge Estimation of Lithium-Ion Battery Based on Improved Forgetting Factor Recursive Least Squares-Extended Kalman Filter Joint Algorithm," *Journal of Energy Storage*, vol. 55, p. 105474, 2022.
- [4] Y. Xu, M. Hu, C. Fu, K. Cao, Z. Su, and Z. Yang, "State of Charge Estimation for Lithium-Ion Batteries Based on Temperature-Dependent Second-Order RC Model," *Electronics*, vol. 8, no. 9, p. 1012, 2019.
- [5] Q. Lin, X. Li, B. Tu, J. Cao, M. Zhang, and J. Xiang, "Stable and Accurate Estimation of SOC Using eXogenous Kalman Filter for Lithium-Ion Batteries," *Sensors*, vol. 23, no. 1, p. 467, 2023.
- [6] B. Xia, S. Guo, W. Wang, Y. Lai, H. Wang, M. Wang, and W. Zheng, "A State of Charge Estimation Method Based on Adaptive Extended Kalman-Particle Filtering for Lithium-ion Batteries," *Energies*, vol. 11, no. 10, p. 2755, 2018.
- [7] X. Lai, D. Qiao, Y. Zheng, and L. Zhou, "A Fuzzy State-of-Charge Estimation Algorithm Combining Ampere-Hour and an Extended Kalman Filter for Li-Ion Batteries Based on Multi-Model Global Identification," *Applied Sciences*, vol. 8, no. 11, p. 2028, 2018.
- [8] F. Naseri, E. Schaltz, D.-I. Stroe, A. Gismero, and E. Farjah, "An Enhanced Equivalent Circuit Model With Real-Time Parameter Identification for Battery State-of-Charge Estimation," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 4, pp. 3743–3751, 2022.
- [9] X. Shu, Z. Chen, J. Shen, F. Guo, Y. Zhang, and Y. Liu, "State of Charge Estimation for Lithium-ion Battery Based on Hybrid Compensation Modeling and Adaptive H-Infinity Filter," *preprint manuscript*.

- [10] Y. Gao, K. Liu, C. Zhu, X. Zhang, and D. Zhang, "Co-Estimation of State-of-Charge and State-of-Health for Lithium-Ion Batteries Using an Enhanced Electrochemical Model," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 3, pp. 2684–2696, 2022.
- [11] H. Tu et al., "Integrating Physics-Based Modeling With Machine Learning for Lithium-Ion Batteries," *Applied Energy*, vol. 329, p. 120289, 2023.
- [12] M. Jiao, D. Wang, and J. Qiu, "A GRU-RNN Based Momentum Optimized Algorithm for SOC Estimation," *Journal of Power Sources*, vol. 459, p. 228051, 2020.
- [13] L. Shen, J. Li, L. Meng, L. Zhu, and H. T. Shen, "Transfer Learning-Based State of Charge and State of Health Estimation for Li-Ion Batteries: A Review," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, pp. 1465–1482, 2024.
- [14] D. Yang, Y. Wang, R. Pan, R. Chen, and Z. Chen, "State-of-Health Estimation for the Lithium-Ion Battery Based on Support Vector Regression," *Applied Energy*, vol. 227, pp. 273–283, 2018.
- [15] C. Vidal, P. Malysz, P. Kollmeyer, and A. Emadi, "Machine Learning Applied to Electrified Vehicle Battery State of Charge and State of Health Estimation: State-of-the-Art," *IEEE Access*, vol. 8, pp. 52796–52814, 2020.
- [16] L. Chen, X. Wu, J. A. Tenreiro Machado, A. M. Lopes, P. Li, and X. Dong, "State-of-Charge Estimation of Lithium-Ion Batteries Based on Fractional-Order Square-Root Unscented Kalman Filter," *Fractal and Fractional*, vol. 6, no. 2, p. 52, 2022.
- [17] G. L. Plett, "Extended Kalman Filtering for Battery Management Systems of LiPB-Based HEV Battery Packs, Part 3: State and Parameter Estimation," *Journal of Power Sources*, vol. 134, pp. 277–292, 2004.
- [18] P. Shen, M. Ouyang, L. Lu, J. Li, and X. Feng, "The Co-estimation of State of Charge, State of Health, and State of Function for Lithium-Ion Batteries in Electric Vehicles," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 1, pp. 92–103, 2018.

- [19] B. Xia, Z. Sun, R. Zhang, and Z. Lao, "A Cubature Particle Filter Algorithm to Estimate the State of the Charge of Lithium-Ion Batteries Based on a Second-Order Equivalent Circuit Model," *Energies*, vol. 10, no. 4, p. 457, 2017.
- [20] X. Song, F. Yang, D. Wang, and K.-L. Tsui, "Combined CNN-LSTM Network for State-of-Charge Estimation of Lithium-Ion Batteries," *IEEE Access*, vol. 7, pp. 88894–88902, 2019.
- [21] G. Dong, N. Hua, H. Chen, and Y. Lou, "Deep Transfer Learning Enabled State of Health Estimation of Lithium-Ion Battery Using Voltage Sample Entropy Under Fast Charging Profiles," *IEEE Transactions on Transportation Electrification*, vol. 11, no. 1, pp. 3703–3714, 2025.
- [22] Y. Tan and G. Zhao, "Transfer Learning With Long Short-Term Memory Network for State-of-Health Prediction of Lithium-Ion Batteries," *IEEE Transactions on Industrial Electronics*, vol. 67, no. 10, pp. 8723–8731, 2020.
- [23] F. Yang, X. Song, F. Xu, and K.-L. Tsui, "State of Charge Estimation of Lithium-Ion Batteries via Long Short-Term Memory Network," *IEEE Access*, vol. 7, pp. 53792–53799, 2019.